Science

Pupil's Book

Standard Four



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Science

Pupil's Book

Standard Four

THE UNITED REPUBLIC OF TANZANIA

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Commissioner for Education

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Introduction

The Science textbook has been prepared in line with the 2023 Science Syllabus for Primary Schools Standard III–VI issued by the Ministry of Education, Science and Technology. Some content in this book has been taken from the earlier books for Science and Technology for Standards Three, Four and Five published in 2018 and Standard Seven published in 2021, in accordance with the curriculum and syllabus of 2016.

This textbook has six chapters namely: Health principles, Diseases, Matter, Combustion, Energy and Computer coding. The content in this book has been presented through text, activities, experiments, exercises and illustrations. This has been done to enable you develop the required competencies from each chapter. Each chapter has exercises to enable you assess your understanding of the respective content. Thus, you are required to do all activities, experiments, questions and exercises. Moreover, consult your teachers, parents or guardians when searching for information from the library, internet and other sources.

Additional learning resources are available in TIE e-Library at *https://ol.tie.go.tz* or *ol.tie.go.tz*

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Chapter One

Health principles

Introduction

Good health is important for our well-being. In this chapter, you will learn about health principles. These include eating a balanced diet, doing physical exercises and maintaining both personal and environmental cleanliness. However, you will also explore safety signs and other important health information in the environment. The competencies developed will enable you to follow health principles. You will also be able to protect your health and the health of others.



Think Things you can do every day to take care of your health

Concept of health principles

Following health principles is important for staying healthy. Good health means being strong in both body and mind. To be healthy, you need to eat a balanced diet, do physical exercises regularly and ensure personal and environmental cleanliness. It is also important to follow safety signs and health information in the environment. Figure 1 shows examples of health principles.

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(a) Doing physical exercises

(b) Eating a balanced diet

Figure 1: *Following health principles*

Activity 1

1. Use the library or reliable online sources to learn about health principles.
2. Share your findings with your peers in the class for discussion.

Balanced diet

A balanced diet is a meal with all the food nutrients in the correct proportions. It is made up of foods from three main groups. The groups are energy-giving foods, body-building foods, protective foods. Energy-giving foods have carbohydrate and fat nutrients. Body-building foods have protein nutrients. Protective foods have vitamins and minerals nutrients. Eating a balanced diet helps to make the body's immune system strong to fight off diseases. Figure 2 shows examples of main food groups in a balanced diet.

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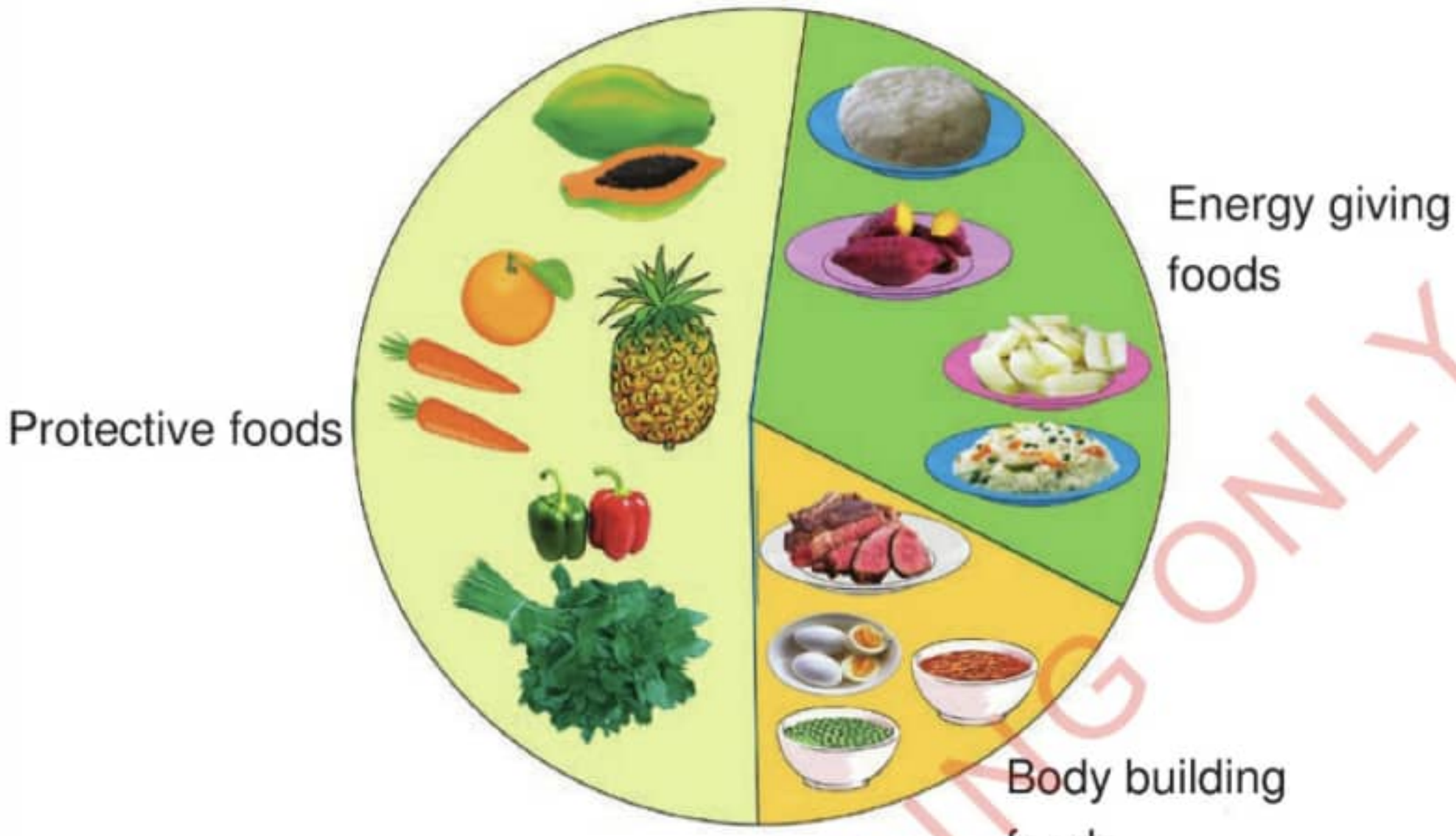


Figure 2: *Example of main food groups in a balanced diet*

Activity 2

1. Identify different foods available in your environment.
2. Sort the identified foods into energy-giving, body-building, or protective food groups.

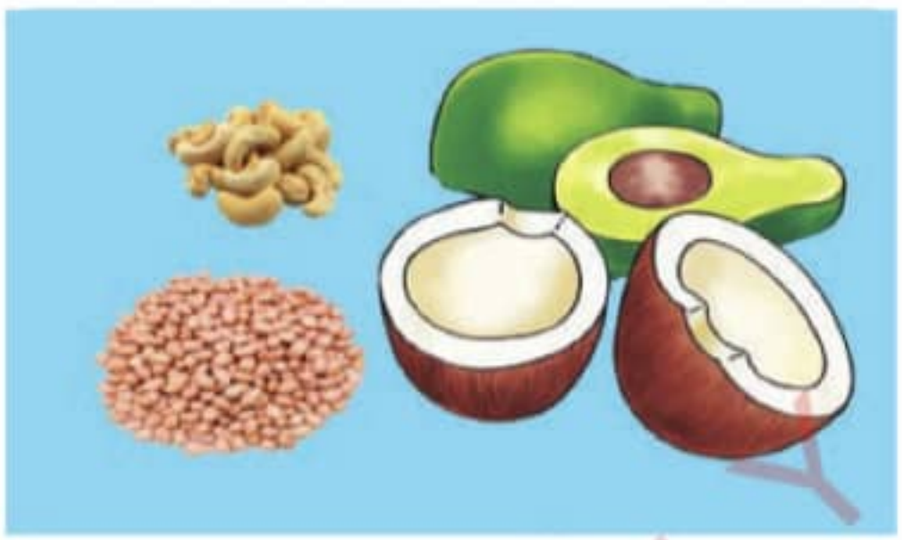
Energy-giving foods

Foods that give the body energy include those rich in carbohydrates and fats. Carbohydrate-rich foods are maize, potatoes, cassava, bananas, rice and bread. Foods rich in fats include coconut, peanuts, sunflower seeds and avocados. These foods provide the energy needed for daily activities like playing, farming and walking. Fats also help keep the body warm. Figure 3 shows examples of foods rich in carbohydrates and fats.

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(a) Foods rich carbohydrate



(b) Foods rich in fats

Figure 3: ***Foods rich in carbohydrates and fats***

Body-building foods

Body-building foods are rich in protein. These foods include fish, milk, meat, eggs, beans and peas. They help build the body and promote proper growth. They also help make new cells to replace old or dead ones. Figure 4 shows examples of body building foods.



Figure 4: ***Body building foods***

Protective foods

Protective foods are rich in vitamins and minerals. These foods support overall body health and help the body defend itself against diseases. Also, they help to build and strengthen the body's muscles and bones. Vitamins and minerals are found in vegetables like cabbage, spinach, eggplant, amaranth

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and carrot. They are also found in fruits like papaya, orange, avocado, grape and mango. Figure 5 shows examples of foods rich in vitamins and minerals.



Figure 5: *Foods rich in vitamins and minerals*

Water

The body needs enough water to work properly. But, water is not a main food group in a balanced diet. Water has minerals that help the body systems work properly. It helps in the process of breaking down food in the digestive system. Also, water regulates body temperature and helps in nutrient absorption. It also helps remove waste and harmful substances from the body.

Activity 3

1. Plan a balanced lunch using different types of food available in your environment.
2. Why did you choose each food in the meal?

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Balanced diet for different groups of people

Nutritional needs are different for each group of people. These groups are based on age, like children, adolescents and the elderly. Also, it depend on work people do, like manual workers, and health status, like the sick.

Children

Children need foods rich in protein because they are growing. This include milk, meat, fish, beans, peas, lentils and eggs. These foods help to build the body, strengthen the immune system and support brain development. Children also need energy-giving foods to provide strength needed for doing activities like playing. Protective foods are needed to make their immune system strong. Figure 6 shows a balanced diet for a child.

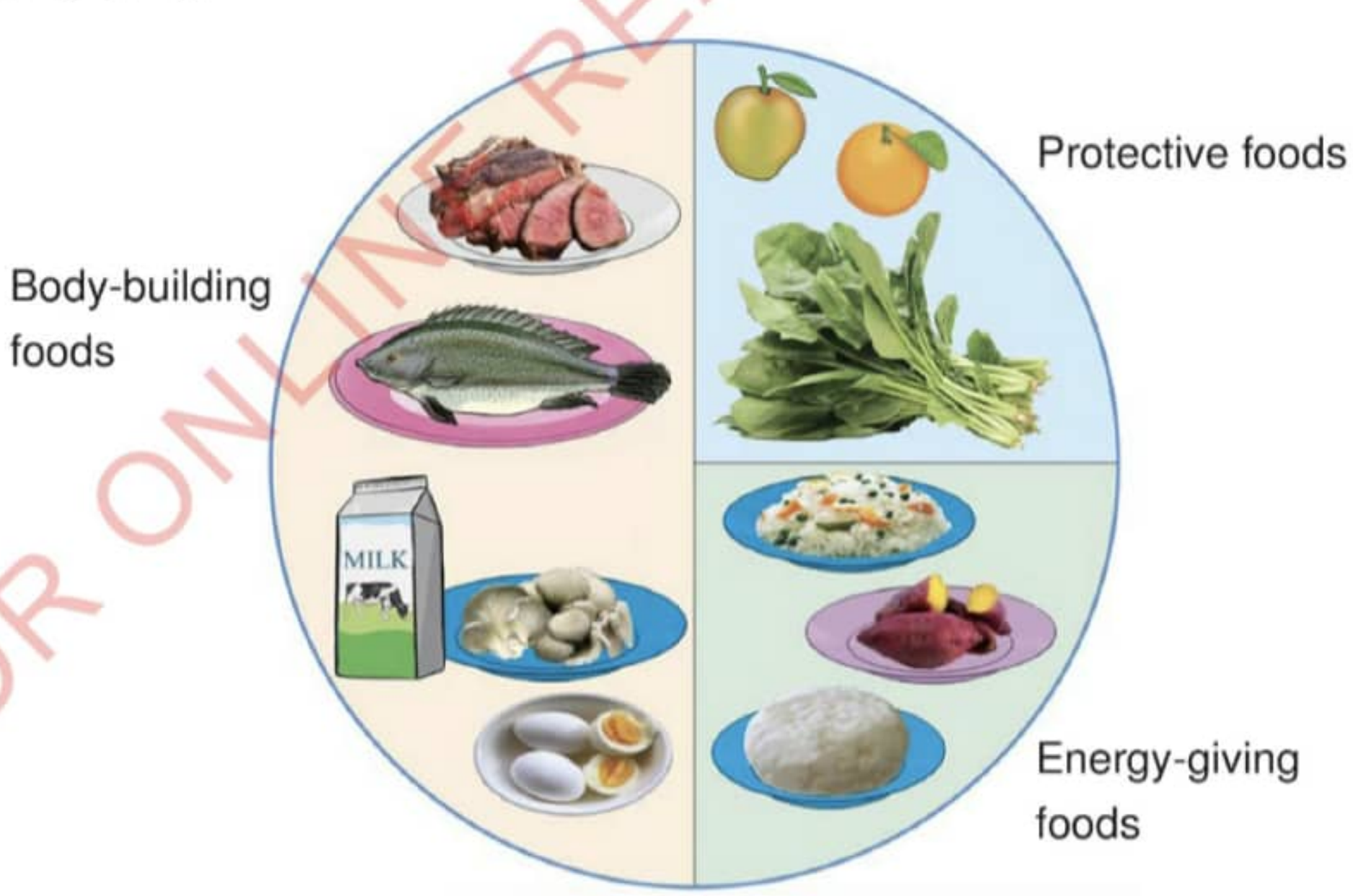


Figure 6: *Balanced diet for a child*

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Adolescents

During adolescence, the body grows fast and goes through changes. Adolescents need a diet rich in body-building, protective and energy-giving foods. At this age, they are involved in many activities like sports, farming and studying for long hours. Because of this, their diet should include plenty of energy-giving foods to provide enough energy for these activities. See Figure 7.

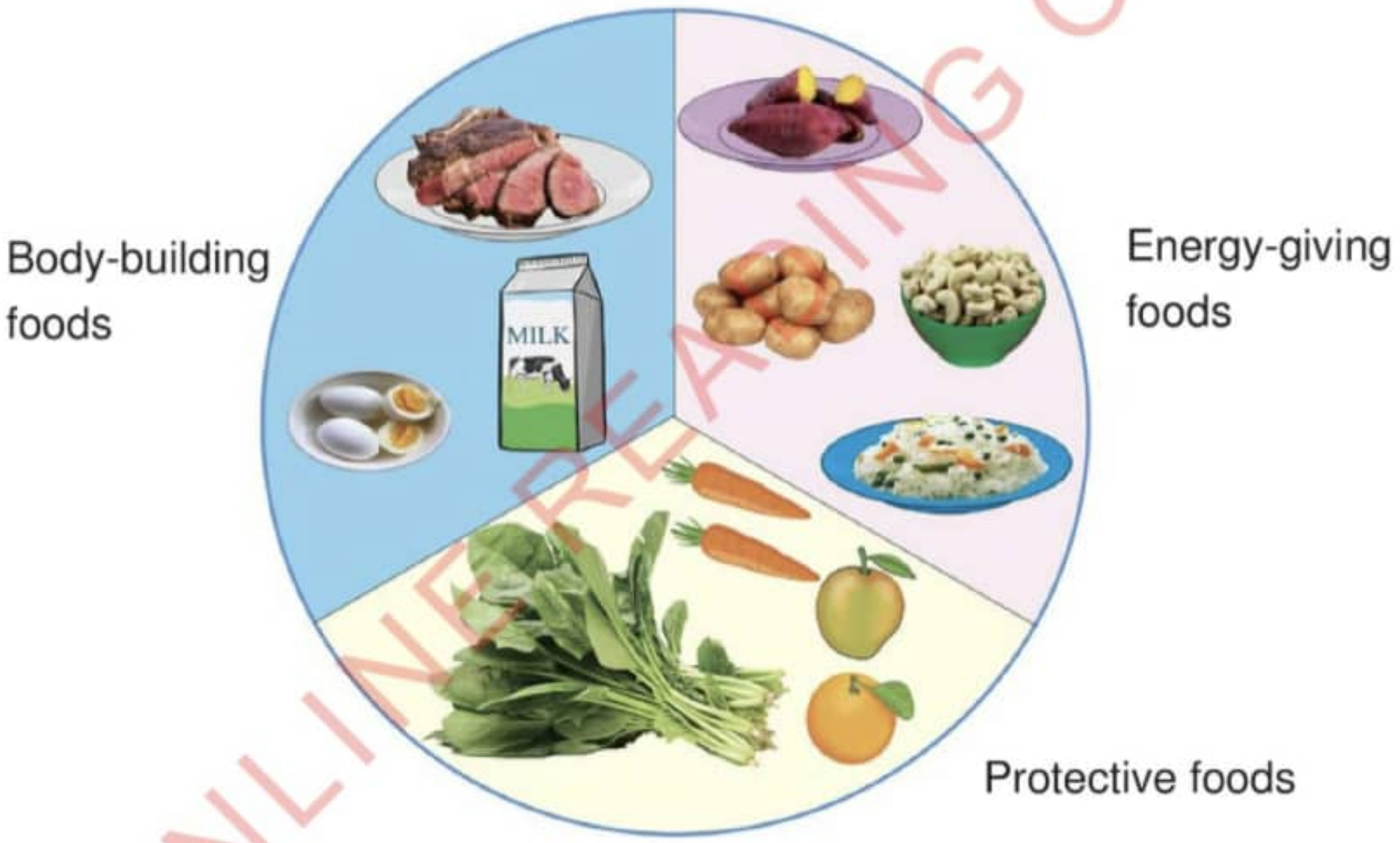


Figure 7: *Balanced diet for an adolescent*

Elderly

The elderly need foods rich in vitamins to protect them from illnesses. Their immune system gets weaker with age. They should also eat foods rich in minerals to keep bones and muscles strong. A small amount of energy-giving foods is needed to support light activities. This helps them avoid gaining too much weight. Body-building foods are also important to replace old or damaged cells. Figure 8 shows a balanced diet for elderly.

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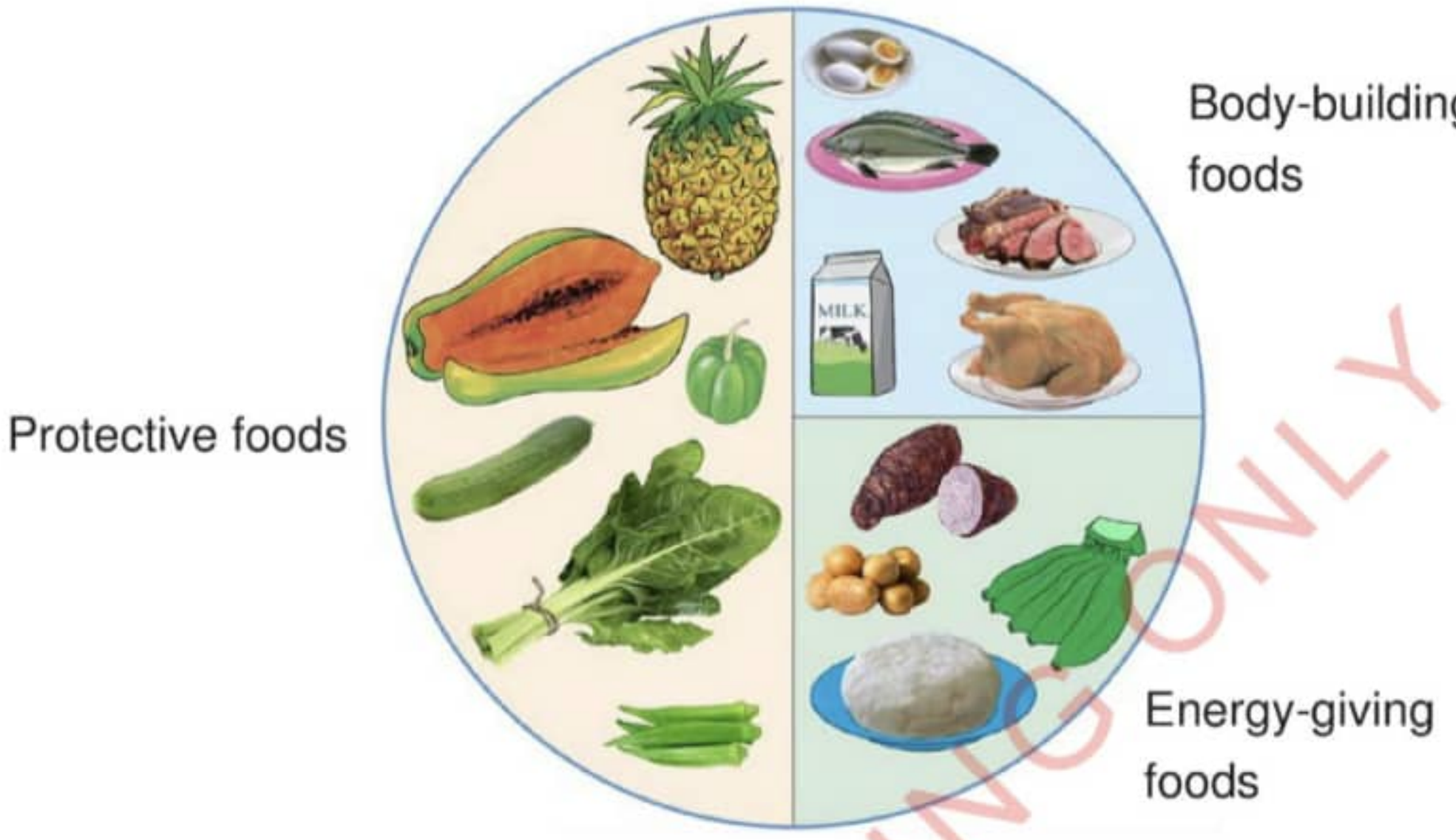


Figure 8: *Balanced diet for eldely*

Manual workers

Manual workers include porters, cart pushers, mechanics and construction workers. Manual workers need a large amount of energy-giving foods. These foods provide enough energy for heavy tasks. They should also drink plenty of water to aid digestion and replace water lost through sweat. Figure 9 shows a balanced diet for manual worker.

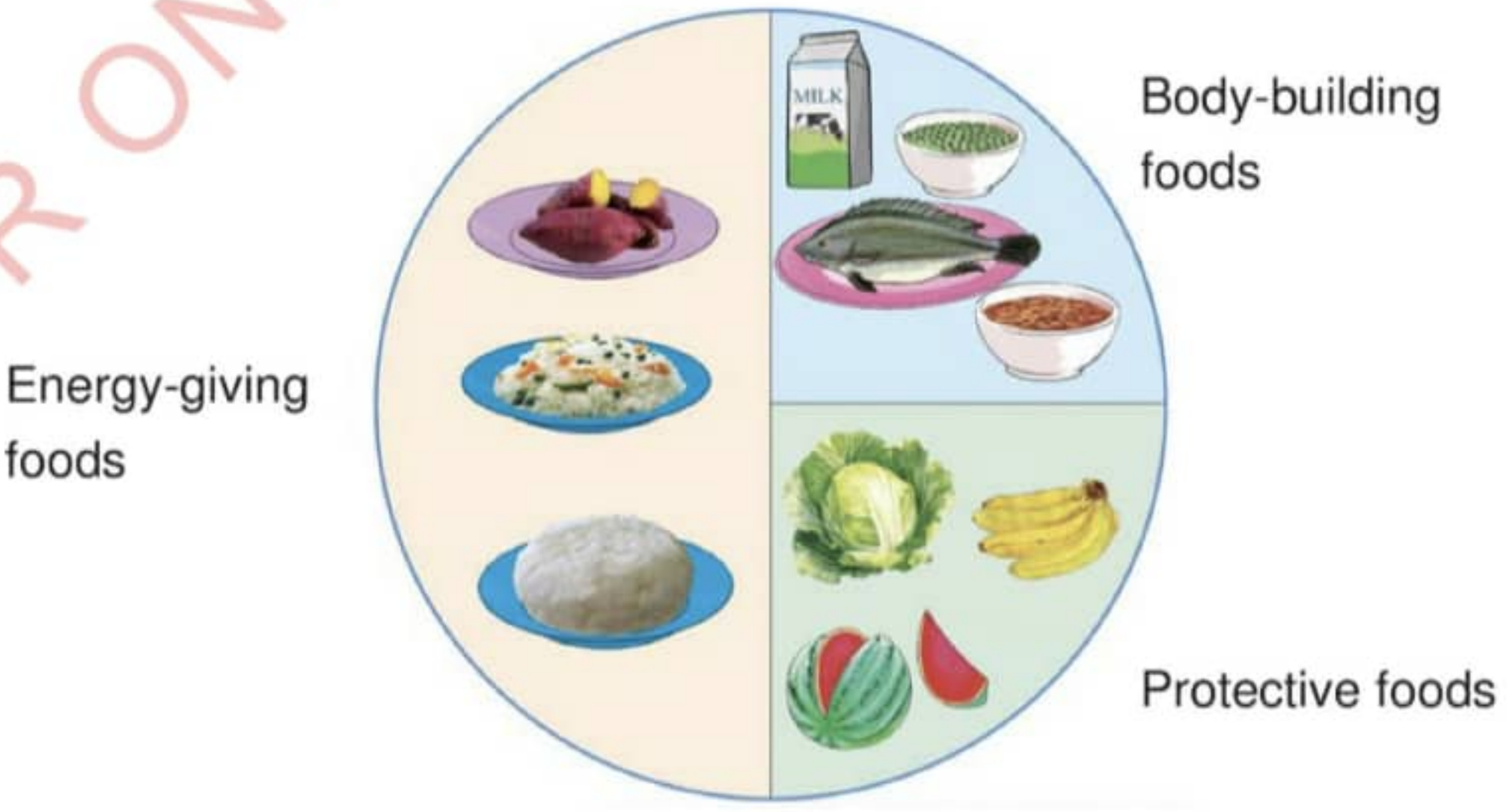


Figure 9: *Balanced diet for a manual worker*

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Activity 4

1. Use the library or reliable online sources to search for information about a balanced diet for manual workers.
2. Plan balanced diet for breakfast, lunch, and dinner for a manual worker.

Sick people

A balanced diet is important for a sick person because it helps the body fight diseases. This diet also helps the sick person to gain strength and important nutrients. Sick people should eat more foods that build and protect the body. These foods help repair damaged parts, like cells and tissues, due to illness. They also make the muscles, bones and immune system strong. However, the food needs may differ depending on the type of illness. Therefore, sick people should eat a balanced diet as advised by health professionals. Figure 10 shows a balanced diet for a sick person.

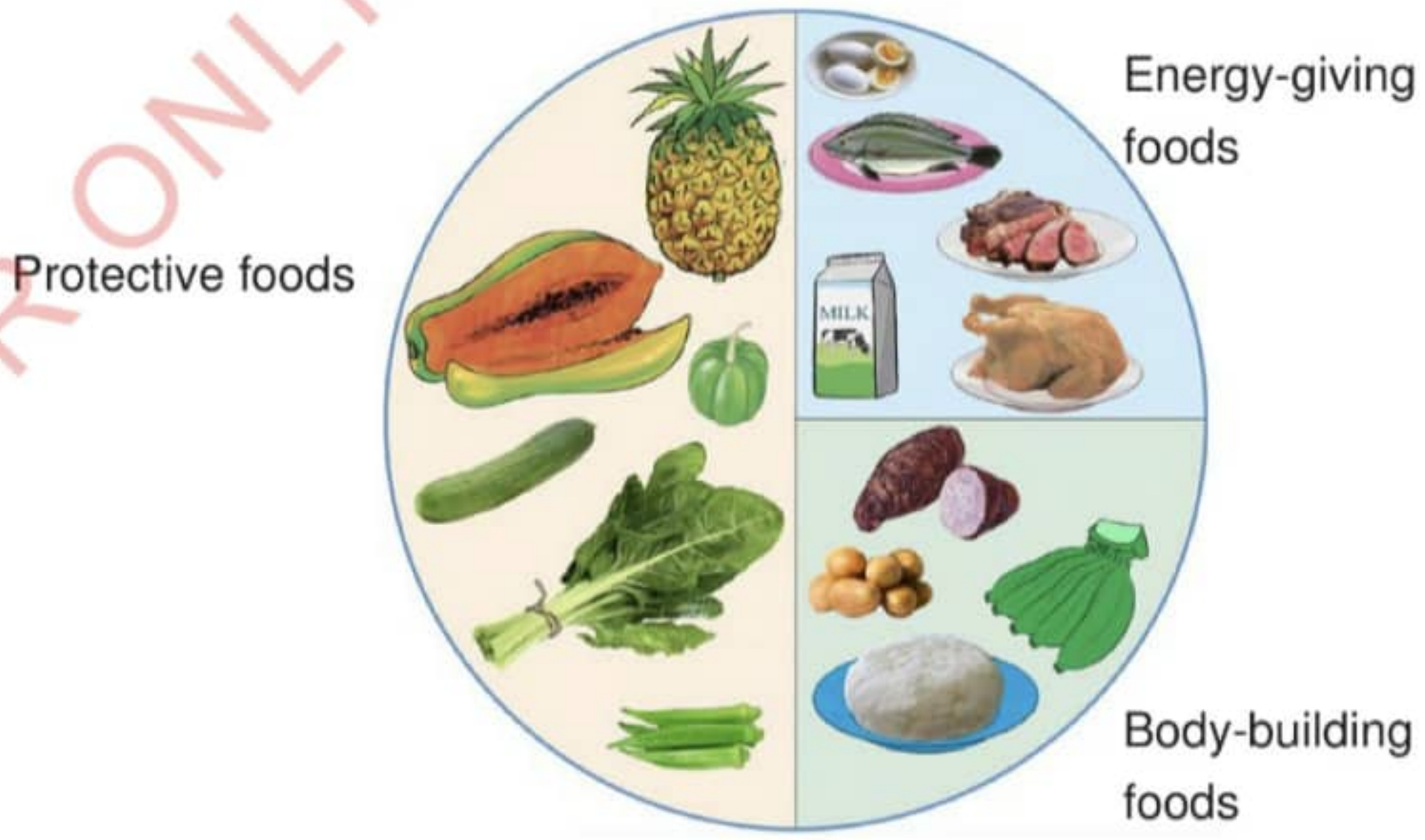


Figure 10: *Balanced diet for a sick person*

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Exercise 1

1. Maria ate rice and meat last Monday. Was this a balanced meal? Explain.
2. Plan a balanced meal for a sick person using foods available at home. Give reasons for including those types of foods in the meal.
3. Explain the differences in dietary needs between a child and a manual worker.
4. What happens if you do not eat a balanced diet?
5. Why are body-building and protective foods important for a sick person?

Physical exercises

Physical exercises are important for keeping the body healthy. Physical exercises include playing football, running, jumping and walking. Figure 11 shows pupils doing physical exercises.



Figure 11: *Pupils doing physical exercises*

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Activity 5

Use the library or reliable online sources to find information about different types of physical exercises. Then, do the exercises that you are able to perform.

Benefits of exercising

It is good to do physical exercises regularly. Regular physical exercises help to:

(a) Stay healthy and strong;

(b) Keep body systems like digestion and excretory systems working well;

(c) Strengthen body immunity to fight different diseases;

(d) Reduce stress and depression; and

(e) Refresh the mind and have fun.

Effects of not doing exercises

Lack of physical exercises can cause problems like:

(a) Having poor health and weak body;

(b) Gaining too much weight, leading to obesity;

(c) Getting chronic diseases like high blood pressure, diabetes, and heart problems;

(d) Feeling stressed and depressed; and

(e) Having weak muscles and bones.

Personal and environmental hygiene

Maintaining hygiene of the body and the environment is important for good health. Body and environment must be kept clean all the time.

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Personal hygiene

Personal hygiene is important for our health. The body and clothes should be cleaned using proper tools to avoid diseases. Personal hygiene includes cleaning the mouth, skin, hair, nails and clothes.

Oral hygiene

The mouth should be cleaned using a toothbrush, toothpaste, and clean, safe water. Brushing removes food particles from the teeth. If the food particles remain, bacteria in the mouth feed on them and produce acid. This acid damages the teeth and causes cavities. It is important to brush your teeth at least twice a day. Brush your teeth in the morning when you wake up and at night before going to bed. This helps keep your teeth healthy.

Activity 6

1. Use the library or reliable online sources to search for information about oral hygiene and care.
2. Share the information obtained on oral hygiene to your peers for discussion.

Bathing

Clean water and soap should be used for bathing. It is good to bathe every day, in the morning and evening to avoid skin diseases. If we do not bathe, our bodies can smell bad. After bathing, dry the body with a clean towel. Apply body oil to keep the skin soft and prevent it from drying. We should also wash our faces with clean water and soap after waking up. This helps remove rheum and prevent eye diseases.

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Activity 7

1. Use the library or reliable online sources to search for information about the tools and steps for bathing.
2. Share the steps for bathing to your peers for discussion.

Hair hygiene

Hair should be cared for by regularly washing with clean water and soap. It is also important to care the hair by braiding, shaving or combing to look clean and neat. Keeping hair clean helps prevent dandruff and pests like lice. This also helps to prevent scalp infections such as ringworms.

Nail hygiene

Fingernails and toenails should be kept short and clean by cutting them regularly. Long nails can hold dirt with germs that cause diseases. Washing hands and nails with clean running water and soap helps remove and kill these germs.

Clothing hygiene

Clothes should be washed with clean water and soap to remove dirt. Wearing dirty clothes can cause body odour and lead to skin problems like itching and rashes. After washing, clothes should be properly dried and ironed. Ironing helps kill germs and keeps the clothes looking neat.

Activity 8

1. Use the library or reliable online sources to search for information about hair, nail and clothing hygiene.

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2. Observe at your classmates' hair, nails and clothes and give advice on maintaining personal hygiene.

Environmental hygiene

Dirty environment are harmful to our health. Dirty places attract insects like houseflies and mosquitoes that can spread diseases like diarrhoea, cholera and malaria. Breathing polluted air can also cause respiratory diseases. These diseases make our bodies weak. To stay healthy and strong, we must keep our surroundings clean. Figure 12 shows pupils cleaning the environment.



Figure 12: *Pupils cleaning the environment*

Safety signs and important information in the environment

There are signs or symbols that guide us in our environment. These signs can be words or pictures and give instructions to keep everyone safe.

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Activity 9

Walk around the school compound and identify the safety signs found there.

Types of safety signs

Safety signs are divided into four main types: warning signs, prohibition signs, mandatory signs and emergency signs. Each type has its own colour and shape to help people understand their meaning, based on international safety rules.

Warning signs

Warning signs give alerts about dangers that might happen. They can be seen in different places like on roads, in the school compound or on things like waste bins.

Warning signs have a yellow colour background. They are triangular in shape, with a black picture and black borders. Figure 13 shows examples of warning signs.



Slippery floor



Toxic



Lift correctly



Danger

Figure 13: *Examples of warning signs*

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Activity 10

1. Use the library or reliable online sources to search for information about warning signs and their meanings.
2. Among the signs you observed, which ones are common in your environment?

Prohibition signs

These signs are also known as regulatory signs. They show things that are not allowed. The sign has a black picture on a white background with a red circular border. Some of these signs also have a red diagonal line crossing the circle. Examples include signs like "No Entry" and "No Eating or Drinking". Figure 14 shows examples of prohibition signs.



Figure 14: *Examples of prohibition signs*

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Emergency signs

These signs give instructions on what to do, especially in emergencies like accidents. They show safe places nearby or directions to get to safety or find helpful equipment. These signs include things like emergency phones and first aid kits. They are rectangular with white pictures on a green background or green pictures on a white background as seen in Figure 15.



Emergency exit



Emergency assembly point



Emergency telephone



Eye washing station

Figure 15: *Examples of emergency signs*

Activity 11

1. Use the library or online sources to search for information about prohibition and emergency signs, and their meanings.

2. Among the signs you observed, which ones are found in your surroundings?

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Mandatory signs

These signs give directions on what you should do in a certain area. These are seen in places like chemical storage areas. The signs are blue and have a circular shape with a white picture in the middle as indicated in Figure 16.



Wear face mask



Wear high visibility clothing



Wash your hands



Wear face and eyeshield

Figure 16: *Examples of mandatory signs*

Activity 12

Identify key areas in your school environment that require safety signs. Then, work in groups to create those signs and place them in the appropriate areas.

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Exercise 2

Match the signs in Section A with the correct category in Section B.

No.	Section A	Section B
1.		(a) Prohibition
2.		(b) Warning
3.		(c) Emergency
4.		(d) Mandatory

Important information for protecting our health

It is important to pay attention to important details about products we use every day. This can help us avoid health problems. Some of the key things to consider are expiration

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date, instruction for using the product, ingredients and storage methods.

Expiration date

An expiration date is the last day a product is safe for use. It tells when a product is no longer safe to use. Products should not be used after this date, as they may be harmful to health. The expiration date is usually written on the product in different ways, like “Expiry date” or as short forms like “ED,” “EXP” or “E.” Figure 17 shows an example of the expiration date of a product.

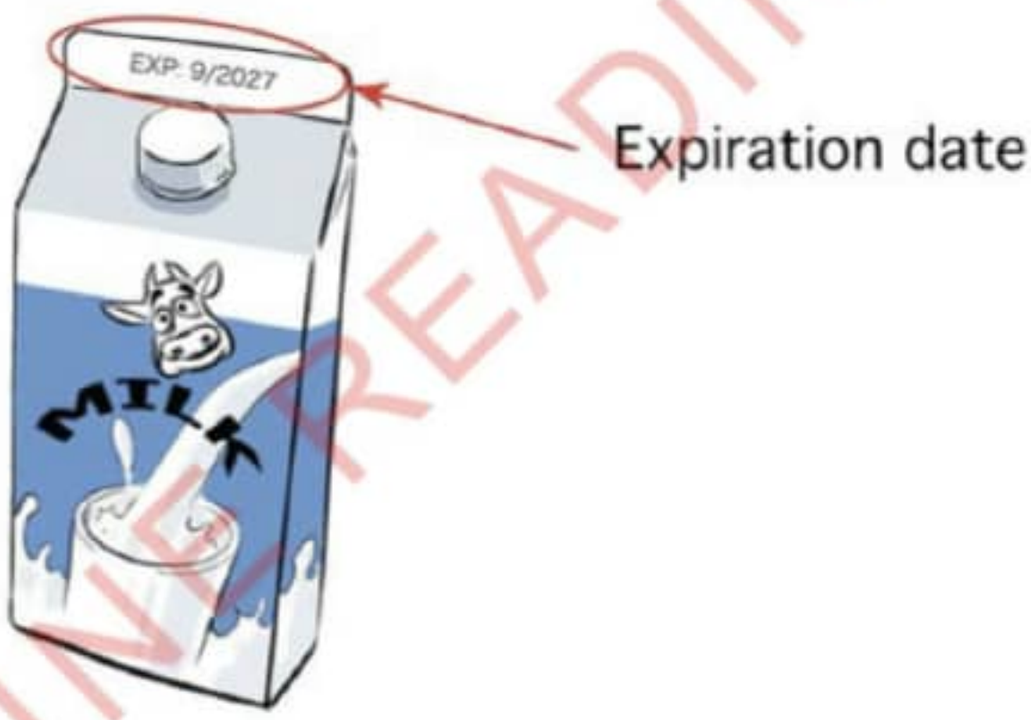


Figure 17: *Example of product expiration information*

Instruction on how to use the product

It is very important to read and understand the instructions before using any product. This helps yo use it in the right way and stay healthy. For example, products like human medicine should be used in the right amount to minimise risks to human health and the environment. Figure 18 shows instructions on how to use a product.

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Figure 18: *Example of instructions of using a product*

Ingredients of the product

Products contain various ingredients. It is important to know what is in a product to prevent health issues like allergies. Figure 19 shows example of ingredients in a product.



Figure 19: *Ingredients in a product*

Storage methods

Storing products in the right way helps keeps them safe and healthy to use. It is important to follow storage instructions to avoid problems. This information is usually written on the packaging. Figure 20 shows storage instructions in a product.

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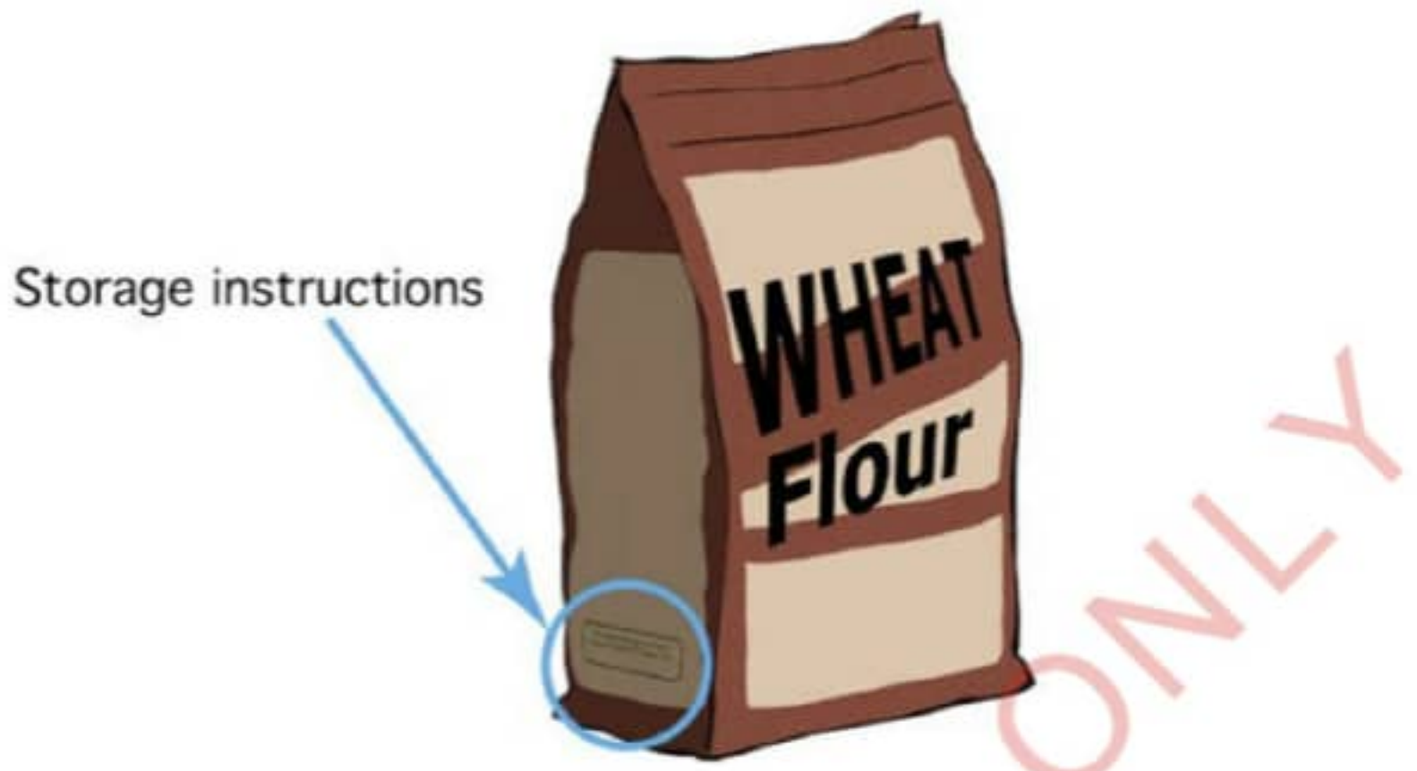


Figure 20: *Storage instructions*

Activity 13

Observe different products found at home and school. Then, check on the following:

1. Expiration;
2. Ingredients;
3. Instructions for using the product; and
4. Storage methods.

Revision exercise

Section A: Choose the correct answer.

1. What is one benefit of checking the information on food products?

   (a) to help us to be careful

   (b) to protect our health

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(c) to cause health problems

(d) to promote environmental awareness

2. Which word means the end of a product's good quality?

(a) Rot

(b) Decay

(c) Freeze

(d) Expire

3. A balanced diet is a meal containing _____.

(a) energy and fat giving foods

(b) plenty of meat and fruits

(c) the right proportion of nutrients

(d) enough vitamins

4. Which food nutrients gives the body energy?

(a) Protein

(b) Minerals

(c) Vitamins

(d) Carbohydrates

5. Using products that have expired can cause _____.

(a) health problems

(b) improved safety

(c) improved body performance

(d) improved product effectiveness

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Section B: Write **TRUE** for a correct statement and **FALSE** for an incorrect statement.

6. Foods rich in minerals help to build the body.

7. The main purpose of protein-rich foods is to protect the body from diseases.

8. Exercising keeps the body strong and helps control body weight.

9. Safety signs indicates a product's expiry date.

10. Warning signs are often triangular and yellow.

Section C: Short answer questions

11. Explain the importance of different safety signs in our environment?

12. Explain any three physical activities that are important for your health.

13. Fill in the table by writing the meaning of each sign (a-d).

No.	Sign	Meaning
(a)		
(b)		

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No.	Sign	Meaning
(c)		
(d)		

Vocabulary

Adolescent someone who is no longer a child but is not yet an adult

Cell smallest unit that makes living organism

Expiration when a product is no longer safe to use because its time has run out

Nutrients substances in foods that help our bodies work properly

Obesity being too overweight

Pathogens living things that can cause diseases

Pesticides chemicals used to kill or stop harmful bugs and plants that can harm crops and animals

Rheum watery or sticky liquid that collects in the corners of the eyes, especially when sleeping

Scalp skin covering the top of the head where hair grows

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Chapter Two

Diseases

Introduction

Diseases make people unhealthy. In this chapter, you will learn about human diseases. You will also learn how diseases spread, their causes and how to prevent them. You will also learn the life cycle of some insects that spread various diseases. The competencies developed will enable you to protect yourself and others from diseases.



Think The effects of diseases on human health

Concept of diseases

Disease is an abnormal condition or disorders in a living organism that disrupts normal bodily function. Many diseases are caused by germs found in our surroundings. These germs are found in air, soil, water, unsafe food and animal bodily fluids like blood. Germs can enter the body through eating contaminated food, drinking unsafe water, breathing in or touching an infected person. However, some diseases are not caused by germs, such as hereditary diseases.

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Exercise 1

1. Identify various areas where germs are commonly found.
2. Explain the different ways in which germs can enter the human body.
3. Mention any three diseases that you know in your environment.

Types of diseases

There are different types of diseases that affect the human body. These diseases are grouped into two main types namely: communicable diseases and non-communicable diseases.

Activity 1

1. Use the library or reliable online sources to find information about communicable and non-communicable diseases.
2. List five examples for each group of diseases you have identified.

Communicable diseases

Communicable diseases are diseases that spread from one person to another. These diseases are caused by pathogens like bacteria and viruses. Examples of communicable diseases are malaria, cholera, chickenpox, tuberculosis, hepatitis and Corona Virus Disease (COVID-19).

Malaria

Malaria is a disease caused by parasites called plasmodium. The disease is spread by a female mosquito called Anopheles.

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The mosquito transmits malaria by injecting the parasites into the human body when it bites a person. Figure 1 shows how mosquitoes transmit malaria.

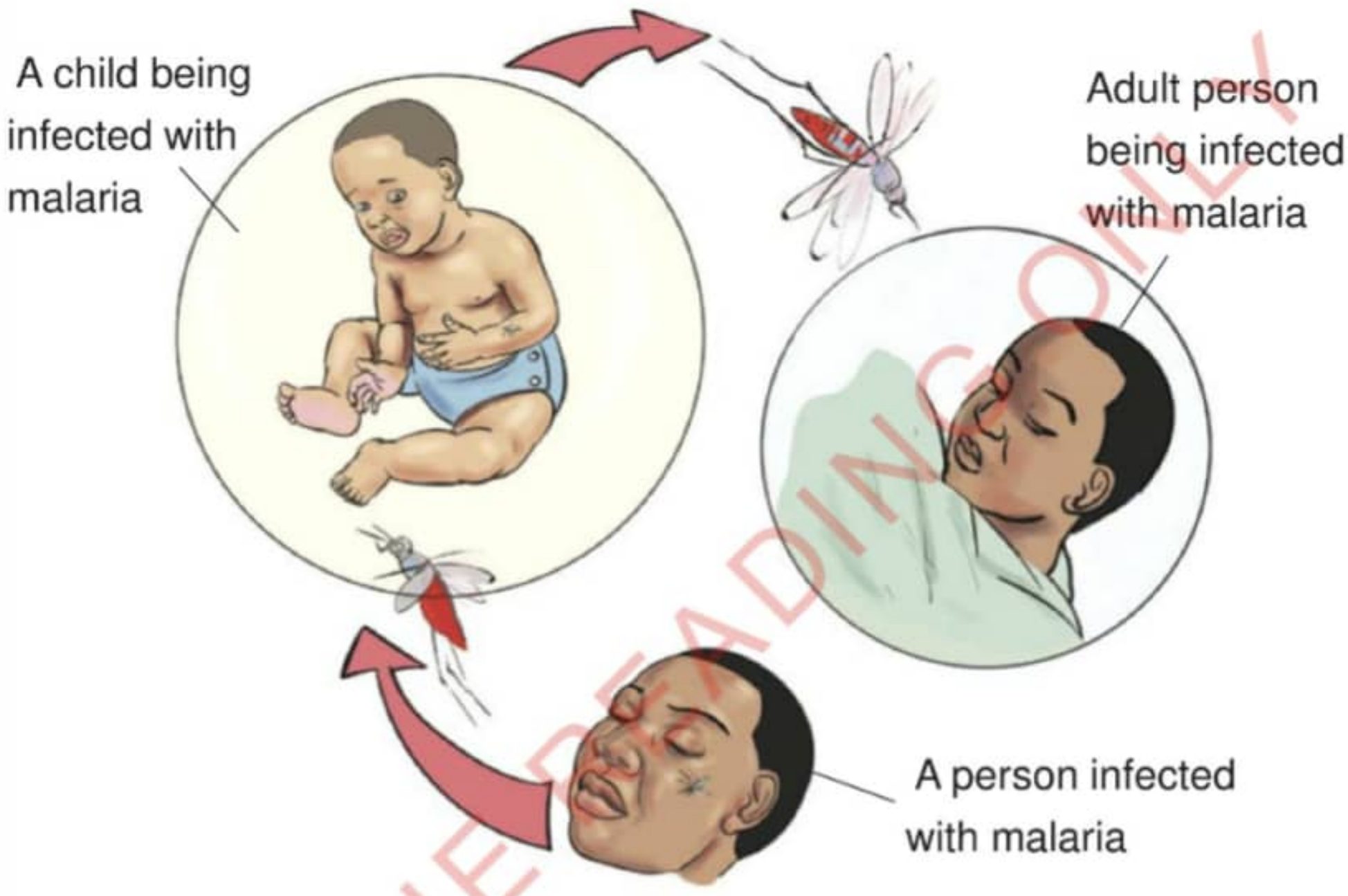


Figure 1: *Malaria transmission*

Symptoms of malaria

Symptoms of malaria include the following:

(a) Having a high fever due to the raising of body temperature;

(b) Feeling cold and shivering;

(c) Headache;

(d) General body weakness;

(e) Vomiting; and

(f) Loss of appetite.

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Control and prevention of malaria

Malaria can be controlled by destroying places where mosquitoes breed. These places include bushes, long grass, and stagnant water in puddles. Tyres, coconut shells and cans that hold water can also serve as breeding areas for mosquitoes. Kerosene or larvicide can be spread on puddles to kill mosquito larvae. Also, coconut shells and other breeding areas should be destroyed. Malaria can also be prevented by sleeping under a treated mosquito net. Other ways include using mosquito sprays, coils and repellents. Also, to put nets on the windows of the house to prevent mosquitoes from entering. If someone has malaria symptoms, they should go to the hospital for diagnosis and treatment.

Exercise 2

1. List four symptoms of malaria.
2. Explain why it is important to sleep under mosquito nets.
3. Describe how a person can be infected with malaria.
4. How would you prevent yourself and your family from malaria?

Cholera

Cholera is a disease caused by bacteria. It spreads when people eat food or drink water that has germs. Houseflies can also spread cholera by carrying bacteria from faeces or vomit to food or water, as shown in Figure 2.

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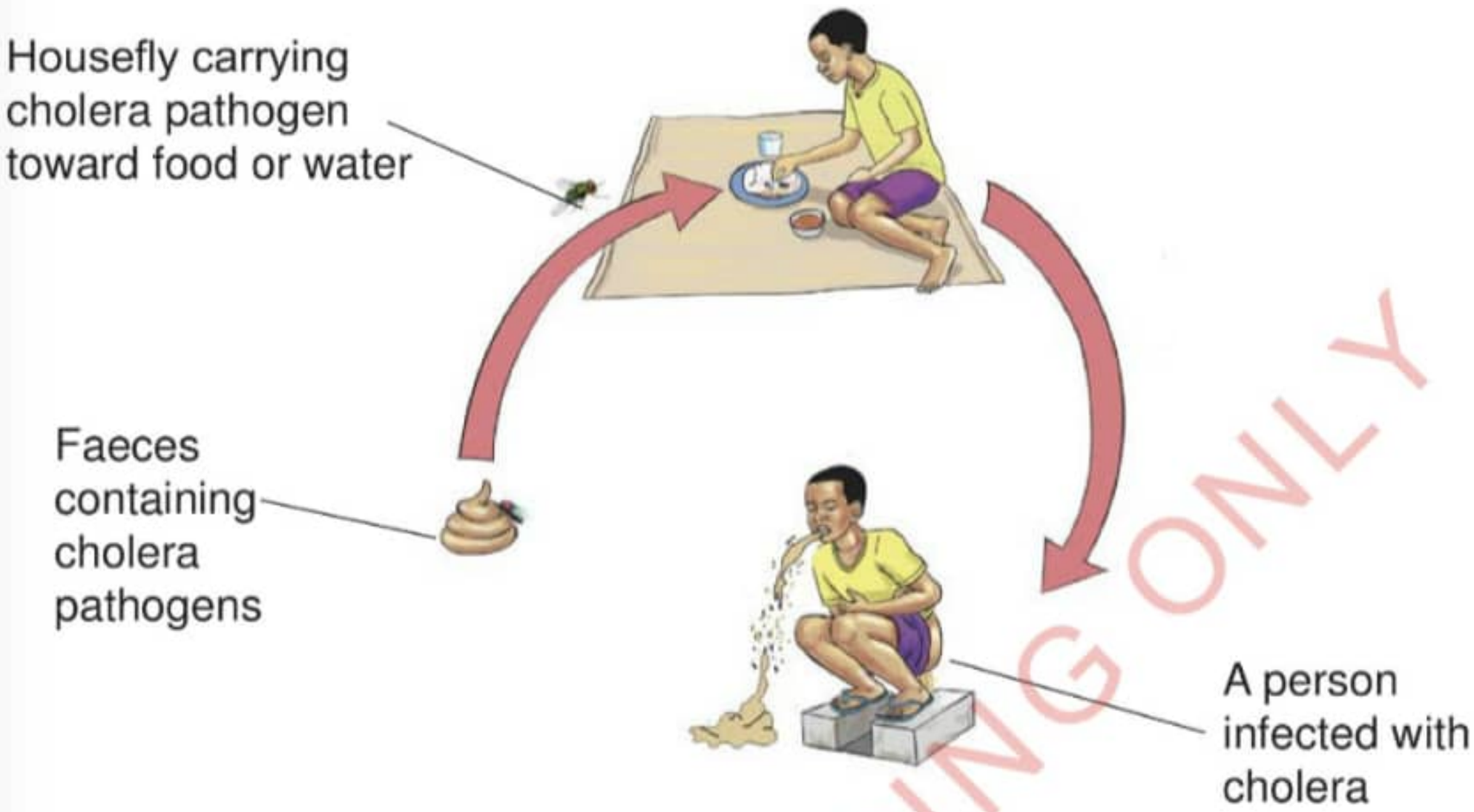


Figure 2: *Transmission of cholera*

Symptoms of cholera

Cholera symptoms include the following:

(a) Continuous diarrhoea and vomiting;

(b) Passing stool that looks like rice water;

(c) Yellowish urine; and

(d) General body weakness and loss of strength.

Prevention of cholera

Living in a clean environment and maintaining cleanliness is important. Eat clean and safe food, and drink clean and safe water. Always use the toilet and use it properly. Wash your hands with clean running water and soap after using the toilet and before eating. A person with cholera should be taken to the hospital or a cholera treatment center immediately.

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Exercise 3

1. Explain how cholera is spread.
2. How can you identify a person with cholera?
3. What can you do to prevent cholera?
4. Explain the importance of washing your hands with soap and clean water after using the toilet.

Chickenpox

Chickenpox is a disease caused by a virus. It spreads through skin contact with an infected person. It can also spread by sharing clothes with an infected person.

Symptoms of chickenpox

The symptoms of chickenpox include a rise in body temperature and headache. It also causes itchy rashes that turn into blisters with pus or fluid. Sometimes, this disease leaves permanent scars on the skin. See Figure 3.



Figure 3: *Chickenpox patient*

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Prevention of chickenpox

Chickenpox can be prevented by avoiding sharing clothes with an infected person and by keeping the body and clothes clean. Also, a patient should use antiseptic when washing the body and clothes. This will help to reduce risk of infection for those who came in contact with the washed clothes.

Exercise 4

1. How does chickenpox spread from one person to another?
2. How can you identify a person with chickenpox?
3. What can you do to stop the spread of chickenpox?

Tuberculosis

Tuberculosis (TB) is a disease caused by bacteria. It mostly affects the lungs. However, it can also affect other parts of the body, such as the spine, bones and kidneys. TB spreads through the air when an infected person coughs or sneezes. Spitting can also cause the bacteria to spread.

Symptoms of tuberculosis

Symptoms of tuberculosis include the following:

(a) Coughing for more than two weeks;

(b) Chest pain;

(c) Fevers and sweating at night;

(d) Rapid loss of weight;

(e) Coughing up blood; and

(f) Loss of appetite.

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Prevention of tuberculosis

Tuberculosis can be prevented by vaccinating infants. Other ways to prevent it include keeping rooms well-ventilated, practicing good hygiene and avoiding populated areas. Do not share utensils with an infected person. People with TB can be treated with medicine by a health care provider.

Exercise 5

1. How do coughing, sneezing and spitting spread tuberculosis?
2. How can you identify a person with tuberculosis?
3. Apart from the lungs, what other body parts can be affected by tuberculosis?

HIV and AIDS

Acquired Immunodeficiency Syndrome (AIDS) is caused by the Human Immunodeficiency Virus (HIV). HIV attacks white blood cells that protect the body. Once in the body, HIV increases in numbers and attacks white blood cells. This weakens the body and makes it unable to fight off infections. A weak body is likely to get other diseases.

Ways of HIV Transmission

HIV is found in blood and other bodily fluids. HIV can spread from one person to another in the following ways:

(a) Sharing toothbrushes and sharp objects like needles, razors, pins, earrings and scissors as shown in Figure 4. Sharing these objects with an infected person can cause infection with HIV.

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Figure 4: *Objects that can spread HIV*

(b) Receiving blood from an infected person (blood transfusion). Blood must be tested to make sure it is HIV-free before transfusion. Figure 5 shows a patient receiving blood.



Figure 5: *A patient receiving blood in a health facility*

(c) From a mother to a child. A baby can get HIV from an HIV-positive mother, especially during childbirth. A baby can also get HIV from the mother while breastfeeding.

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(d) Sexual intercourse with someone who is infected. Children should not be involved in sexual activities. They should also avoid risky places where they might be raped, like bars, clubs and abandoned houses. If anyone is sexually abused, it is important to tell parents or teachers immediately to get proper help.

Prevention of HIV and AIDS infections

HIV is dangerous and has no vaccine or cure. However, people living with HIV get drugs called antiretroviral to fight the virus and live a long and healthy life.

HIV can be prevented by avoiding the sharing toothbrushes and sharp objects like needles, scissors, nail cutters and syringes. Young people should avoid sexual acts to stay safe from HIV. A person with HIV should eat a healthy diet. They should also exercise regularly.

Exercise 6

1. Mention three ways the HIV can spread.
2. Why do sharp objects spread HIV?
3. How does HIV affect the body's immune system?
4. How can you care for and support people with HIV?

Non-communicable diseases

Non-communicable diseases do not spread from one person to another. They are not caused by germs. These diseases can be inherited from one generation to another or be caused by an unhealthy lifestyle. Examples of non-communicable diseases are cancer, diabetes and asthma.

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Cancer

Cancer is a disease that affects the cells in the body. These cells begin to divide and make new cells in a disorganised way. The new cells keep multiplying quickly and can form a tumor. Tumors may develop into cancer. Cancer can occur in any part of the body, like in blood, intestines, brain and bones.

Causes of cancer

Cancer is caused by different things that harm the cells of the body. These include exposure to strong radiation and chemicals that cause cancer. These chemicals can come from food, air, cosmetics and medicines.

Prevention of cancer

Cancer can be prevented by eating health foods, drinking safe beverages and using medicines properly. It is also important to avoid alcohol and smoking, as these can cause cancer. If a person is taken early for check-ups, cancer can be treated.

Activity 2

1. Use the library and reliable online sources to search for information about various types of cancer.
2. Identify the types of cancer that are common in Tanzania.

Diabetes

Diabetes occurs when the body cannot control blood sugar levels. It happens when a pancreas organ does not produce enough insulin. Insulin is a hormone needed to control blood sugar levels in the blood. A person can be born with diabetes

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from their parents. They can also develop diabetes later in life due to unhealthy habits such as eating unhealthy foods and lack of physical exercises.

Symptoms of diabetes

Symptoms of diabetes include:

(a) Frequent urination;

(b) Rapid weight loss;

(c) Feeling thirsty even after drinking water;

(d) Feeling tired all the time;

(e) Poor eyesight vision;

(f) Wounds that heal slowly; and

(g) Numbness in hands or feet.

Prevention of diabetes

Diabetes can be prevented by living a healthy lifestyle. This includes eating properly and controlling body weight. Regular exercise can also lower the risk of diabetes.

Exercise 7

1. What three symptoms that indicate person has diabetes?
2. How can regular exercise help reduce the chance of getting diabetes?
3. Why do healthy eating habits prevent diabetes?

Asthma

Asthma is a disease that affects the respiratory system. A person can be born with asthma from their parents or get it later in life due to environmental factors. Asthma can be triggered

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by allergies caused by environmental factors such as smoke, dust, perfume, soap and pollen. The allergy causes the airways to swell and produce thick mucus. This swelling makes it hard to breathe and a person gets an asthma attack.

Symptoms of Asthma

Symptoms of asthma include:

(a) Coughing, sneezing and a runny nose;

(b) Tightness and pain in the chest and ribs; and

(c) Shortness of breath with a wheezing or whistling sound.

Prevention of Asthma

Asthma is a chronic condition with no cure. Asthma attacks can be prevented by:

(a) Avoiding asthma triggers like pollen and perfumes; and

(b) Getting treatment immediately after getting asthma symptoms.

Exercise 8

1. How can different lifestyles cause diabetes?
2. List three things that can trigger asthma.
3. How can cancer be prevented?

Life cycle of insects which spread diseases

Insects like houseflies, mosquitoes and cockroaches spread diseases. These insects have different growth stages through which they change their shape and behaviour. These changes are called metamorphosis. There are two types of metamorphosis namely: complete and incomplete metamorphosis. Complete

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metamorphosis has four stages of development which are egg, larva, pupa and adult. Examples of insects with complete metamorphosis include houseflies and mosquitoes. Incomplete metamorphosis has three development stages which are egg, nymph and adult. This type of metamorphosis occurs in insects like cockroaches and lice.

Life cycle of housefly

The stages of housefly development are egg, larva, pupa, and adult fly. The female housefly lays eggs in dirty places where it can find its food. These places include areas with faeces, food waste, and dead and decayed organisms. Depending on environmental conditions, housefly eggs can hatch into larvae within 24 hours. The larva can develop for about seven days before becoming a pupa. The pupa does not feed and may take about six days to become a fully grown housefly. See Figure 6.

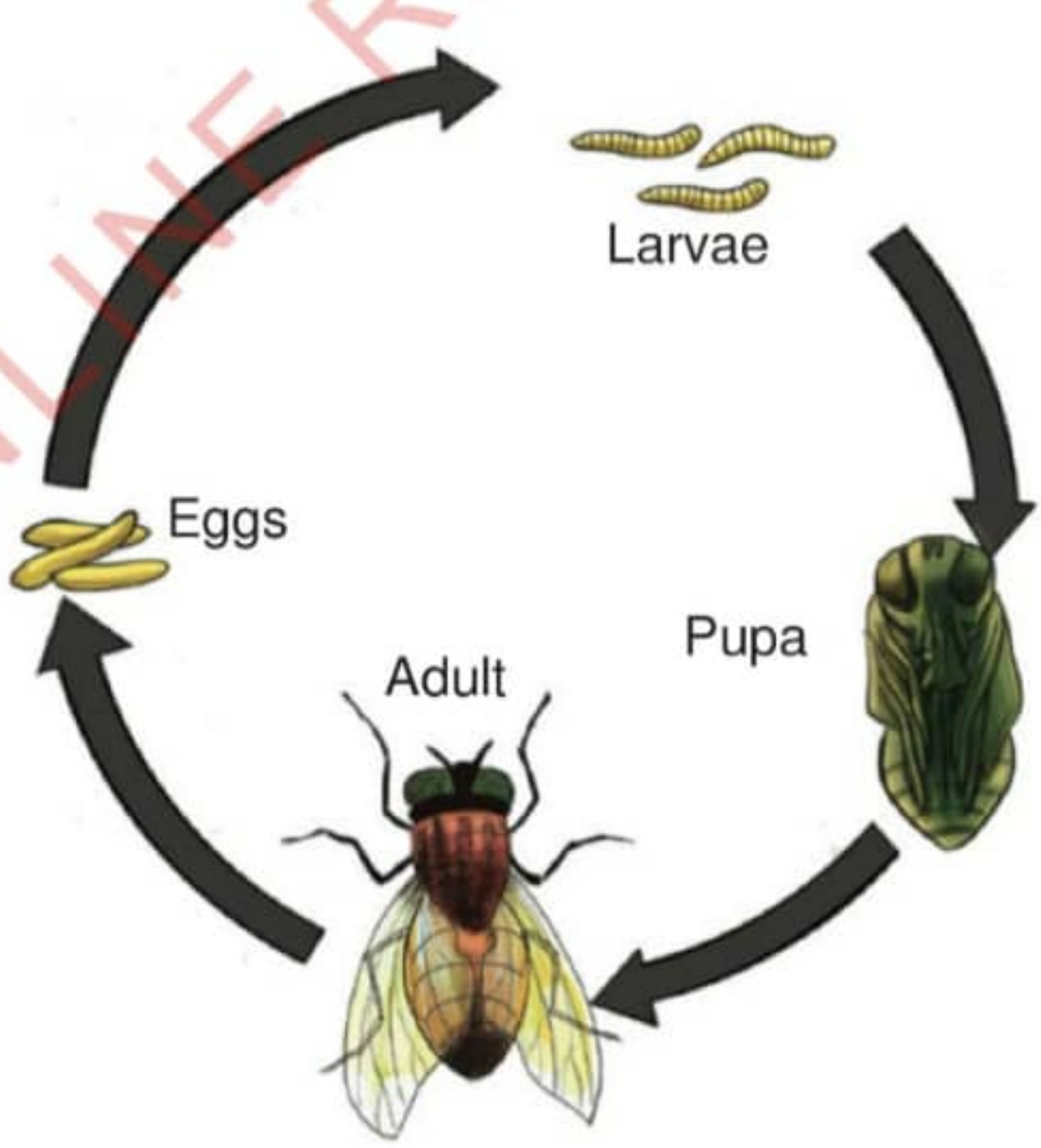


Figure 6: *Stages of housefly development*

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Activity 3: *Investigating the development stages of a housefly*

Materials: A hand lens, a piece of raw meat, plastic wrap, pins and a plate

Procedure

1. Place the piece of meat on the plate.
2. Place the plate with the meat in an open space where houseflies can land on it.
3. Use a hand lens to look at the meat on the plate. Check for any living organisms. What do you see?
4. After 24 hours, observe how the eggs have changed into larvae.
5. Continue to observe periodically how larvae change into pupae by answering the following questions:
   (a) What kind of shield or casing covers these creatures?
   (b) After how many days did these creatures change?
   (c) Are these creatures moving?
6. Count ten pupae and leave them on the plate.
7. Cover the plate with a plastic wrap. Make small holes in the plastic wrap with a pin.
8. Continue observing the pupae and write down your observations guided by the following questions:
   (a) How many pupae are left on the plate?
   (b) What happened to the other pupae?
   (c) What other organisms can you see on the plate?

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Activity 4: *Investigating the stages of mosquito development*

Materials: Hand lens, a bottle or coconut shell, fine mesh net

Procedure

1. Collect water from a stagnant pool that has been there for a long time. Make sure the water has mosquito eggs or larvae.
2. Place some mosquito eggs or larvae from the water into a bottle or coconut shell.
3. After two days use a hand lens to observe water for changes of eggs into larvae.
4. Cover the bottle or coconut shell containing water with larvae with a fine mesh net. Make sure it is securely closed.

Questions

(a) What did you observe during your investigation?

(b) Draw a diagram of mosquito development stages.

Life cycle of a cockroach

Activity 5

Observe the stages of cockroach development in their natural environment. Share your observations with your classmates for discussion.

Cockroaches lay eggs in a hard casing. They hide their eggs in places that cannot be easily seen. The eggs hatch into nymphs

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within two months depending on the environmental conditions such as temperature. Nymphs are small with soft, outer layer and do not have wings. After three months, the nymphs grow into adult cockroaches. See Figure 7.

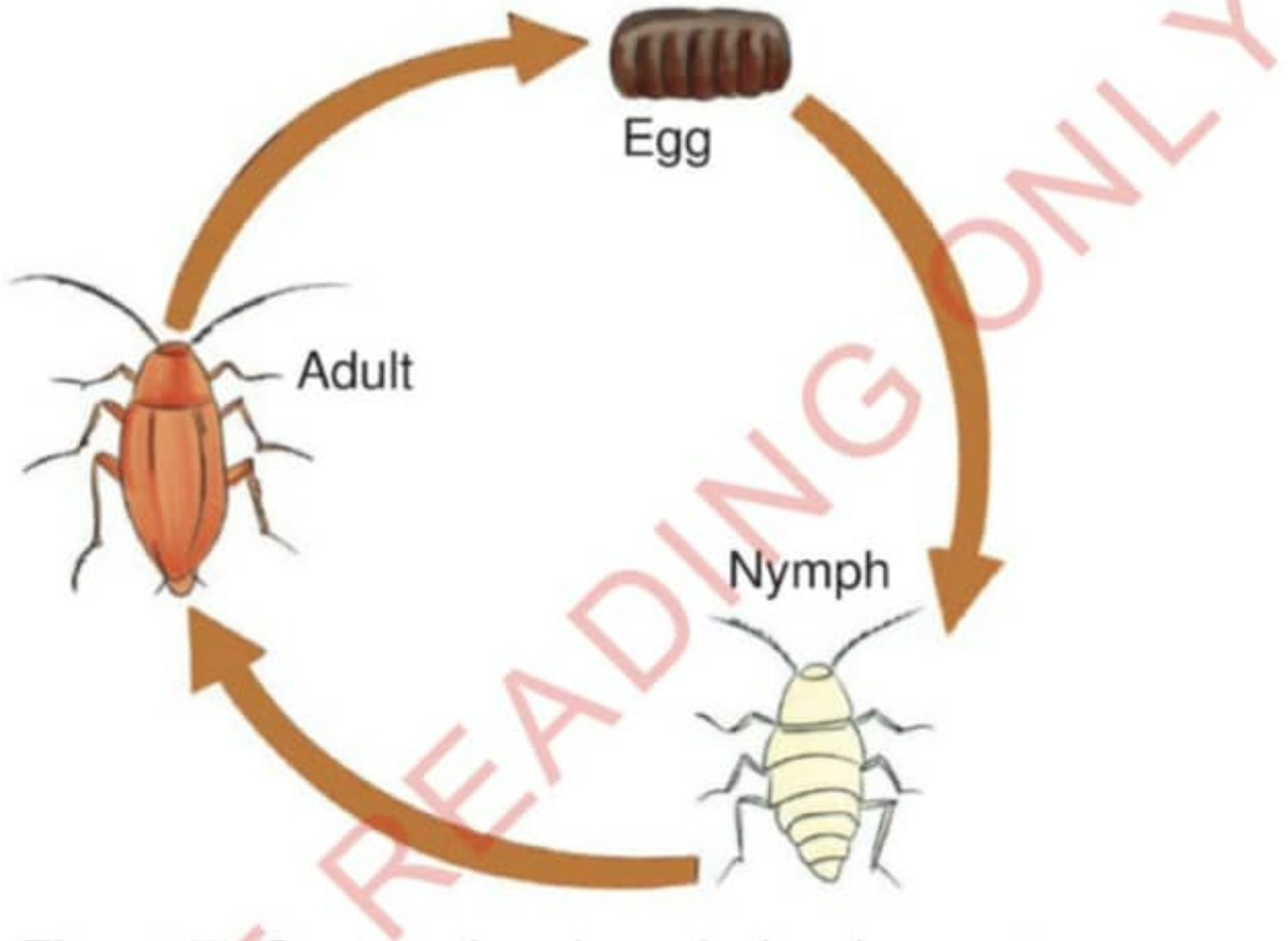


Figure 7: *Stages of cockroach development*

Regular cleaning and organising of clothes and other items can remove cockroach breeding and hiding places. You can also use insecticides to kill cockroaches.

Revision exercise

Section A

1. Match the disease in Section A with its symptoms in Section B.

No.	Section A	Section B
(a)	Malaria	(i) Causes rashes that turn into blisters
(b)	Chickenpox	(ii) Coughing for two weeks or more
(c)	Asthma	(iii) Frequent urination

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No.	Section A	Section B
(d)	Tuberculosis	(iv) Continuous vomiting and diarrhoea
(e)	Diabetes	(v) Makes a whistling or wheezing sound when breathing
(f)	Cholera	(vi) High fever and joint pain

Section B

2. What is the difference between communicable and non-communicable diseases?

3. How are communicable diseases spread?

4. What does metamorphosis mean?

5. Give an example of an insect that goes through:

   (a) Complete metamorphosis; and

   (b) Incomplete metamorphosis.

6. List the stages of insect development in:

   (a) Complete metamorphosis; and

   (b) Incomplete metamorphosis.

7. Why are houseflies, cockroaches, and mosquitoes dangerous to human health?

8. What can you do to reduce houseflies breeding near your home?

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Vocabulary

Antiseptic	a chemical used to kill germs in clothes and other areas
Allergy	a condition where the body reacts to things like dust, pollen, or certain smells that do not harm others
Blisters	fluid-filled bumps that appear on the skin
Germs	very small living organisms that cause diseases and cannot be seen with the naked eye
Genes	the units of inheritance found in cells of living organisms
Opportunistic	diseases recurrent diseases that affect a person with a weakened immune system
Pathogen	harmful microorganisms such as bacteria, virus, fungus or parasite that causes diseases
Replicate	to make something again in the same way
Transfusion	to receive blood from another person usually in the hospital
Tumour	swelling of a part of the body, generally without inflammation, caused by an abnormal growth of cell

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Chapter Three

Matter

Introduction

The environment comprises substances that occupy space and have mass. Such substances are called matter. In this chapter, you will learn about the states of matter. You will also conduct simple experiments to investigate their properties. The competencies developed will enable you to use various types of matter in everyday life.



Think Various things that have mass and occupy space

Concept of matter

Matter is anything that has mass and occupies space. Substances like rocks, water and air are examples of matter.

Activity 1

1. Use library or reliable online sources to find information on the concept of matter, then provide other examples of matter.

2. Collect various matter samples, observe their properties, and categorise them into the three states of matter: solid, liquid, and gas.

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States of matter

There are various states of matter. In this section, you will learn about the three states of matter, which are solid, liquid and gas.

Solid

A solid is a state of matter that is firm and hard. Solids have a definite shape. Solids include stones, firewood and cooking pots as shown in Figure 1.



Stones



Cooking pots



Firewood

Figure 1: *Examples of solid matter*

Liquid

Liquid has no fixed shape but conforms to the shape of the container it is placed in. When you pour water into a container, the water takes up space and conforms to the shape of that container. Examples of liquids include water, juice and soda. Figure 2 shows a liquid being poured into a glass



Figure 2: *Liquid being poured into a glass*

Gas

Gas has no fixed shape and is usually lighter than solids and liquids. An example of a gas is air. Air is stored in solid objects.

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Figure 3 shows balloons filled with air. As you fill a balloon with air, its size increases.



Figure 3: *Balloons filled with air*

Investigating the properties of matter

Matter can change from one state to another depending on changes in temperature. The following experiments 1-4, will demonstrate the three states of matter using water.

Experiment 1: *Investigating the properties of water in the solid state*	
Aim:	To demonstrate the properties of water in the solid state
Materials:	Six equal-sized ice cubes, a saucepan, an ice box, a flat surface such as tray, a jug and a heat source (such as a stove)
***Note*:**	Ensure the ice is from safe and clean water.

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Procedure

1. Take one ice cube and place it in the saucepan. Observe the colour of the ice and press it with your fingers. Does it deform?

2. Take another ice cube, place it in the saucepan, and observe its shape. Transfer the ice to the jug and observe its shape over a short period. Did the shape and size of the ice change after being placed in the jug?

3. Light the stove, then place a saucepan with another piece of ice on the stove. What do you observe?

4. Place another piece of ice from the ice box in the melted water. What did you notice?

5. Record the results of your observations for each step.

6. Take a piece of ice from the ice box and place it on a flat surface, such as a tray. Gently tilt the surface and note the observation. Does the ice flow like liquid water?

Results

Water in its solid state (ice) is colourless, firm when pressed, maintains its shape regardless of the container it is placed in, and does not flow like liquid water. Solid ice floats in liquid water, and when exposed to warmer conditions, it melts into liquid water.

Conclusion

The experiment confirmed that water in its solid state is firm, retains a definite shape and volume, and does not flow like liquid water.

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Experiment 2:	*Investigating the properties of water in the liquid state*
Aim:	To demonstrate the properties of water in the liquid state
Materials:	A jug, a saucepan, drinking water, a plastic bag and a clean cup

Procedure

1. Pour the drinking water into a clean cup, then smell and taste it. What do you notice?
2. Examine the colour of the water. What colour did you see?
3. Place some water into the jug and some into the saucepan. Put the remaining water into the plastic bag and seal it.
4. Observe the shape of the water in the jug, saucepan, and plastic bag.
5. Tilt the jug or saucepan to observe how the water flows.
6. Pour the water from the jug into the plastic cup and observe the volume. Then, pour the water from the plastic bag into the plastic cup and compare the volume with the initial amount.
7. Record your observations.

Results

Water in its liquid state has no colour, taste or smell. It has no definite shape, taking the shape of the container it is placed in while maintaining a fixed volume regardless of the container. Additionally, water flows freely from one place to another.

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Conclusion

The experiment confirmed that water in its liquid state has no fixed shape but maintains a definite volume.

Experiment 3: *Investigating the properties of water in its gaseous state*

Aim: To demonstrate the properties of water in a gaseous state

Materials: Water, a saucepan, and a heat source (stove), mirror and thermometer

***Note*:** This activity should be carried out under the supervision of the subject teacher or laboratory technician.

Procedure

1. Pour water into the saucepan.
2. Turn on the stove, then place the saucepan with water on the stove. Allow the water to boil.
3. Observe what you see as the water starts to boil.
4. Let the water boil for about 10 minutes. What do you notice?
5. Hold a mirror above the steam rising from the boiling water. What do you notice on the mirror?
6. Record your results.

Results

As the temperature rises to 100 ℃, water boils and produces water vapour, which is water in its gaseous state. Water vapour

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has no colour, smell and cannot be touched. Water vapour does not have a fixed shape or volume; it spreads out to fill any available space. When cooled, the water vapour turns back into liquid water.

Conclusion

The experiment showed that water in its gaseous state has no fixed shape or volume, and when cools down, it changes back into liquid water.

Experiment 4: *Changing gas to a liquid state*

Aim: To demonstrate changing of gas to liquid state

Materials: Mirror or glass, water, a kettle and a heat source (stove)

Procedure

1. Pour water into the kettle.
2. Turn on the stove, then place the kettle with water on the stove. Allow water to boil.
3. Observe what you see at the kettle's spout as the water boils.
4. Hold the mirror or glass in front of the kettle spout has shown in Figure 4.
5. Observe what happens in a mirror or glass. Describe your observation.

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Figure 4: *Changing gas to liquid state*

Results

As the liquid water boils, water vapour escapes from the kettle's spout. When the water vapour comes into contact with the cool surface of the mirror or glass, it condenses into small water droplets. This demonstrates that water vapour changes back into liquid water when it cools.

Conclusion

The experiment demonstrated the process of changing gas (water vapour) into a liquid state (liquid water) through condensation.

Water can exist in various states such as solid, liquid and gas. When the temperature of water drops below 0 ℃ (32 ℉), it turns into a solid form called ice. As the temperature rises above 0 ℃ (32 ℉), the ice slowly melts and turns into liquid water. At 100 ℃, water boils to form water vapour. When the vapour comes in contact with cool surface, it changes to a liquid as

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shown in Figure 4. For example, the vapour coming from the spout of a kettle with boiling water is warm and in a gaseous state. When this gas comes into contact with a cooler mirror or glass, it condenses and appears as mist. This mist continues to cool and turns into a liquid state. This demonstrates that when gas is cooled, it changes into a liquid. The process is called condensation. Therefore, condensation is the process of changing gas to liquid state.

Exercise 1

1. What is matter?
2. Give two examples for each state of matter and explain their uses in our daily lives.
3. Based on shape and volume, how does a solid state differ from a liquid and a gas state?
4. How does water exist in three states of matter?

Freezing point of water

Water freezes when it reaches a specific temperature. The freezing point of water is the exact temperature at which a liquid (water) freezes and changes into a solid (ice). Ice has many uses, such as preserving food and beverages. The cold helps to slow bacterial growth and prevent spoilage of some foods.

Experiment 5:	*Investigating the freezing point of water*
Aim:	To demonstrate the freezing point of water
Materials:	Refrigerator, table, thermometer, water, and a plastic cup
***Note*:**	Do not remove the thermometer from the water while reading the temperature.

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Procedure

1. Place a plastic cup filled with water on the table.

2. Take the thermometer and immerse it in the water, holding it from above as shown in Figure 5.



Figure 5: *Measuring water temperature*

3. Read the temperature. Make sure you get help reading the water temperature if needed. Record the temperature.

4. Place the cup of water in the freezer. Measure the temperature of the water every 10 minutes. Continue measuring the temperature until ice starts to form. Record the observations.

5. Compare the temperature of the water in its solid state to its liquid state.

Results

When the temperature of water drops to 0 ℃ (32 ℉), the water begins to freeze. Therefore, water freezes at a specific temperature known as the freezing point. The freezing point,

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0 ℃ (32 ℉), is the exact temperature at which water changes from a liquid to a solid (ice).

Conclusion

This experiment verified that the freezing point of water is 0 °C (32 ℉), at which it changes from a liquid to a solid.

Boiling point of water

Water boils when it reaches a specific temperature. The boiling point is the exact temperature at which a liquid (water) boil and turns to water vapour. The boiling point is important because it ensures safe drinking water. Some foods are cooked better when the water has reached its boiling point.

Experiment 6: *Investigating the boiling point of water*

Aim: To demonstrate boiling point of water

Materials: Heat source (stove), water, saucepan and thermometer

***Note*:** Do not remove the thermometer from the water while reading the temperature.

Procedure

1. Place the saucepan with water on the stove that is already switched on.
2. Measure the temperature of the water as it begins to boil.
3. Continue measuring the temperature of the water until it boils vigorously.
4. Record the results.

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Results

When water reaches 100 °C (212 °F), it boils. This temperature is known as the boiling point of water.

Conclusion

The experiment showed that boiling point of water is 100 °C.

Revision exercise

Section A: Choose the most correct answer.

1. Below 0 °C water exist in a state known as _____.

   (a) fluid

   (b) steam

   (c) ice

   (d) gas

2. A state of matter that has definite shape and volume is called _____.

   (a) solid

   (b) liquid

   (c) plasma

   (d) gas

3. Water can exist in three states of matter, which are _____.

   (a) solid, gas and stone

   (b) solid, ice and gas

   (c) stone, air and ice

   (d) solid, liquid and gas

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4. Water temperature at 100 ℃ or 212 ℉ is called _____.

   (a) the freezing point of water

   (b) the boiling point of water

   (c) the melting point of water

   (d) the steam point of water

5. When water is heated, it boils and changes into _____.

   (a) steam

   (b) milk

   (c) ice

   (d) liquid

Section B: Write **TRUE** for a correct statement and **FALSE** for an incorrect statement.

6. Water obtained by condensing steam is clean and safe.

7. Hail is often common in very low temperature environment.

8. Water that has boiled to its boiling point is safe for drinking.

9. Frozen food can last longer without spoiling.

10. When water freezes, it turns into gas immediately.

Section C: Answer the following questions:

11. How can the liquid water be changed into steam and turned into a liquid state?

12. How can you use water in its solid, liquid, and gaseous states?

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13. Explain the importance of the freezing point of water and the boiling point of water.

14. Differentiate the properties of water in its solid, liquid, and gaseous states.

Vocabulary

Celsius (℃)	unit of measurement for temperature based on the freezing point of water at 0 ℃ and the boiling point of water at 100 ℃
Degree	unit of measurement used to express how hot or cold something is
Fahrenheit (℉)	unit of measurement for temperature based on the freezing point of water at 32 ℉ and boiling point of water at 212 ℉
Temperature	measure of how hot or cold something is

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Chapter Four

Combustion

Introduction

Combustion is a common process in our everyday lives. In this chapter, you will learn about the meaning, causes and effects of combustion. You will also learn various ways to protect yourself from accidents that can occur due to combustion. Similarly, you will also perform simple experiments to investigate the importance of air (specifically oxygen) in combustion. The competencies developed will enable you to produce heat and light safely for various uses. It will also enable you to protect yourself and others from accidents caused by burning materials.



Think The importance and effect of combustion in daily life

Concept of combustion

Combustion is a process in which a flammable substance reacts rapidly with oxygen. This reaction releases energy in the form of heat and light, often producing a flame. Flames are visible and feel hot during burning. For example, when firewood burns in a fire, it combines with oxygen. This reaction releases heat and light. Additionally, combustion produces carbon dioxide and water vapour. This process is essential in many activities. These include cooking food, melting metal and incinerating waste.

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Activity 1

1. Use library or reliable online sources to identify the necessary conditions for combustion.
2. Describe any daily combustion events you have witnessed relating to what you have searched in (a).

Necessary conditions for combustion

For combustion to occur, three components are needed: heat, oxygen gas and fuel.

Heat

Heat increases the temperature of the fuel. For combustion to begin, sufficient heat must ignite the fuel. A spark, a flame or another heat source can cause this heat.

Oxygen

Oxygen is the gas required for combustion. It is available in the air. Without oxygen, combustion cannot take place.

Fuel

Fuel can be solid, liquid or gaseous. Examples of fuel include wood, charcoal, gas, oil or other materials that can catch fire and burn. The relationship between oxygen gas, heat and fuel forms the fire triangle, as shown in Figure 1. If any of these components is missing, combustion cannot occur.



Figure 1: *Fire triangle*

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Experiment 1: *Investigating the importance of oxygen gas in combustion.*

Aim: To demonstrate the importance of oxygen gas in combustion

Materials: Matchbox, two candles, a transparent container with one open side or an air-tight jar, a watch and a table

Procedure

1. Take the two candles and label them A and B.
2. Light both candles.
3. Cover candle B with a transparent container, preventing air from entering.
4. Observe the results of both candles for one minute.
5. Write the results of your observation for each candle.

Question

Write the reasons for the results of the observation.

Results

Candle A continues to glow because there is enough oxygen to support the combustion process. The burning of candle B slows down and eventually goes out due to a lack of sufficient oxygen to sustain combustion.

Conclusion

The experiment demonstrated that oxygen is essential for combustion.

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Types of combustion

There are various types of combustion, including complete combustion and incomplete combustion.

Complete combustion

Complete combustion occurs when fuel burns with sufficient oxygen gas, completely consuming the fuel. The fuel buns clearly with a bright flame.

Experiment 2: *Investigating complete combustion*

Aim: To demonstrate complete combustion

Materials: Wood, dry leaves or pieces of paper, a fire source, and an open environment

Caution: Take precautions when conducting this experiment to avoid burning your body parts or damaging the surrounding environment.

Procedure

1. Arrange the wood in an open prepared space.
2. Ignite the wood using dry leaves or pieces of paper.
3. Allow the wood to burn completely.
4. Observe what has happened after the wood has burnt?

Results

The experiment was conducted in an open space to ensure there was enough oxygen for the wood to burn completely. When the firewood was ignited using dry leaves or pieces of paper, it caught fire and began to burn. As the wood continued to burn, it produced a bright flame. After the firewood burned

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completely, only ash remained, with no unburnt wood. Burning firewood or charcoal during cooking is an example of complete combustion.

Conclusion

The experiment confirmed complete combustion, as the firewood burned entirely to ash due to the presence of sufficient oxygen. In this experiment, the firewood burned to release heat and light. All firewood turns into ashes due to the presence of enough oxygen gas.

Incomplete combustion

Incomplete combustion occurs when fuel burns with insufficient supply of oxygen gas. In this type of combustion, the fuel does not burn completely. It is also produces more smoke and soot.

Experiment 3: *Investigating incomplete combustion*

Aim: To demonstrate incomplete combustion

Materials: Firewood, dry leaves or pieces of paper, a fire source, a large metal container, and a tool for digging soil, an open environment

Note: Caution is required when conducting this experiment.

Procedure

1. Arrange the firewood in a prepared open space.
2. Ignite the firewood using dry leaves or pieces of paper.
3. Keep the fire glowing until the wood catches fire properly.
4. Cover the fire with the prepared metal container.

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5. Using soil, cover all areas that might allow air to enter.

6. Wait for the burning overnight, then uncover.

7. Observe what has happened.

Questions

1. Explain the difference between the results in Experiments 2 and 3.

2. What caused different results in these experiments?

Results

When firewood is burned under a metal container, the limited supply of oxygen causes the materials to burn incompletely. Upon uncovering the set up the next day, it is observed that the firewood has not fully combusted, indicating incomplete combustion. Charcoal making is an example of incomplete combustion; when wood burns in a pit covered with soil, it turns into charcoal due to the limited oxygen supply.

Conclusion

The experiment demonstrated incomplete combustion caused by insufficient oxygen, resulting in unburned charcoal. In this experiment, burning firewood in a limited amount of oxygen often releases heat without a light. The firewood changed into charcoal due to the absence of enough oxygen gas.

Effects of combustion

The combustion process can affect living organisms and the environment. These effects can be good or bad.

Advantages of combustion

The following are some of the benefits of the combustion process:

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(a) Combustion produces energy for cooking, heating, and running various machines. For example, gas combustion in stoves provides energy for cooking food, as shown in Figure 2;



Figure 2: *Cooking*

(b) Combustion is used in industries for various activities. It includes smelting iron and producing items such as steel bars and plastics. Fire is also used in smelting gold;

(c) The heat generated from the combustion helps to dry and preserve foods and crops; and

(d) The heat generated from the combustion helps us during cold weather. For example, we can warm our bodies with heat from fire during cold weather as shown in Figure 3.

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Figure 3: *Warming the body*

Disadvantages of combustion

The combustion can lead to various disadvantages as follows:

(a) Fire can cause significant environmental damage, such as destroying vegetation and animal habitats.

(b) Forest burning can cause soil erosion and loss of soil fertility;

(c) Fire can destroy buildings, causing significant property loss as shown in Figure 4.



Figure 4: *Burning of property*

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(d) Burning substances such as charcoal and petrol contribute to the increase in harmful gases in the environment.

Protection against burning

Protecting yourself from accidents involving burning materials requires a combination of safety practices and preparedness. Here are some strategies which will protect you and others from accidents caused by burning:

Keeping flammable materials properly

Keep flammable items (like paper, cloth, and chemicals) away from heat sources and open flames to prevent accidental ignition.

Using proper tools

Always use tools specifically designed for handling burning materials, such as tongs or long-handled utensils. This practice minimises the risk of accidents, reassures you and boosts your confidence in fire safety practices.

Wearing protective gear

When handling burning or hot materials, it is crucial to wear protective clothing, such as heat-resistant gloves and aprons. This provides a layer of safety and reduces the risk of injuries.

Supervising open flames

Never leave open flames, such as candles, fireplaces and outdoor grills, unattended.

Monitoring the burning

If you are burning waste, never leave it unattended and ensure it is completely extinguished before leaving the area.

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Note: Your safety is a priority. Seek medical help if severe burns or injuries have occurred.

Activity 2

Visit the kitchen in your home place or school and observe how the cook protects from burning when handling hot substances.

Revision exercise

Section A: Choose the most correct answer

1. What does combustion mean?
   (a) The process of solidifying substances
   (b) The process of deteriorating substances
   (c) The process of changing substances into a liquid state
   (d) The process of combining fuel with oxygen to produce heat and light
2. What is needed for combustion to take place?
   (a) Air, water and light
   (b) Fuel, water and oxygen
   (c) Fuel, heat and oxygen
   (d) Water, heat and oxygen
3. What is the role of oxygen in the combustion process?
   (a) To provide heat
   (b) To prevent the fire

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(c) To support the ignition of the fire

(d) To convert the fire into water

4. What happens during the combustion process?

(a) Substances turn cold

(b) Substances lose their original nature

(c) Substances release cold energy

(d) Substances remain unchanged

Section B: Write **TRUE** for a correct statement and **FALSE** for an incorrect statement

5. Combustion helps us iron our clothes.

6. It is advised to minimise or stop using firewood because it leads to environmental degradation.

7. Wet pieces of cloth can be used to handle hot containers.

Section C: Short answer questions

8. What do you understand about the combustion process?

9. Briefly explain the conditions necessary for combustion to take place.

10. Briefly explain four effects of combustion.

11. How can combustion pollute the environment?

12. What is complete combustion?

13. Explain the condition necessary for incomplete combustion.

14. Explain the safety measures you will use when handling a hot charcoal burner.

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Vocabulary

Factories	places where products are manufactured
Fuel	substances that can burn to produce heat and light
Transparent	a state that allows light to pass through

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Chapter Five

Energy

Introduction

There are various forms of energy used in our daily lives. In this chapter, you will learn about the forms and uses of heat, light, and sound energy. Additionally, you will conduct simple experiments on some types of energy. The competencies developed will enable you to identify sources of energy and use energy efficiently in everyday life.



Think How life could be without energy

Meaning of energy

In our environment there are various sources of energy. These energies can cause work to be done. Therefore, energy is the ability that enables various tasks to be accomplished.

Forms of energy

There are various forms of energy that we use in our daily live including heat, light and sound.

Heat energy

Heat is a form of energy that is transferred between objects with different temperatures. It comes from natural sources such

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as stars including the sun and lightning. It can also come from non-natural sources like firewood, charcoal, coal, electricity, oil, friction and gas. These sources are used to produce heat for various daily applications. Examine Figure 1, then read the explanations that follow.



(a) Sun



(b) Firewood stove



(c) Charcoal



(d) Electric cooker



(e) Gas cooker

Figure 1: *Sources of heat energy*

Figure 1(a) shows the sun, which is the primary natural source of heat energy. Figure 1(b) and 1(c) show cooking activities using firewood and charcoal as sources of heat. Figure 1(d) and 1(e) show cooking activities using electricity and gas respectively.

Activity 1

Identifying a source of heat

Materials: One dry stick and a piece of dry wood

Procedure

1. Rub your palms together.
2. Write down what you feel while rubbing your palms.

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3. Take a thin dry stick.
4. Rub the stick on a piece of dry wood.
5. Touch the part of the rubbed stick and write down your observation.

There are various sources of heat, such as the sun, electricity, burning wood and friction. Additionally, some forms of energy can be converted into heat energy. For example, sunlight can be converted into heat energy, electrical energy and chemical energy. Rubbing two objects together also generates friction, which produces heat energy, as observed in Activity 1.

Heat transfer

Heat energy can travel from a hot source to a cooler area. It can move through solids, fluids (liquids and gases) and empty space (vacuum).

Experiment 1: Investigating transfer of heat through solid objects

Aim: To demonstrate the transfer of heat through solid objects

Materials: A piece of metal or a spoke, a piece of wood, heat source such as a stove, matchstick and wax or a candle

Caution: Be careful when using fire to avoid fire related hazards.

Procedure

1. Lay the piece of metal on top of the piece of wood as shown in Figure 2(a).

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2. Place wax or a candle at one end of the metal as shown in Figure 2(b).

3. Position a piece of stiff paper between the wax and wood. This prevents direct heat from the stove from reaching the wax as shown in Figure 2(c).

4. Place the charcoal stove at the other end of the metal piece as shown in Figure 2(c).



Figure 2: *Heat transfer in solids*

5. Light the charcoal stove and wait. What happens to the wax?

6. Write down what happened to the wax after lighting the stove.

Results

After lighting the stove, the wax melted. This implies that, heat was transferred from the source to the metal and wax.

Conclusion

This experiment proves that heat travels in solids by *conduction*.

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Experiment 2: *Investigating good and bad conductors of heat*

Aim: Identify good and bad conductors of heat through solids

Materials: A metal spoon, a plastic spoon, a wooden stick, a cup of hot water and a few small pieces of butter

Caution: Teacher supervision is required.

Procedure

1. Place a small piece of butter at the tip of metal spoon, plastic spoon and the wooden stick. Make sure the butter sticks well to the tip of each object.
2. Dip the bottom end (the part without butter) of each spoon and the wooden stick into the cup of hot water at the same time.
3. Observe which piece of butter melts the fastest and record your observation.

Results

The metal spoon will conduct heat quickly from the hot water, causing the butter to melt fast. The material whose butter melts first is the good conductor of heat. The plastic spoon and wooden stick will conduct heat much slower, and the butter may take longer to melt or not melt at all. The one whose butter melts last is the bad conductor of heat.

Conclusion

This experiment helps us identify how different solids (metal, plastic, wood) conduct heat, allowing us to differentiate between good and bad conductors of heat.

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Experiment 3: *Investigating transfer of heat in liquids*

Aim: To demonstrate transfer of heat in liquids

Materials: A cooking pot, a stove a matchstick, water, small pieces of A4 paper and a thermometer

Caution: Exercise caution throughout the experiment to avoid fire hazards.

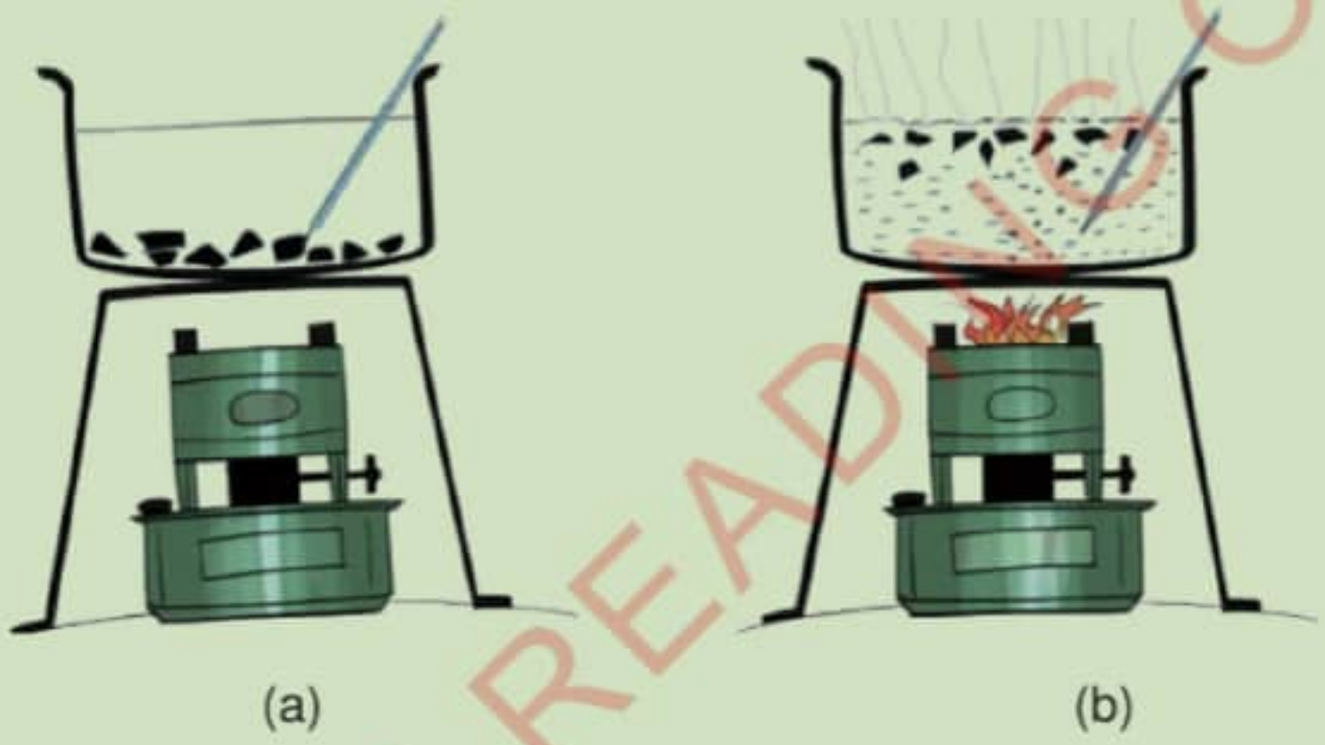
Procedure

1. Take small pieces of paper and put them in the cooking pot.
2. Add cold water into the cooking pot and leave it for five minutes as shown in Figure 3(a). What happens to the small pieces of paper?
3. Measure the temperature of the water using a thermometer and record it.
4. Place the pot with the pieces of paper and water on the lit stove as shown in Figure 3(b).
5. Wait for ten minutes, then measure the water temperature using a thermometer and record it.
6. What happened to the pieces of paper and the water temperature?
7. Write the results as in Table 1.

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Table 1: *Heat in liquids*

No.	Pieces of paper	Time (Min.)	Temperature (℃)	Results
1.	Before lighting the stove	5		
2.	After lighting the stove	10		



(a) (b)

Figure 3: *Heat transfer in liquids*

Results

The paper pieces stayed at the bottom after five minutes in water. This shows the water did not rise, and the temperature stayed the same. After heating, the water and paper pieces began to rise together. The temperature increased after the water started boiling. Heat transferred from the stove to the pot and water by convection method.

Conclusion

In this experiment, it showed that, as water heats up, it rises and transfers heat through *convection*.

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Experiment 4: *Investigating transfer of heat in empty space*

Aim: To demonstrate transfer of heat in *vacuum space*

Materials: A cloth, water, a clothesline, pegs and a basin

Procedure

1. Fill the basin with water.
2. Take a dry cloth and place it in the water. What does the cloth look like?
3. Remove the cloth from the basin. Wring it out slightly, then hang it in the sun for one hour as shown in Figure 4.
4. After an hour, remove the cloth and observe it. How does the cloth look like?



Figure 4: *Heat transfer in empty space*

Results

The dry cloth became wet when placed in water. When the wet

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cloth was exposed to the sun, it dried due to the heat from the sun.

Conclusion

In this experiment, we learned that the wet cloth dried due to the sun's heat. The sun's heat transfers to objects through empty space by *radiation.*

In the experiments you conducted, you found that heat can be transferred in three main ways: *conduction, convection* and *radiation*.

(a) *Conduction* is the transfer of heat energy in solids, such as metal.

(b) *Convection* is the transfer of heat energy in fluids such as liquid and gases.

(c) *Radiation* is the transfer of heat energy in empty spaces. In the experiment, air was considered as empty space because it behaves almost as empty space when it comes to radiation transfer. The three ways of heat transfer are illustrated more in Figure 5.



Figure 5: *Three types of heat transfer*

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Uses of heat energy

Heat energy can be used for various purposes as shown in Table 2.

Table 2: *Uses of heat energy*

Picture	Explanation
	Using heat to stay warm against the cold.
	Ironing clothes to kill germs and make them look neat.
	Drying crops to remove moisture that could cause spoilage.

Other uses of heat include; cooking food, boiling drinking water, melting metals in industries and generating electricity. Additionally, heat is used in the production of various products such as metals, plastics and glasses.

Exercise 1

Section A: Write **TRUE** for the correct sentence and **FALSE** for an incorrect sentence.

1. Firewood, charcoal, coal and gas are non-natural sources of heat.

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2. Metal and wood are good conductors of heat.

3. Heat can travel through solids, liquids and gases.

4. We dry crops using heat energy.

5. Heat and temperature have the same meaning.

Section B: Questions with short answers

6. Name any three materials that are good conductors of heat.

7. Explain the importance of heat energy at home.

8. Why are the handles of cooking pots or irons made of plastic or wood?

9. Why do some clothes take long time to dry during the rainy season?

10. Why are we advised to wear sweaters and use blankets during cold weather?

Light energy

Light energy is a type of electromagnetic radiation. There are two main types of light energy sources: natural and artificial light sources. Examples of natural light sources are stars including the sun, lightning and bioluminescent organisms. The artificial light sources include fire, flashlights, candles and fireworks. Figure 6 shows sources of light energy.

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(a) Kerosene lamp



(b) The sun



(c) A bulb



(d) A torch

Figure 6: *Sources of light energy*

Activity 2

Use reliable sources of information, including online libraries, videos or digital labs to complete this activity.

1. List examples of light energy sources found in your environment
2. Draw one example of the light sources you listed in number (1) and write its use.

Examples of sources of light in our environment are like the sun, lamp and flash light. The sun helps plants to grow, the lamp helps us read and flashlight helps us see in the dark.

Objects that allow or block light from passing through

Activity 3

1. Use a library, reliable online sources, or other sources of information to identify objects that transmit light, block light, or partially transmit light.

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2. Using the Information obtained in number (1) examine the objects in Table 3 below and mark an "√" in a respective space.

Table 3: *Indicating objects that allow or block light from passing through*

Picture	Allows light (Transparent)	Does not allow light (Opaque)	Partially allows light (Translucent)
 Ice block			
 Wood			
 Glass window			
 Bowl			
 Foggy bottle			

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Picture	Allows light (Transparent)	Does not allow light (Opaque)	Partially allows light (Translucent)
 Cooking pot			
 Car windshield			

In our environment some objects allow light rays to pass through. These objects are called *transparent.* For example, glass and some plastic. Objects that allow only portion of light to pass through are called *translucent*. Examples include foggy bottle. Objects that block light completely are called *opaque*. For example, wood, books and walls.

From the reliable sources of information, researched you have observed how light passes through various objects. By marking "√" on appropriate statement and diagram it shows that:

(a) Light passes easily through a transparent object, such as a car windshield glass bowl and clean window glass.

(b) Light passes partially through an object if it has limited transparency, such as foggy glass and an ice block

(c) Light cannot penetrate the opaque object. Examples include pieces of wood and cooking pot.

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Experiment 5: *Investigating interaction of light rays with surfaces of various objects*

Aim: To demonstrate the ability of different objects to allow light to pass through

Materials: Flashlight, battery, empty glass, black plastic cup, white cloth, piece of eye glass, piece of wood, glass of water and a book

Procedure

1. Take a flashlight, insert the battery, and turn it on to check if it emits light.
2. Enter a dark room with the flashlight on. Arrange the listed items on a table.
3. Direct the flashlight on the book and record your observations.
4. Repeat step 3 with the other items listed above.
5. Record the results for each item based on the following questions:
   (a) Which items allowed light rays to pass through?
   (b) Which items did not allow light rays to pass through?
   (c) What happened to the light when it struck the surfaces of these items?

Results

An empty glass and a piece of eye glasses allow rays to pass through. A white cloth and a glass full of water will allow some light to pass through, while a black plastic cup, a piece of wood and a book will not allow light rays to pass.

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Conclusion

Transparent objects allow light to pass through, translucent objects allow partial light and opaque objects block light completely.

Experiment 6: *Investigating transfer of light in straight line*

Aim: To demonstrate transfer of light

Materials: A glass of water, opaque paper, perforated paper, flashlight and water

Procedure

1. Place the glass of water on a desk.
2. Position the perforated paper in front of the glass.
3. Place the opaque paper on the opposite side of the glass.
4. Direct the flashlight beam at the perforated paper as shown in Figure 7. Can you see the light on the opaque paper?

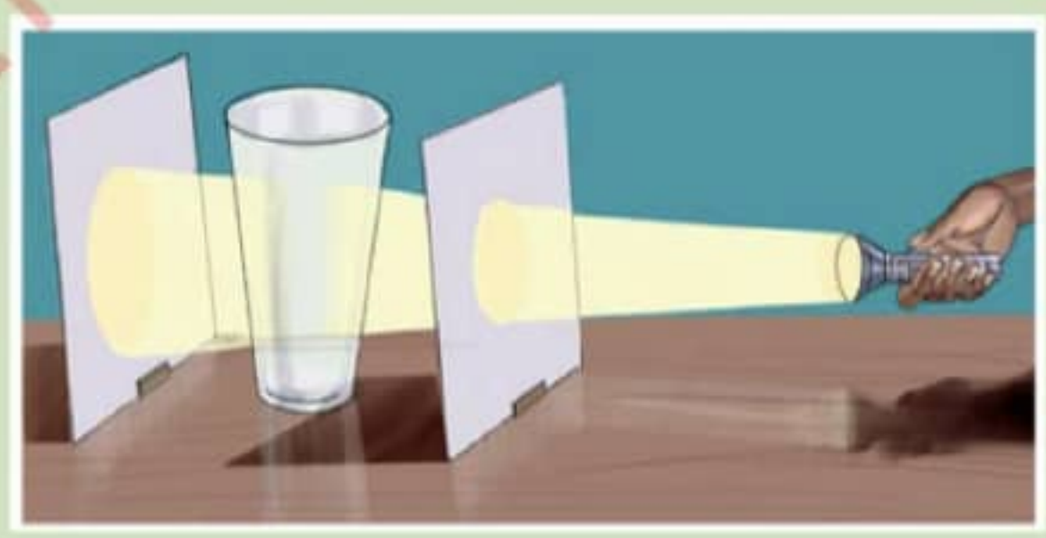


Figure 7: ***How light travels***

5. Record the results of the investigation.

Results

The light beam emanates from the flashlight and passes through the perforated paper. It then travels through water and glass and exits outside on the opaque paper.

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Conclusion

This demonstrates that, light travels in a straight line

Properties of light

The properties of light include traveling in a straight line, forming shadows, bending and reflection.

Activity 4

Investigating transfers of light in the air

1. Observe in the morning as the sun rises. What do you see?
2. Record your observations.

When you wake up early in the morning as the sun begins to rise, you will see light rays. These rays appear to travel in straight lines. Light allows us to see various objects as shown in Figure 8.

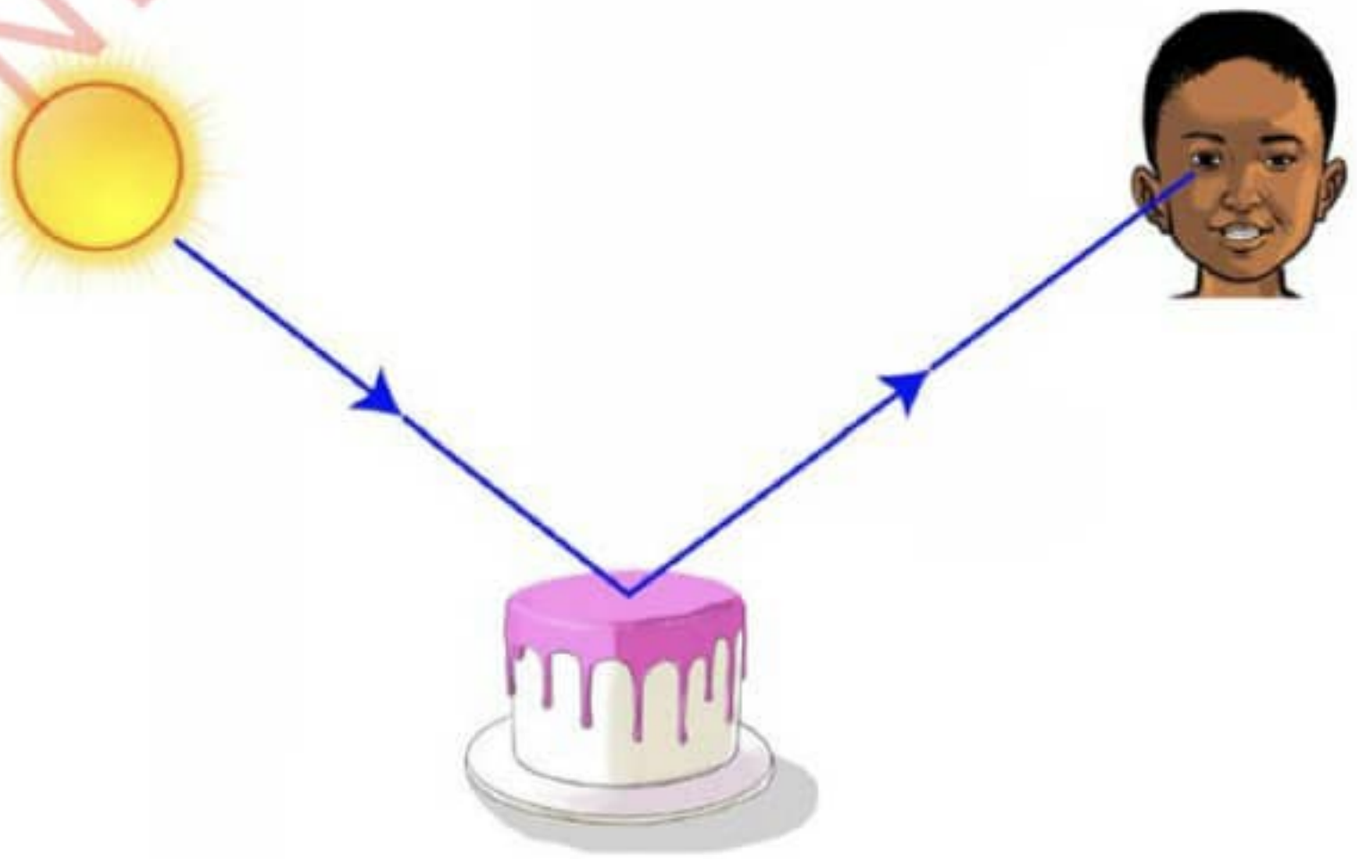


Figure 8: *Light helps us see various objects*

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Experiment 7: *Investigating formation of shadow*

Aim: To demonstrate the formation of a shadow

Materials: Ball, flashlight, cup, chair, sunlight, stool and table.

Procedure

1. Hold the cup or book facing the sunlight. What do you see?
2. Place the table and chair outside the classroom in the sunlight.
3. Look under the table and chair. What do you see?
4. Enter a dark room. Place the ball on the support such as stool and direct the flashlight at the ball as shown in Figure 9. What do you see?
5. Record your observations for steps 1, 3 and 4.

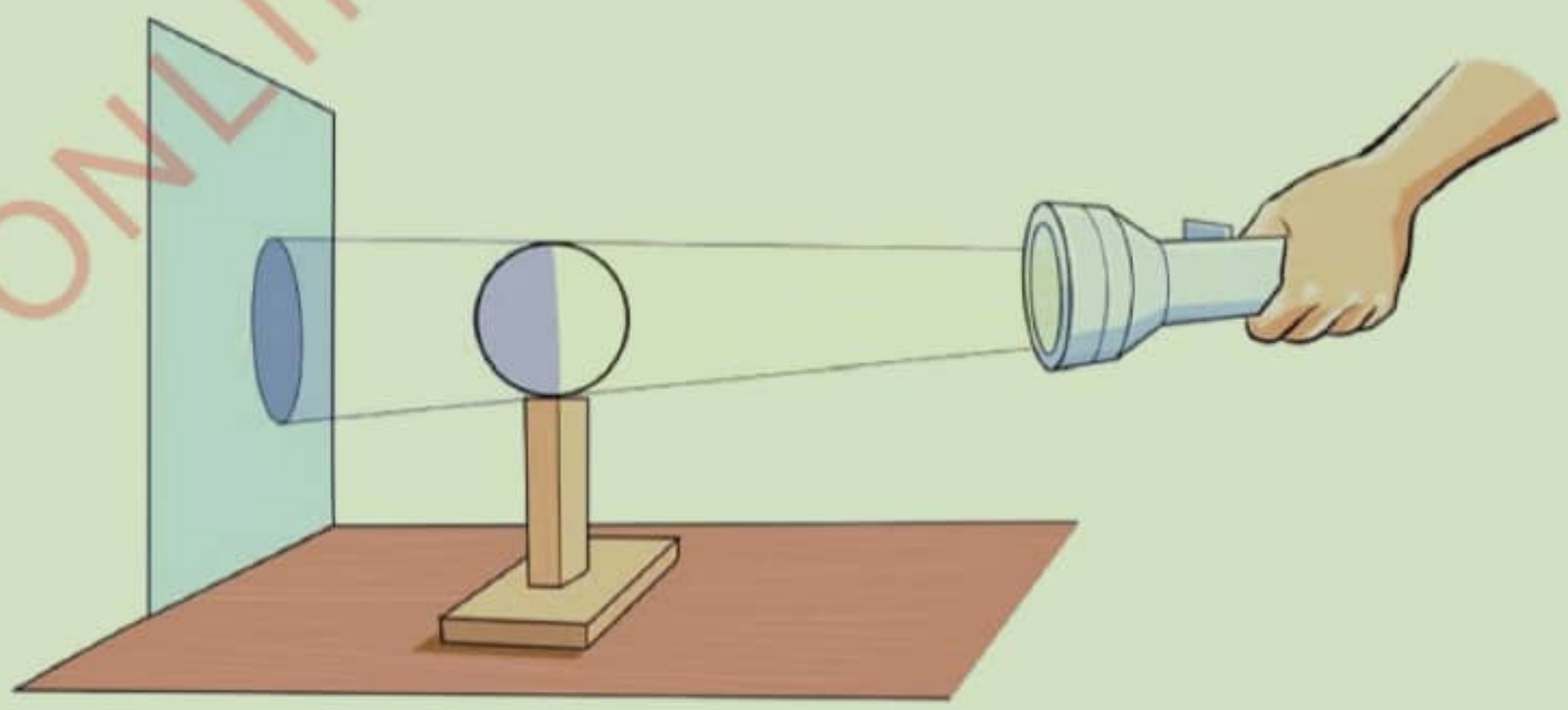


Figure 9: *Flashlight directed at a ball and formation of shadow on the wall*

6. What would happen if there were no light energy?

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Results

A shadow which is a dark area appeared behind the ball on the wall

Conclusion

The shape of the shadow resembles the object that blocked the light.

Examine Figure 10, and then answer questions that follow.



Figure 10: *Formation of shadows*

Questions

1. What do you observe about the shape of the shadow in Figure 10?
2. Can shadows be colourful? Give reasons for your response.

Reflection of light rays

The reflection of light is the process of light rays bouncing back when they hit a smooth and shiny surface. An example of a smooth, shiny surface is a flat mirror. Examine Figure 11, and then complete the following activity.

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Figure 11: *Formation of an image in a plane mirror*

Activity 5

Investigating the reflection of light

Materials: Plane mirror, table and pen

Procedure

1. Take a plane mirror and place it on the table, then look at yourself in the mirror.
2. Take a pen and hold it in front of the mirror.
3. Record the results of your actions. What did you discover from the observation you made?

When light rays hit smooth, shiny surface, they are reflected. The reflection of light rays causes the formation of image of an object or people in the mirror. For example, when you look in the mirror, light from a light source bounces off your body to the mirror. The mirror reflects the light back to your eyes, allowing you to see your image.

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Refraction of light rays

Another property of light is the bending of light rays, which is referred to as refraction. Light rays bend when they move from one medium to another, such as from air to water. For example, when light passes from air into water, the rays bend. We do not see the light rays bending directly but rather observe the effects of this bending. Examine Figure 12, then proceed with the following experiment.



Figure 12: *Effects of refraction of light*

Experiment 8: *Investigating refraction of light*

Aim: To demonstrate the refraction of light

Materials: A transparent bowl or beaker, clean water in a jug and pencil

Procedure

1. Take a clear bowl or beaker. Submerge part of the pencil into the bowl while holding it.

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2. Observe how the pencil appears.

3. Pour water from the jug into the bowl or beaker until it reaches half full, as shown in Figure 12.

4. Observe again the appearance of the pencil in the bowl or beaker after adding the water. What differences do you notice in the appearance of the pencil before and after adding the water?

Results

The pencil appeared to be bent at the water surface due to the refraction of light.

Conclusion

This experiment shows that light rays bend when they move from one transparent medium to another.

Uses of light energy

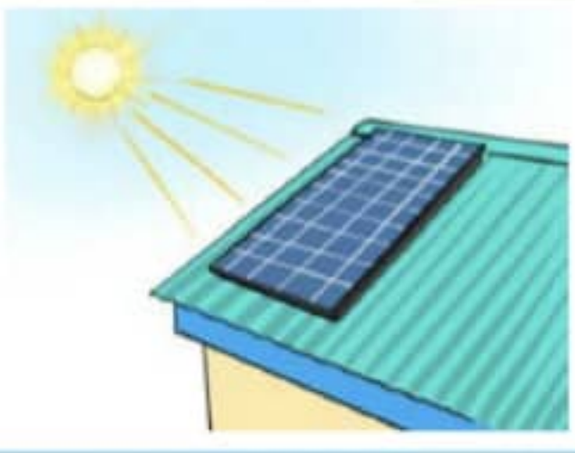
Light energy can be used for various purposes as shown in Table 4.

Table 4: *Uses of light energy*

Picture	Explanation
	Vision: Light helps us see things. Without light, we would not be able to see colours, shapes or objects around us.
	Photosynthesis: Plants use light energy from the sun to make their food.

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Picture	Explanation
	Solar panels: Solar panels capture sunlight and convert it into electricity, which can power homes.
	Lens (magnifying glasses): Lens use light to make objects look larger and clearer.
	Flashlights: Flashlights use light to help us see in the dark.

Sound energy

Sound energy is a type of energy that we can hear. Sound is produced by the vibration of various objects. These vibrations travel as waves. The sound waves move back and forth through materials. When you strike a drum, blow a whistle, or play a guitar, what is produced is sound. Other sources of sound include the keyboard, trumpet, and bell. Additionally, sound is produced when people talk or animals make noise. A chicken calling its chicks or an airplane flying also creates sound. Sound facilitates communication. Figure 13 shows actions that produce sound.

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(a)



(b)



(c)



Figure 13: *Actions of sound production*

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Activity 6

Identifying how sound transfers through air

Observe what happens when a bell rings at school.

1. Where does the sound come from when the bell rings at school?
2. From which direction do pupils gather?
3. In which direction do sound wave travel?
4. What have you discovered on how sound travels through the air?
5. What is the importance of sound energy in our everyday life?

When sound is produced, air particles vibrate and collide. These vibrations cause oscillations, creating sound waves. Sound waves spread through vibrating objects. These objects transfer sound by vibrating surrounding particles. For example, an airplane's sound reaches the ear through vibrating air molecules. The sound is transmitted from the plane to the ear via air.

Reflection of sound

The reflection of sound occurs when sound waves bounce off a surface. A reflected sound is called an echo. This can be observed in spaces where sound encounters reflective surfaces. Examples of reflective surfaces are walls or large structures. The time it takes for sound to return determines the distance of the surface.

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Activity 7

Demonstrating an echo

1. Enter an empty hall and stand at least 17m from one of the walls as shown in Figure 14.
2. Shout loudly. What do you hear?
3. Write the results of the action you performed.



Figure 14: *Demonstrating an echo formation*

When you stand in an empty hall and shout loudly, you hear the sound repeating. For example, if you say 'hello', after a few seconds, you would hear 'hello' again. This repeated sound is called an echo. What happens is that sound waves travel from your mouth through the air in the wall, hit a hard surface like a wall, and bounce back, causing you to hear the sound you initially made. Sound is reflected when it encounters a barrier or a hard surface like a wall or rock.

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Movement of sound in matter

Activity 8

Investigating how sound transfers in water

Materials: Bucket, water, bell or any object that can make a ringing sound

Procedure

1. Take a bucket and fill it with water.
2. Hold a bell in one hand and dip it in water.
3. Keep one of your ears gently on the surface of water, but don't let water get into your ear.
4. Ring the bell inside water as shown in Figure 15.
5. What do you hear and see? Write the results.



Figure 15: *Sound travel in water*

Sound travels through water just like it does in air, however it travels faster in water. When you ring the bell inside the water, you hear sound clearly. The waves demonstrate how sound travels through water.

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Activity 9

Investigating how sound travels through metal

Materials: A long metal rod, a table and a ruler

Procedure

1. Take a long piece of metal and place it on the table as shown in Figure 16.



Figure 16: *Sound travel through metal*

2. Ask your friend to place his/her ear on one end of the metal piece.
3. Tap the metal with a ruler.
4. Ask your friend what he/she hears and write it down.
5. Place your ear on one end of the metal.
6. Ask your friend to tap the metal at another end.
7. Write down what you hear.

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Materials like metal, water, and air can transmit sound. This shows that certain solids, liquids and air can carry sound waves.

Activity 10

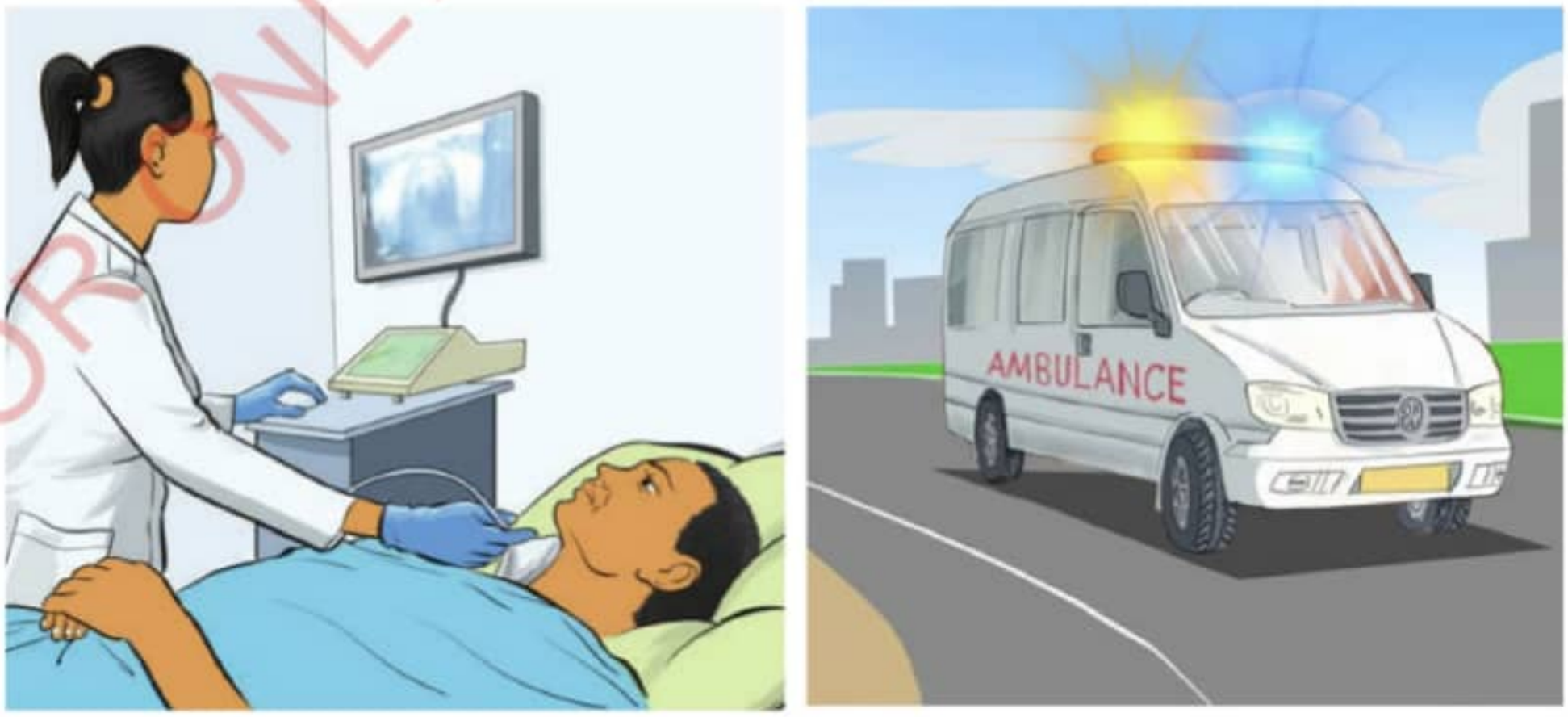
Read online articles or videos, then briefly explain which medium air, water or solids transmits sound the fastest.

Uses of sound energy

Sound energy enables communication in homes, schools and on the road. Devices like phones and tablets produce sound for communication. Phones, tablets, radios, televisions and computers also use sound for information sharing. They provide news, entertainment and various information through sound videos or pictures.

Sound is also used in medical treatment. For example, various medical devices use sound energy for effective diagnosis as well as surgeries. See Figure 17(a).

An ambulance has a siren which uses sound energy to signal medical emergency as in Figure 17(b).



(a) *Sound energy in medical equipment*

(b) *Sound energy in the siren of an ambulance*

Figure 17: *Devices that uses sound energy*

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The relationship between forms of energy

Electrical energy can be converted from one form to another depending on the device used. For example, a light bulb in flashlight, converts electrical energy from batteries into light and heat energy. Carefully examine the actions depicted in Figure 18, then perform Activity11.



(a) (b) (c)

Figure 18: *Uses of light, sound and heat*

Activity 11

Investigating types of energy in the environment and their relationships

Use a library, reliable online sources, or other sources of information.

1. List the forms of energy observed in Figure 18(a), 18(b), and 18(c).
2. Identify the relationship between the types of energy observed in Figure 18(a), 18(b), and 18(c).

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Revision exercise

Section A: Choose the correct answer.

1. When an object is struck, it produces _____.

   (a) stringed instrument

   (b) sound

   (c) drum

   (d) bell

2. When sound hits a wall and returns, it is called _____.

   (a) transmission

   (b) light

   (c) echo

   (d) procession

3. The main primary source of light energy is _____.

   (a) candle

   (b) moon

   (c) sun

   (d) fire

4. Light travels in a _____.

   (a) line that narrows

   (b) straight line

   (c) curved line

   (d) angled line

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5. Heat travels through solid materials like metal and iron by _____.

   (a) conduction

   (b) convection

   (c) refraction

   (d) radiation

6. The speed of sound energy is slowest in _____.

   (a) solids

   (b) liquids

   (c) air

   (d) solids and liquids

7. Energy that can travel from a hot to a cooler place is called _____.

   (a) sound energy

   (b) light energy

   (c) wind energy

   (d) heat energy

Section B: Write **TRUE** for the correct sentence and **FALSE** for an incorrect sentence

8. Radio is a communication device that emits sound energy.

9. Images appear in non-reflective and smooth objects.

10. Sound travels in air, liquids, and solids.

11. Light is refracted as it move from air to water.

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12. Sound energy is used in communication.

13. The handles of a pot and an iron are made of metal to prevent heat transfer.

14. Television produces sound energy only.

Section C: Match the items in **Section A** with items in **Section B** to form the correct sentences.

15.

Section A	Section B
(i) Shadow	(a) Sun and moon
(ii) Firewood	(b) Dark area that appears after light is blocked by an object
(iii) Straight line	(c) Produces light and heat energy after combustion
(iv) Two different transparent media	(d) Results from the vibrations of various objects.
(v) Reflection	(e) The way light rays travel
(vi) Image	(f) The result of a picture or drawing appearing on smooth, reflective surfaces
(vii) Sound	(g) Causes light to bend
(viii) Sun	(h) The main source of light and heat energy
(ix) Echo	(i) The action of light rays bouncing back after hitting a reflective or smooth surface
(x) Guitar, bell and piano	

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Section A	Section B
	(j) Sound that is reflected back after hitting a hard surface (k) Sources of sound energy (l) Hard objects, liquids, and gases

Section D: Questions with short answers.

15. List any two objects that allow light to pass through.
16. Explain how a shadow is formed?
17. What type of energy is produced when you sing a song?
18. Explain the three main methods of heat transfer.
19. What form of energy does a hen provide as it incubate its eggs?
20. What is the difference between the surfaces of objects that reflect light and those that do not reflect light?
21. What happens to a light ray when it travels from one transparent medium to another?

Vocabulary

Bright	the extent to which light is emitted or reflected from a surface
Candle lamp	a lamp that uses a wick and oil to burn
Radiate	emit energy in the form of waves
Reflect	to bounce light off a smooth and shiny surface
Refract	to bend light

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Chapter Six

Computer coding

Introduction

Computers help us do many things in our daily lives. In this chapter, you will learn to draw shapes that can be used in everyday life. You will also learn to arrange shapes logically to create patterns. Additionally, you will learn to create simple games using the Scratch program. The competence developed will enable you to think logically and create simple computer games.



Think

Different tasks that can be performed using a computer

Concept of patterns

Patterns are everywhere around us. A pattern is something arranged in a logical sequence, which make easier to predict the next element. Some patterns are natural such as zebra stripes and others are man-made such as chessboard as shown in Figure 1.

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(a) Natural pattern



(b) Man-made pattern

Figure 1: *Examples of patterns*

Activity 1

Observe Figure 1 and examine the logical sequence of each pattern.

You can make patterns by combining different shapes in a logical way as shown in Figure 2. You should think logically about the sequence of the patterns to identify which shape starts and which shape follows next. Also, you should have an idea of the final pattern you want to make.



Figure 2: *Different shapes logically arranged to form a pattern*

Activity 2

Observe Figure 2 and list the next three shapes that come next in the sequence.

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Drawing and arranging shapes logically

There are various computer programs used for drawing shapes. In this section, you will use a program called *Paint* to draw different shapes and arrange them logically.

Opening the paint program

The steps to open a Paint program using Windows 11 are as shown in Figure 3. The location of the search bar may differ slightly in other versions of Windows.

Steps

1. Click on the *'Windows Start button'*.
2. Type *'Paint'* in the search bar.
3. Click on *'Paint'*.

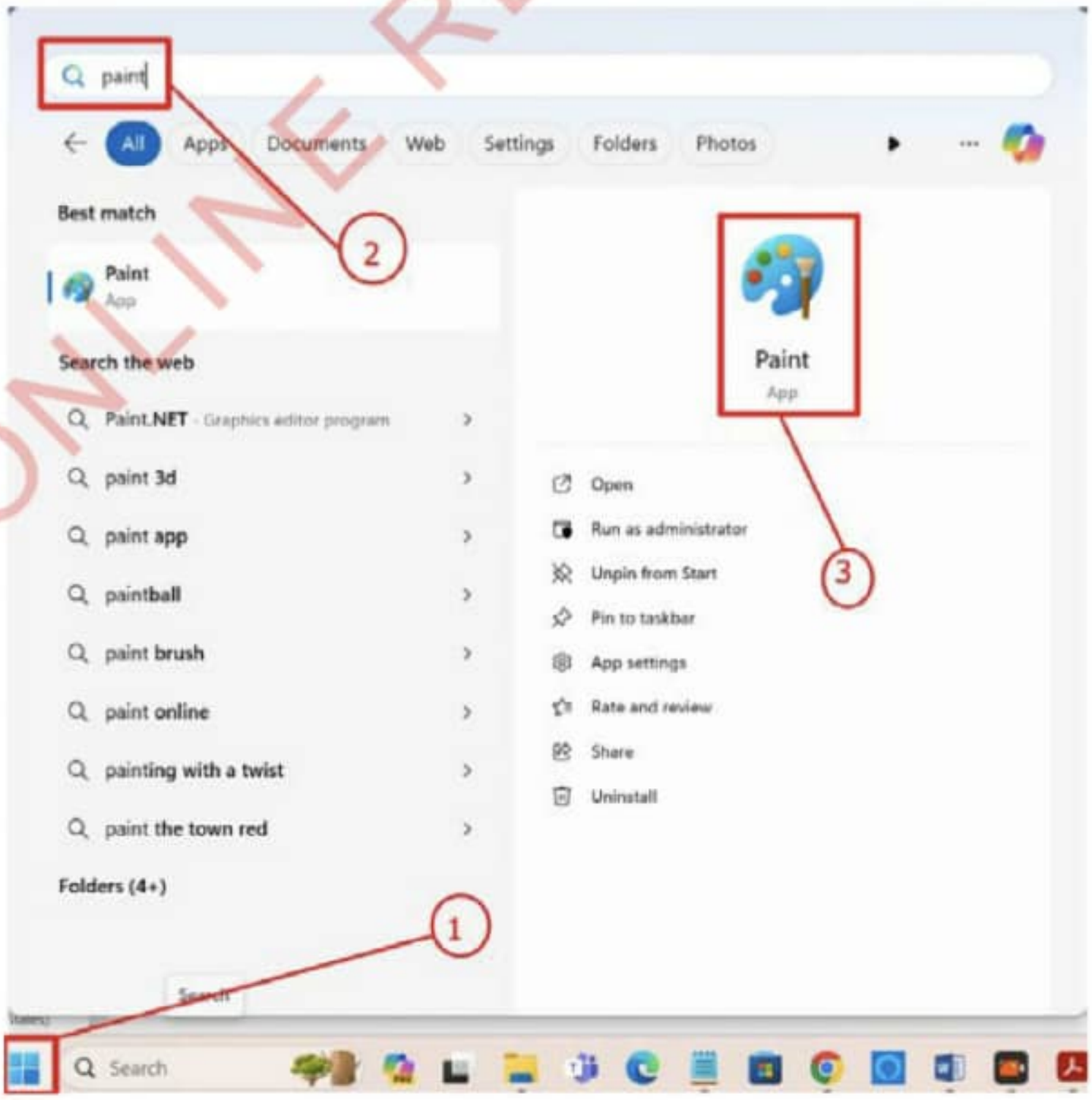


Figure 3: *Opening Paint program*

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The **Paint** program will open. Figure 4 shows parts of the paint window.

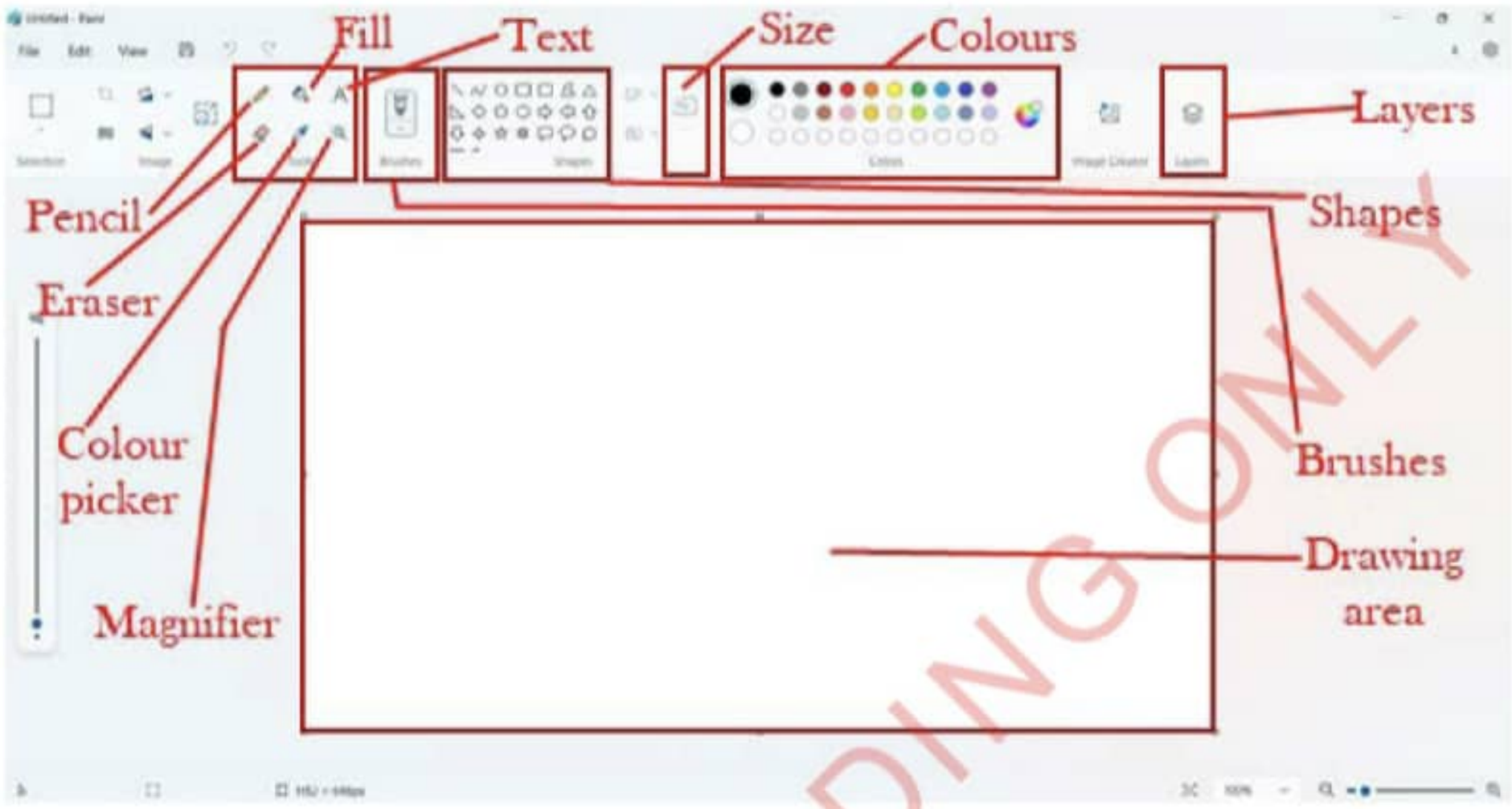


Figure 4: **Parts of the paint program window**

Drawing shapes

Activity 3

Drawing circles on a slope

Steps

1. Click on the **'Right-angled triangle'** tool as shown in Figure 5.

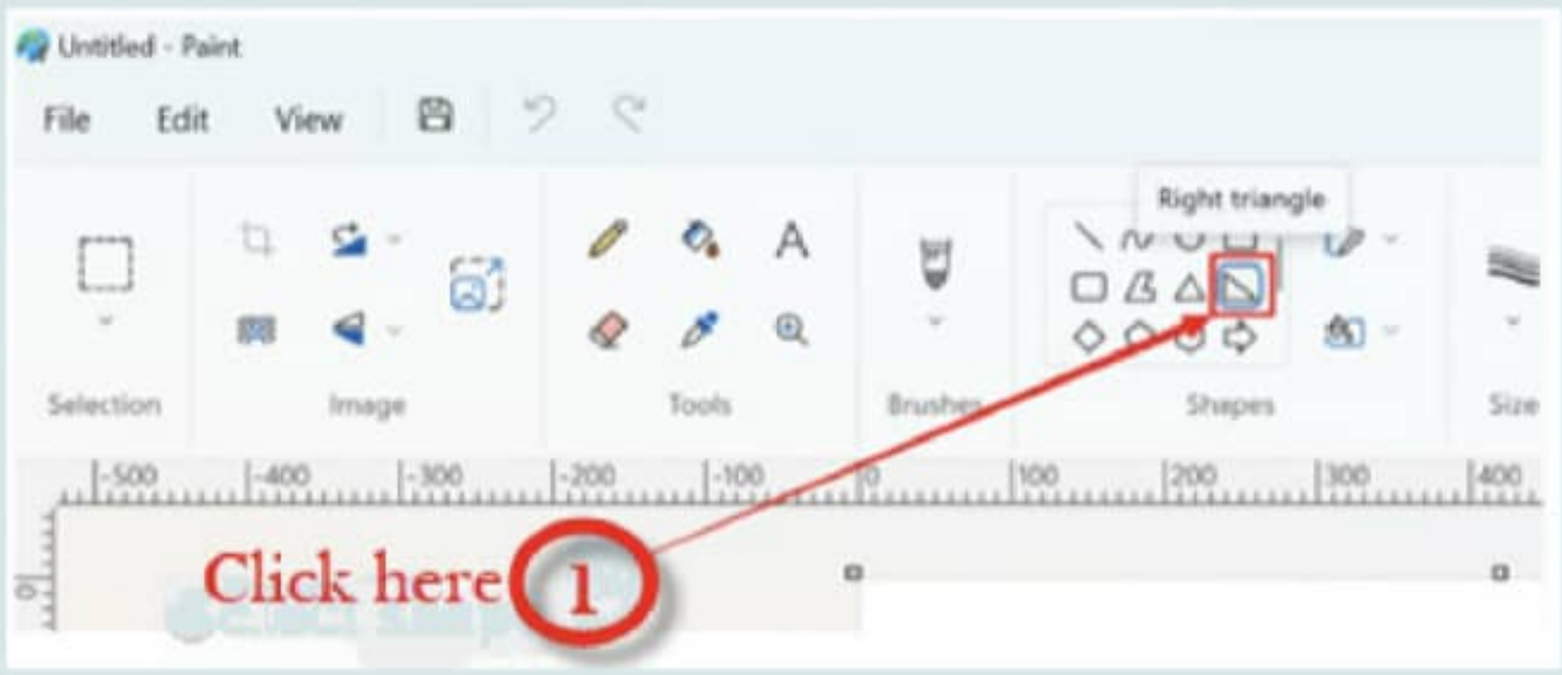


Figure 5: **Selecting right-angled triangle tool**

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2. Place the mouse cursor on the drawing area.

3. Click and hold the mouse, then drag it in the direction as shown in Figure 6.

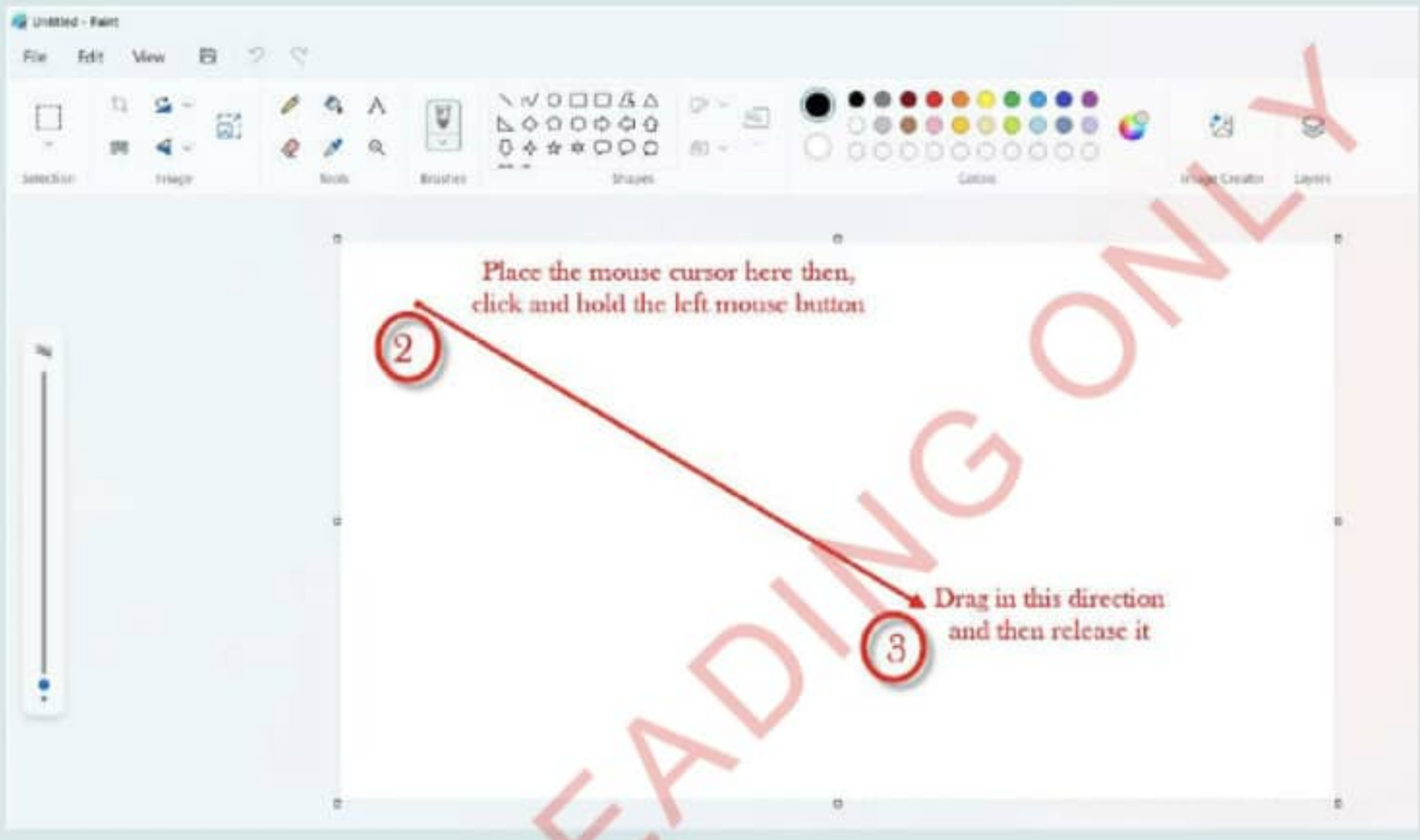


Figure 6: *Drawing a right-angled triangle*

4. You will get a *'right-angled triangle'* shape similar to the one shown in Figure 7.

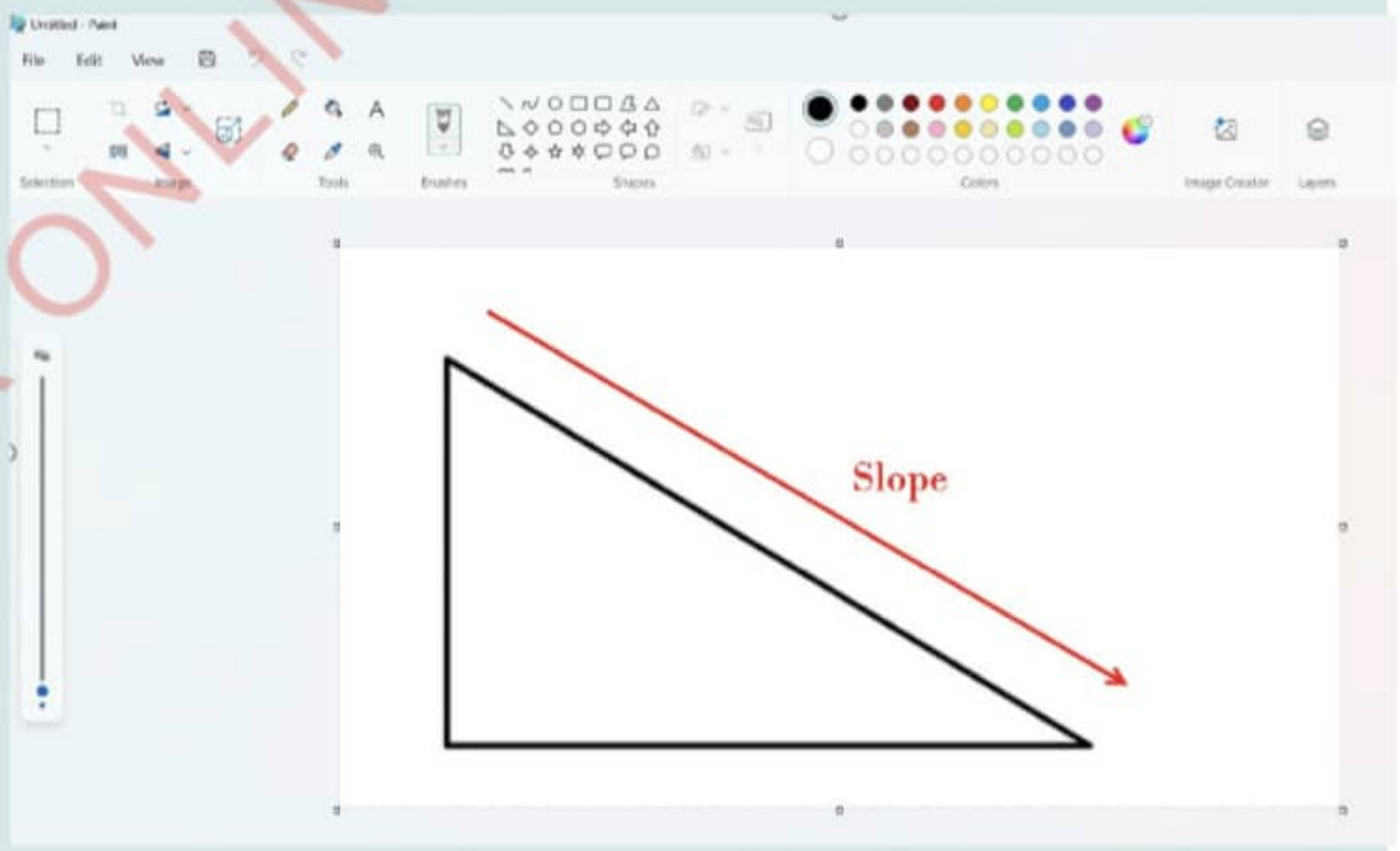


Figure 7: *A right-angled triangle forming a slope*

5. Click on the *'circle'* tool as shown in Figure 8.

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6. Place the cursor on the drawing area, hold and drag the mouse to draw the smallest *circle near the top of the slope*.

7. Place the cursor on the drawing area, hold and drag the mouse to draw the second *circle*, which is larger than the first in the middle of the slope.

8. Place the cursor on the drawing area, hold and drag the mouse to draw the third *circle*, which is larger than the second at the bottom of the slope.

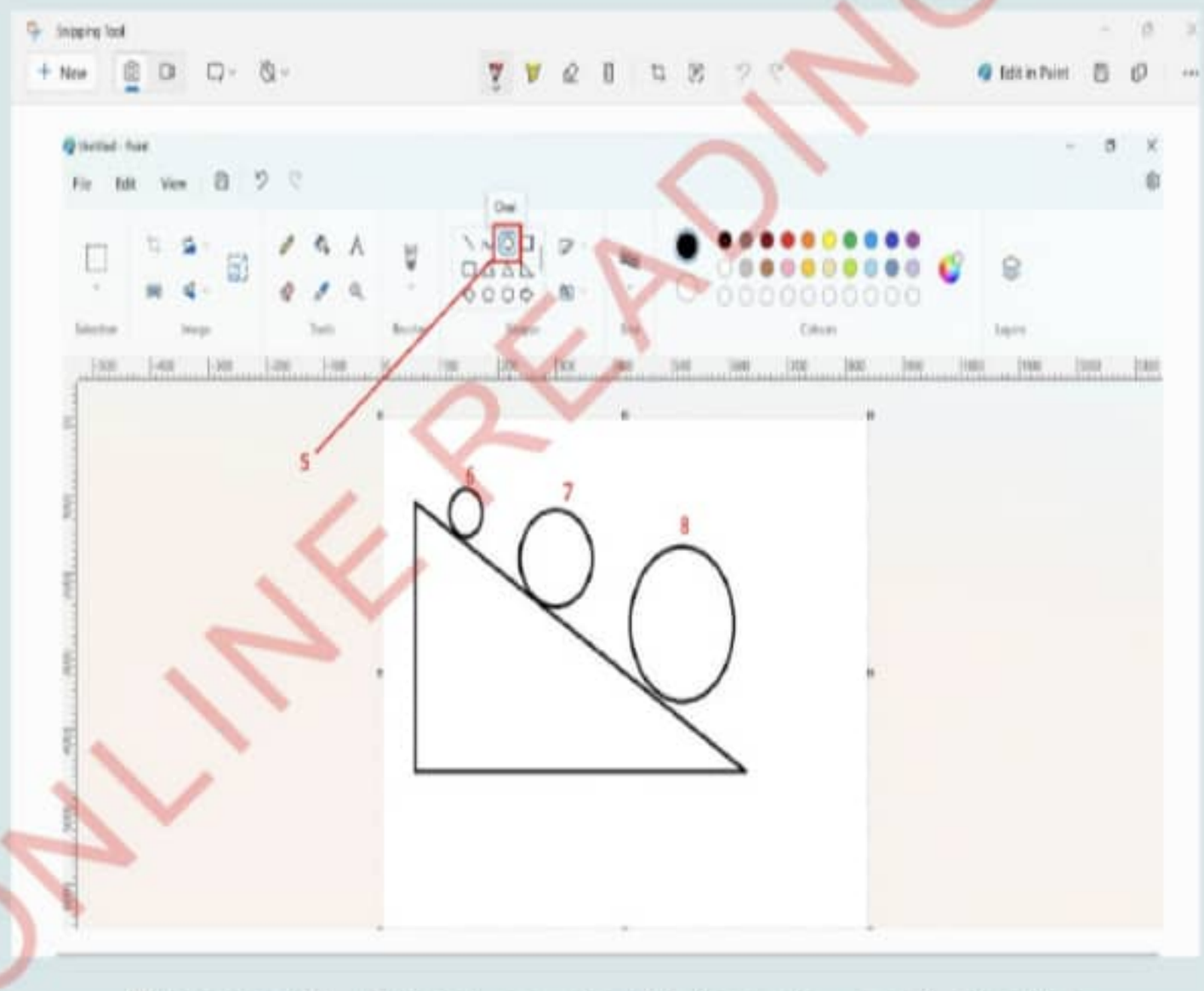


Figure 8: *Right-angled triangle and circles*

Saving your work

To save your work using *Paint program*, follow these steps:

1. Click on *'File'* as shown in Figure 9(a).

2. Click *'Save as'*.

3. Select *'PNG'* picture. You can also choose *'JPEG'* picture or *'BMP'* picture as seen in Figure 9(a).

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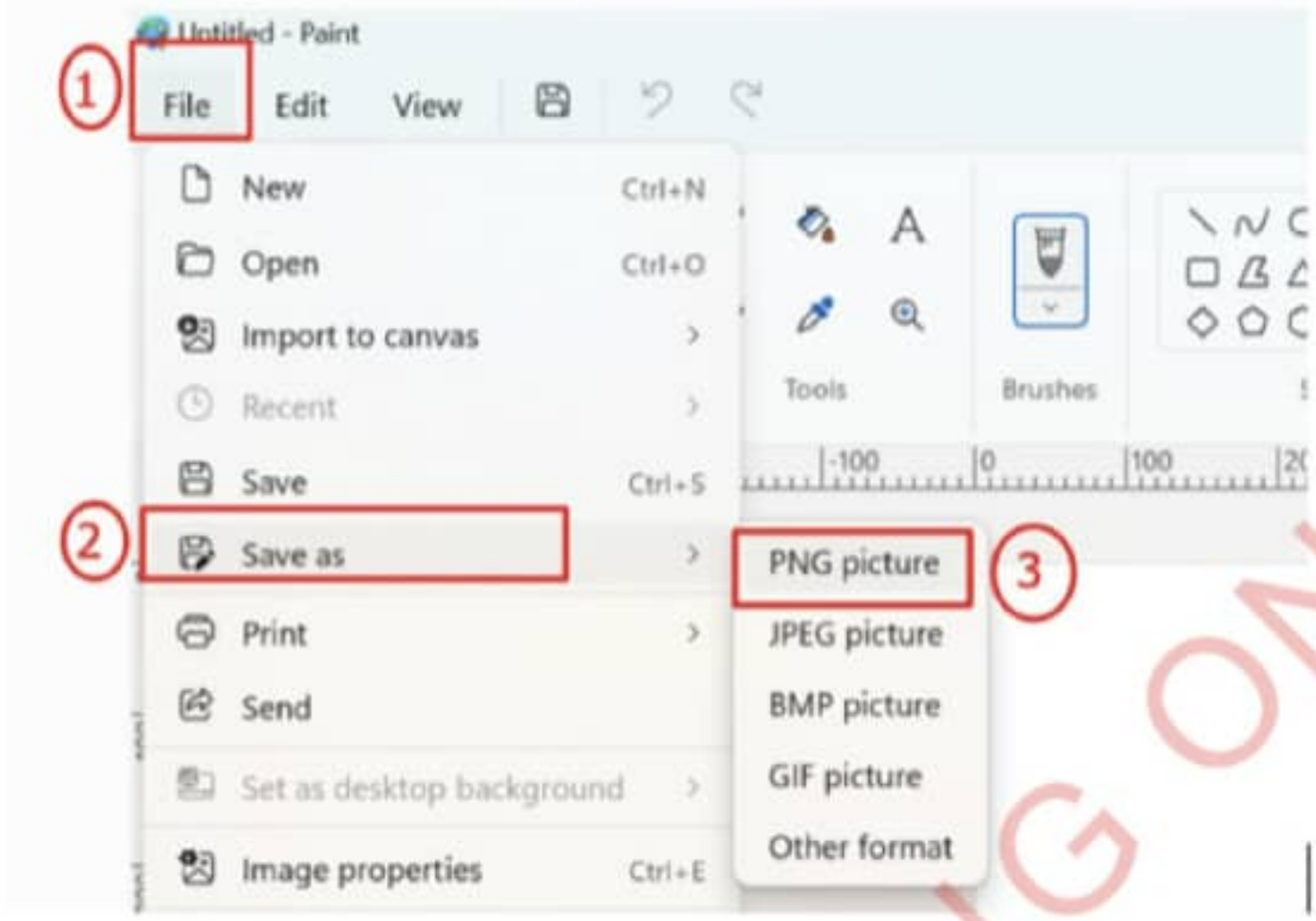


Figure 9(a): *Preliminary steps to save your work*

1. Click on 'Documents' as shown in Figure 9(b).

2. Enter a name for your file. For example, type "Shape 1" as shown in Figure 9(b) .

3. Click 'Save'.

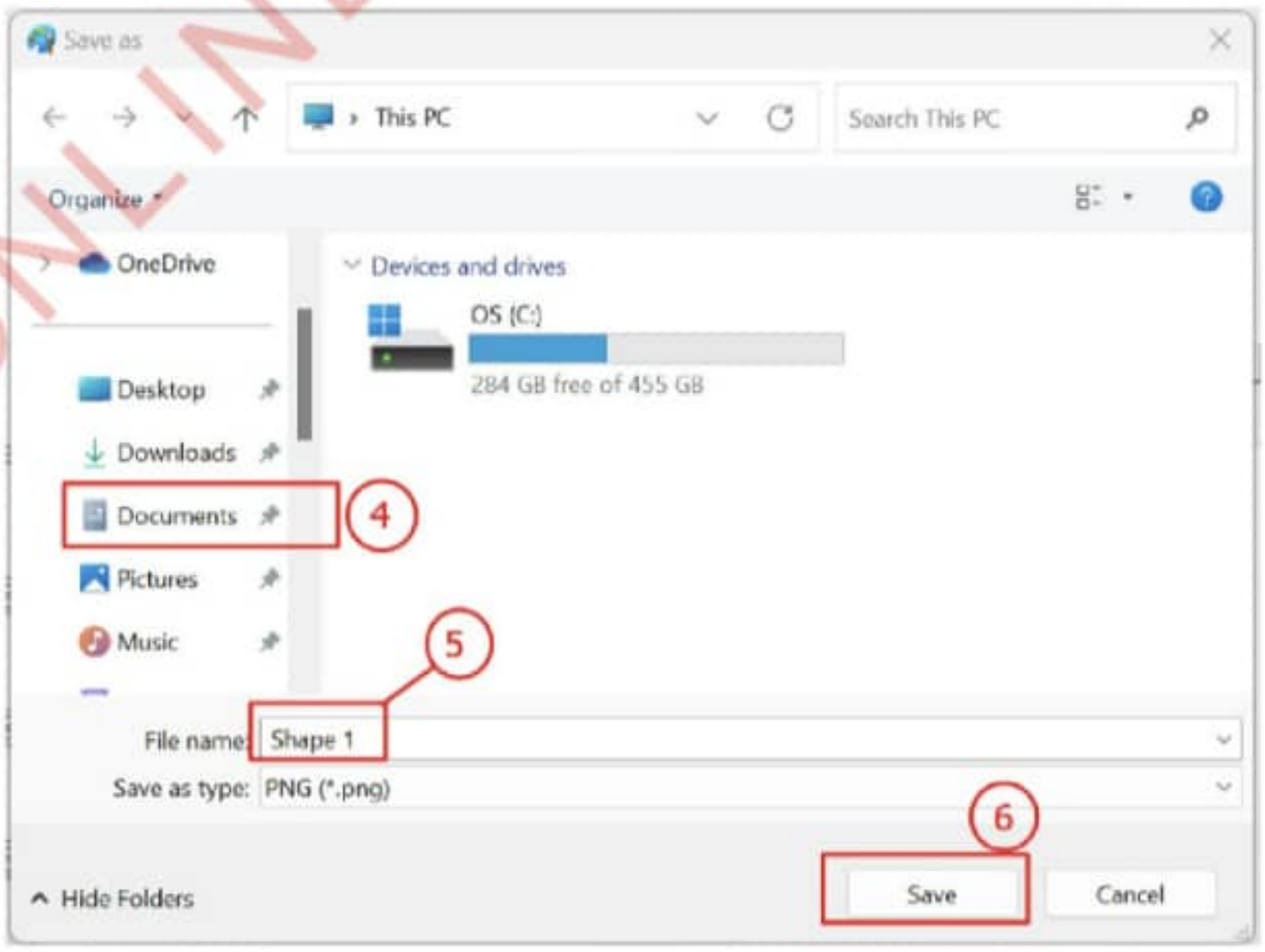


Figure 9(b): *Final steps to save your work*

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After drawing the shapes and saving your work, the next step is to colour shapes.

Colouring shapes

To colour the shapes, use the following steps.

Steps

1. Click on the colour '**Red**' as shown in Figure 10.
2. Click on '**Fill**'.

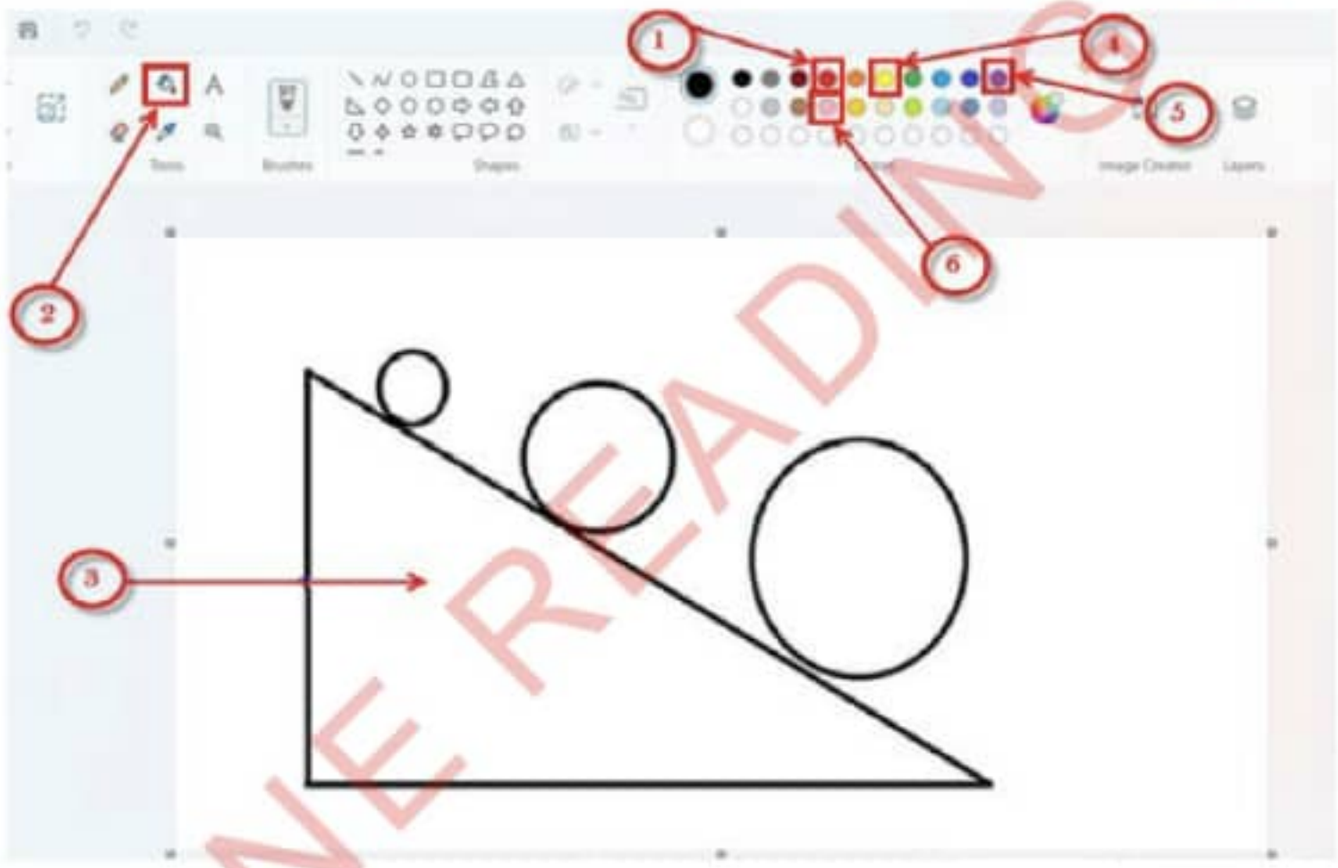


Figure 10: **Steps in colouring a shape**

3. Click inside the right-triangle and apply the '**Red**' colour. The triangle will be in red as shown in Figure 11.

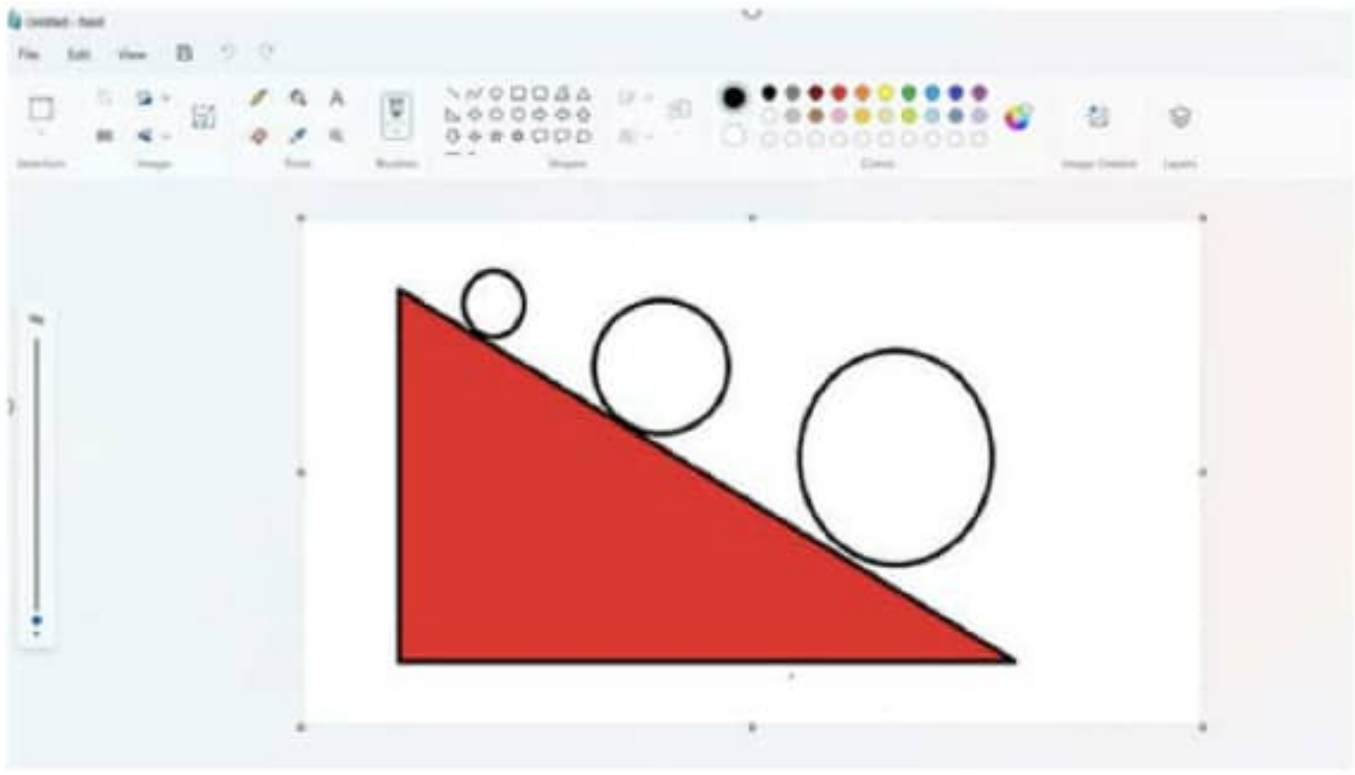


Figure 11: **A Red coloured right-angled triangle**

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4. Click on the colour *'Yellow'*, then click inside the smallest circle.

5. Click on the colour *'Purple'*, then click inside the second circle.

6. Click on the colour *'Rose'*, then click inside the third circle.

The shapes will look similar to the one shown in Figure 12.

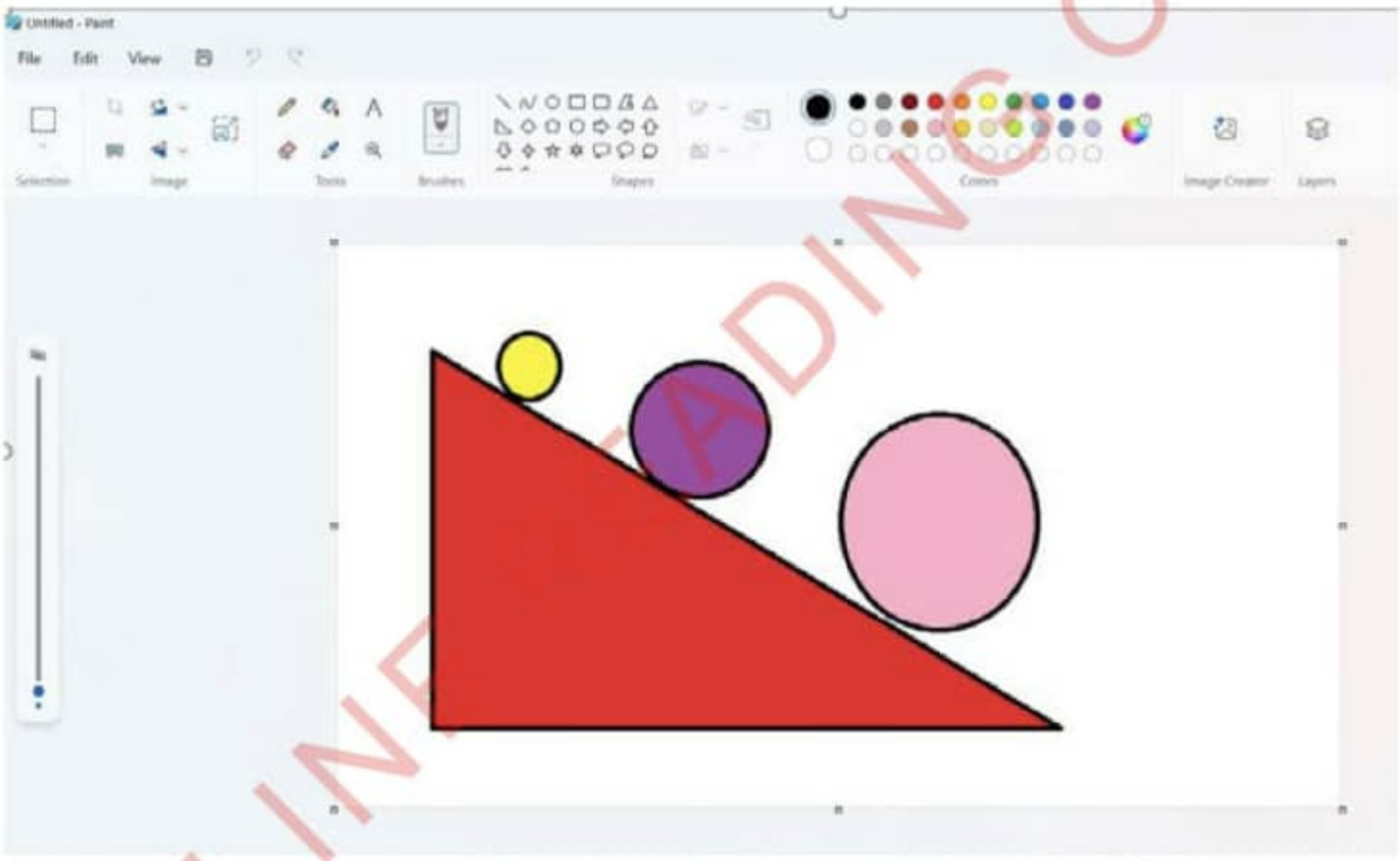


Figure 12: *The coloured right-angled triangle and circles*

Drawing and combining shapes

You can draw shapes and combine them together to form another shape. For example, draw many rectangles and combine them to form a brick wall. Instead of drawing each of the rectangle using the rectangle tool, you can draw one then copy it and paste many times to create the other rectangles. You can use colours to create patterns on the wall.

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Activity 4

Drawing a brick wall with different colour patterns

Steps

1. Select the rectangle tool by clicking on *'Rectangle'* as shown in Figure 13.



Figure 13: *Selection of the rectangle tool*

2. Draw six bricks and arrange them in a row as shown in Figure 14.

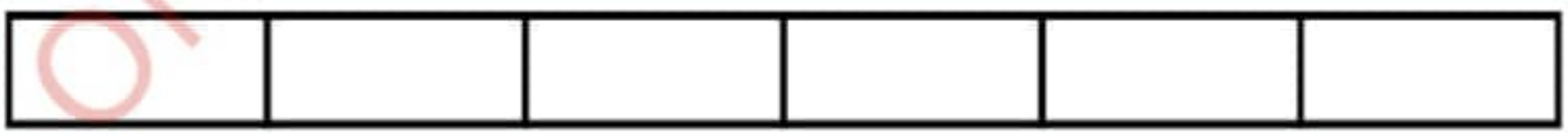


Figure 14: *The first layer of bricks in the wall*

3. Colour the layer with *gray* and yellow as shown in Figure 15.



Figure 15: *After colouring the first layer of the wall*

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4. Draw the second layer on top of the first layer and colour it **orange and blue**. Figure 16 shows the second layer of the wall placed on top of the first layer.

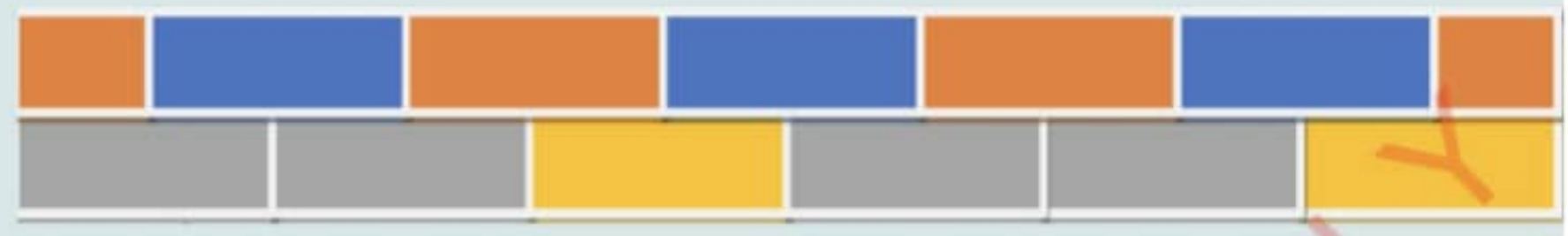


Figure 16: The first and second layers of the wall

5. Continue adding layers until you reach the sixth layer. Ensure the colour patterns in the layers look like the ones shown in Figure 17.

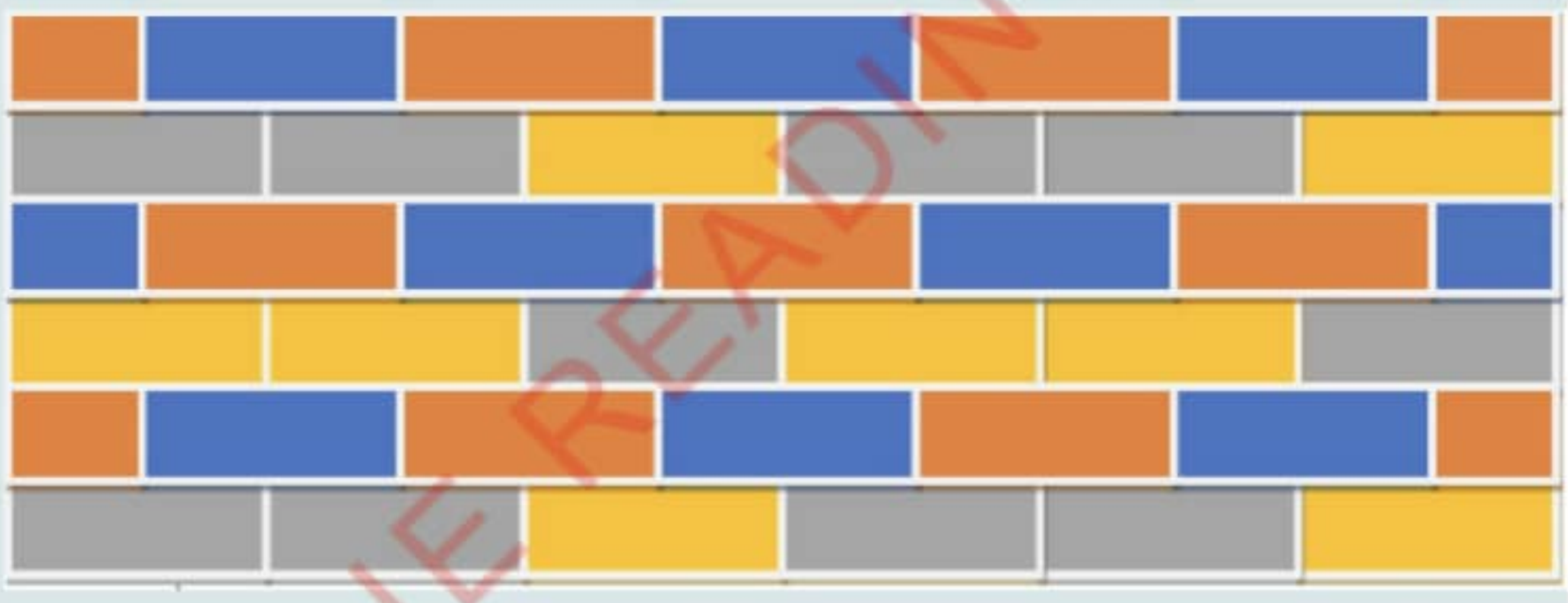


Figure 17: **The completed brick wall showing colour patterns**

Exercise

1. Mention shapes you can draw using the Paint program.
2. Write the steps of colouring a shape using Paint program.
3. The Paint program does not have a square shape. What will you do to ensure you draw a square using the rectangle shape in the Paint program?
4. Identify the colour patterns in the brick wall in Figure 17.
5. Draw a simple house using shapes and lines in the Paint program.

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Creating simple computer games

Creating simple computer games involves writing instructions using a programming language. Some important things to consider when creating a simple computer game include game rules, playtime, creativity and the tools for creating games. Use the following steps to create simple computer games:

1. Choose the type of game: Decide on the type of game you want to create, such as a quiz game, obstacle avoidance, or memory game.

2. Choose game creation tools: Use the Scratch tool to create your game. This tool has a user-friendly interface and provides step-by-step instructions.

3. Create characters and environment: Design the characters for your game and the environment they will play in. This can include drawing pictures or using existing images like 'Sprites' in the Scratch program.

4. Write game rules: Write the rules of the game and how the players should play.

5. Coding: Write instructions using block programming to create the game you want.

6. Test and improve: Play your game to check for errors or areas that need improvement. Fix any mistakes and enhance your game.

Coding

Coding is writing instructions on a computer to perform a specific task. In this section, you will code using the Scratch program. Scratch uses block programming to make it easier to create simple games. This program can be downloaded from https://scratch.mit.edu/download.

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Opening the Scratch program

Follow the steps provided in activity 5 to open the Scratch program.

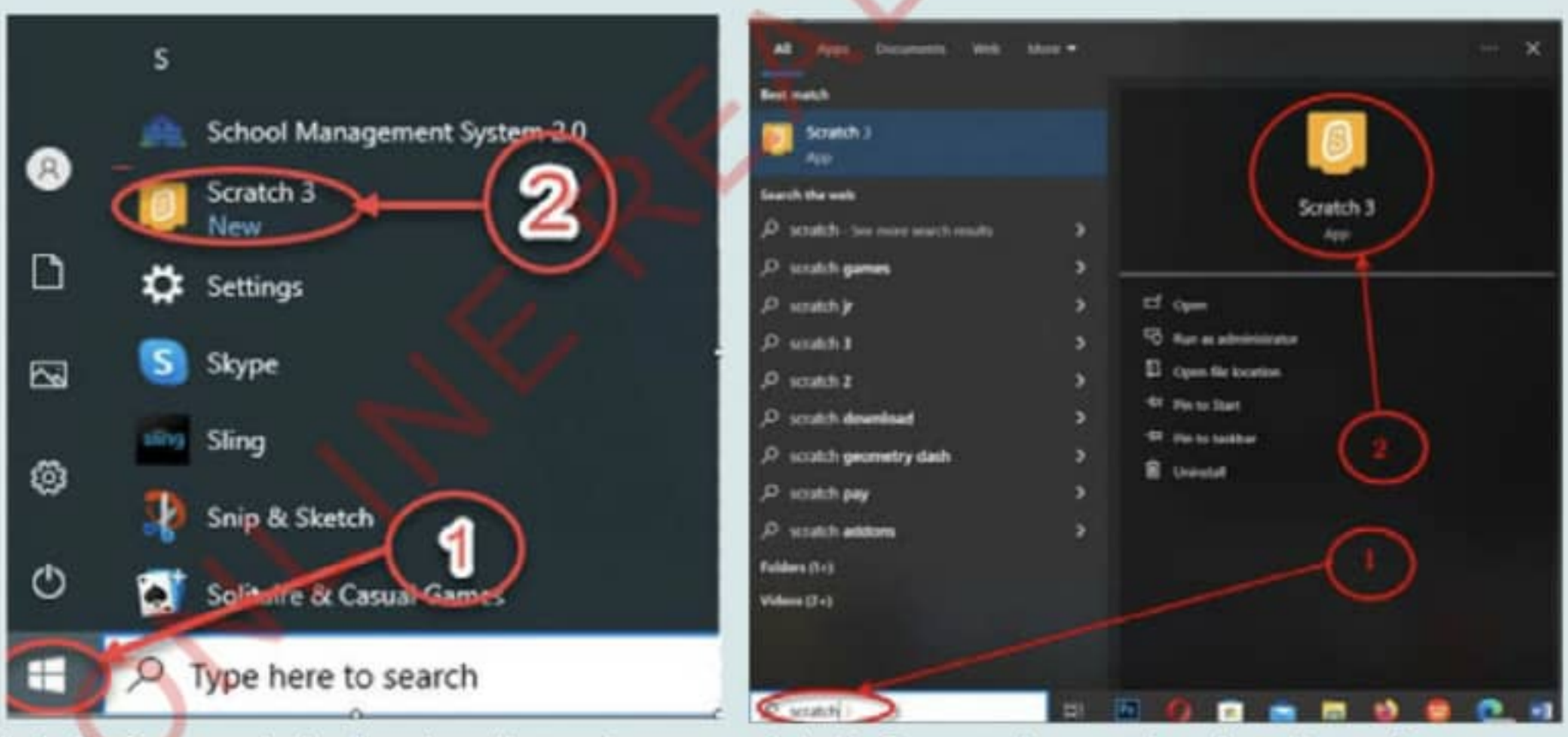
Activity 5

Opening the Scratch program

Steps

To open Scratch you can use one of the options in Figure 18.

1. Click on the ***'Windows button'*** as shown in Figure 18(a).



(a) If Scratch is in the list of programs

(b) If Scratch not in the list of programs

Figure 18: ***Opening a Scratch Program***

2. Select Scratch program click to open the program as in Figure 18(a).

   If Scratch is not in the list of programs, type the word 'Scratch' in the search as shown in Figure 18(b).

3. Click 'Scratch' to open as shown in Figure 19.

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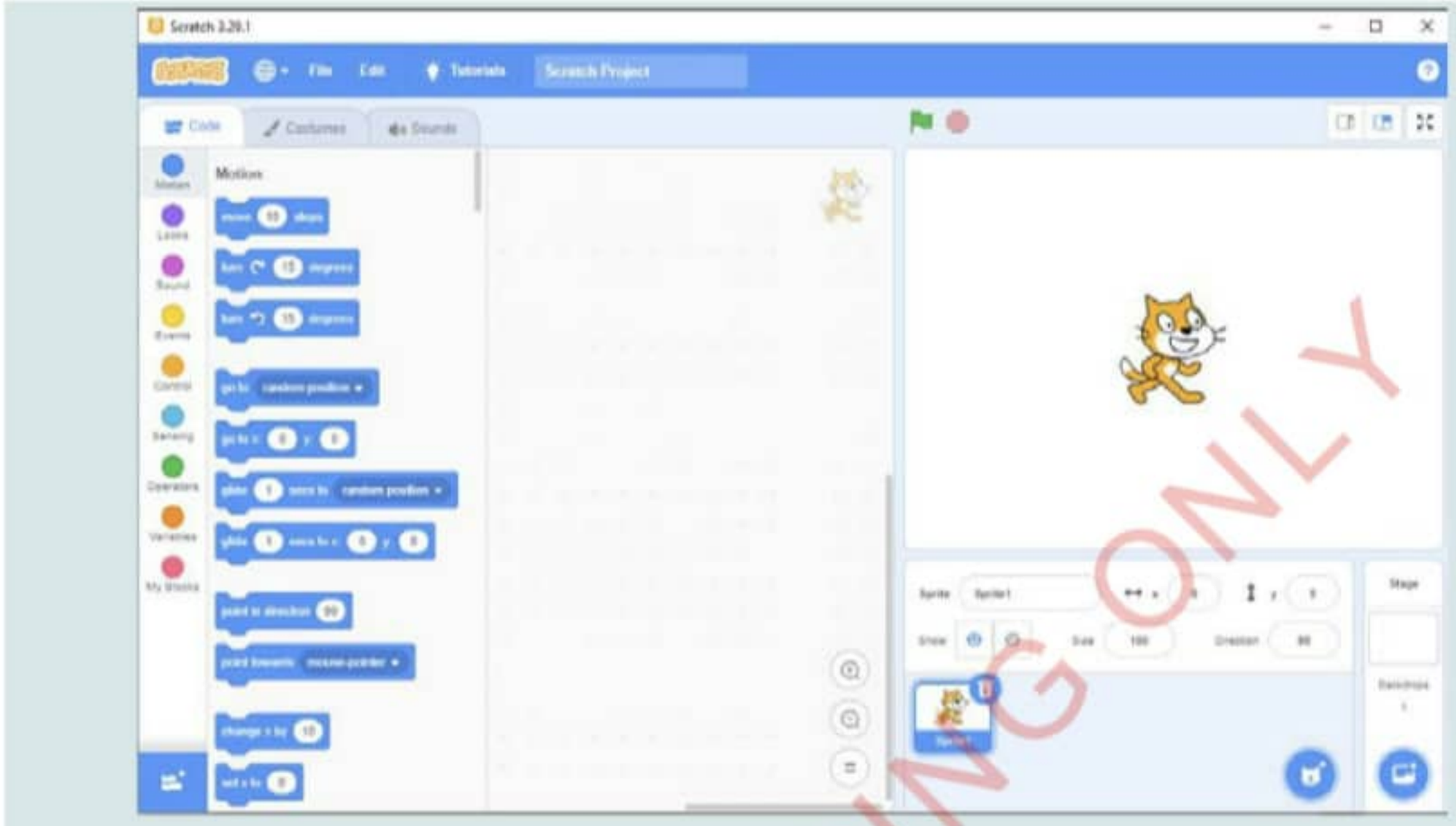


Figure 19: *The Scratch program*

Parts of the Scratch window

The Scratch window has four main parts. These are the script area, block menu, stage and Sprite list area as shown in Figure 20.

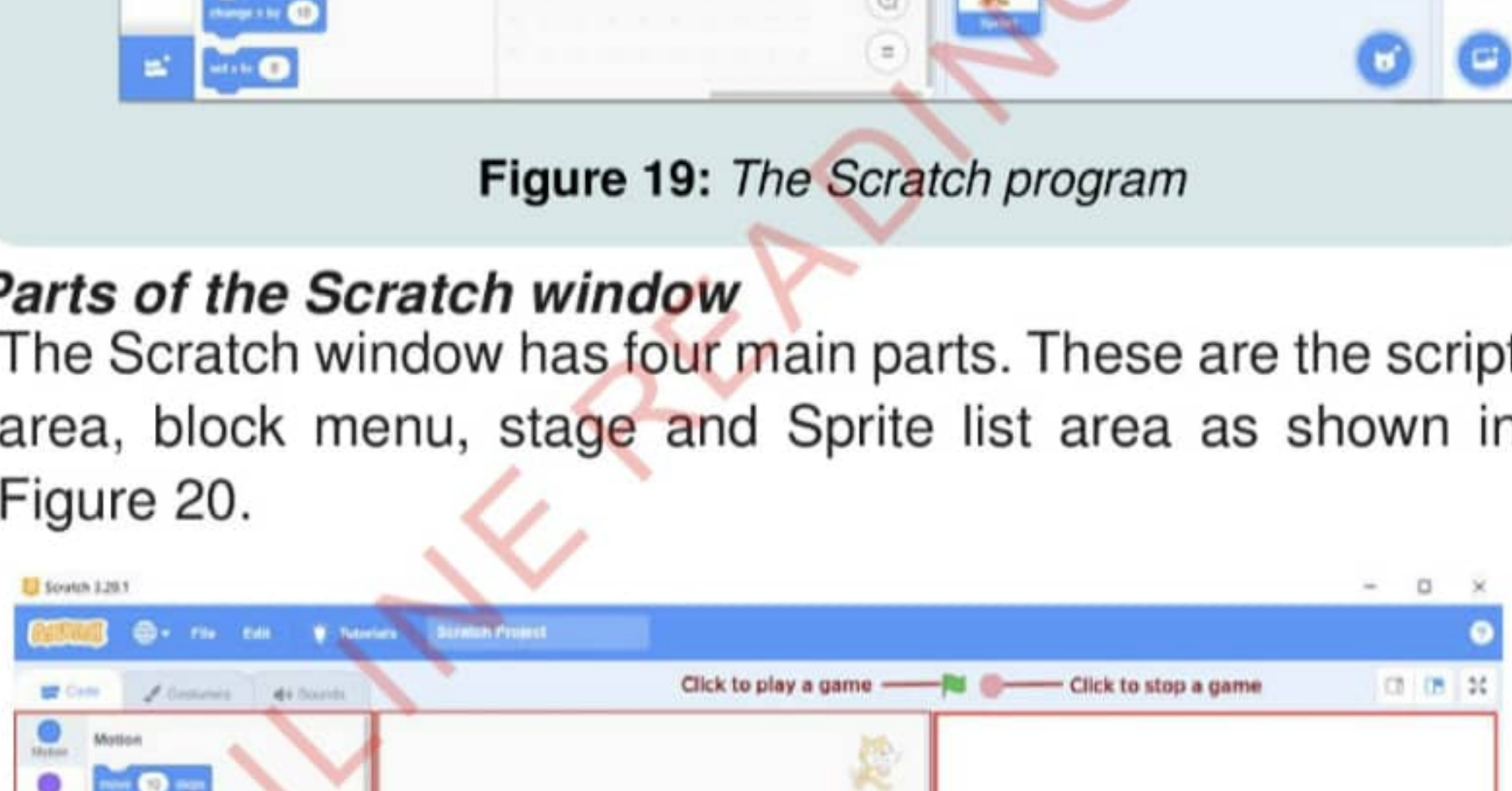


Figure 20: *Parts of the Scratch window*

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(a) *Script area:* The place where you create your programs in Scratch, by putting together different blocks.

(b) *Blocks:* Allows you to access and select blocks to create your program.

(c) *Stage:* The area where a games is displayed.

(d) *Sprite list area:* Represents the objects or characters used to show how the game operates.

(e) *Sprite icon:* A place to choose a 'Sprite'.

Creating simple game

Scratch program is used to create a game in which a sprite can move around on the stage. Follow the steps provided in activity 6.

Activity 6

Creating a game which make a 'Sprite' (Cat) to move forward

Steps

1. Click on the '*Events*' block menu as shown in Figure 21.
2. Drag and drop the '*when clicked*' block on the script area as shown in Figure 21.

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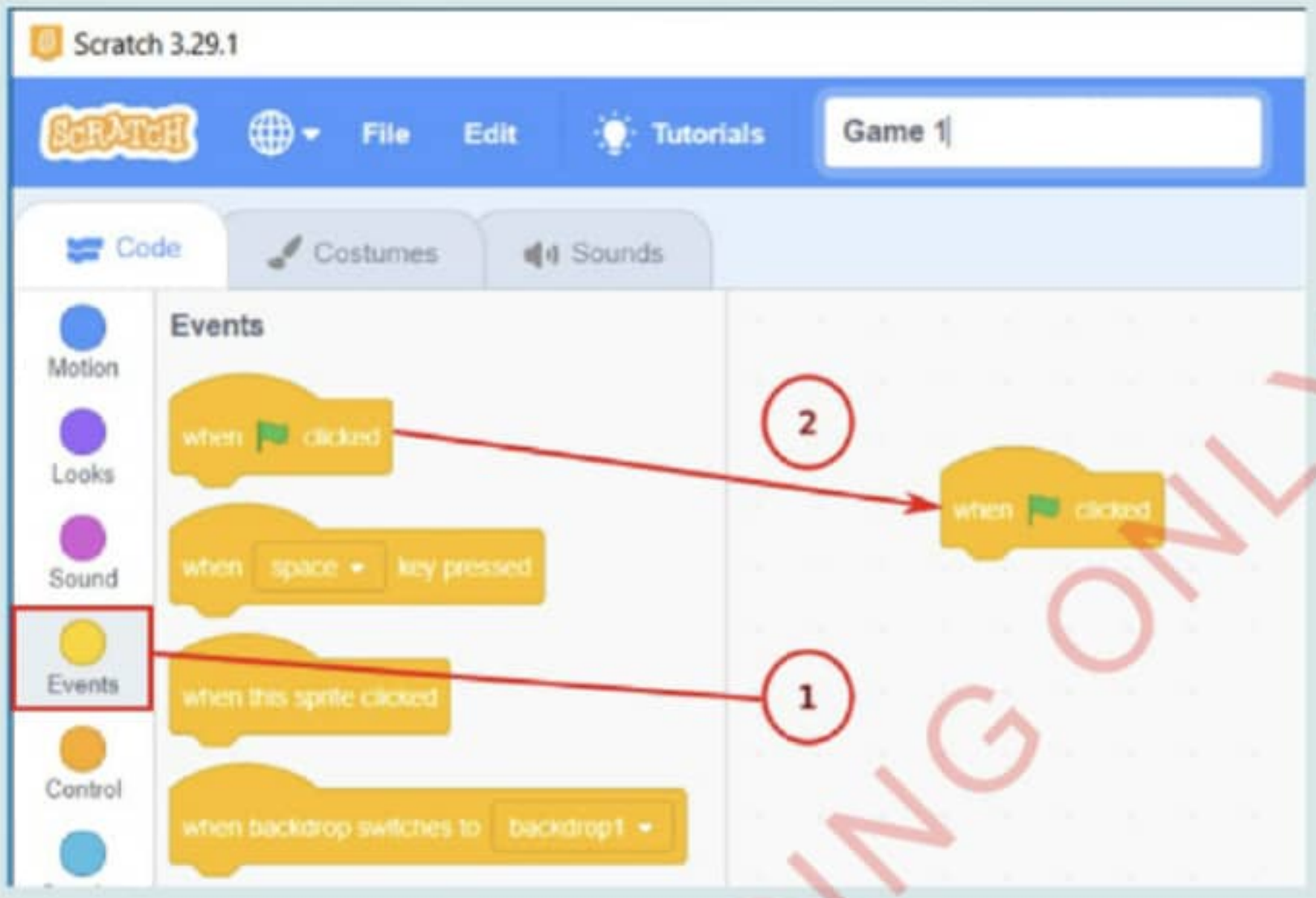


Figure 21: *Dragging the 'when clicked' block to the script area*

3. Click on the '*Motion*' block menu as shown in Figure 22.

4. Drag and drop the '*move steps* ()' block to the script area.

5. Attach the '*move steps* ()' block to the bottom of the '*when clicked*' block as shown in the Figure 22.

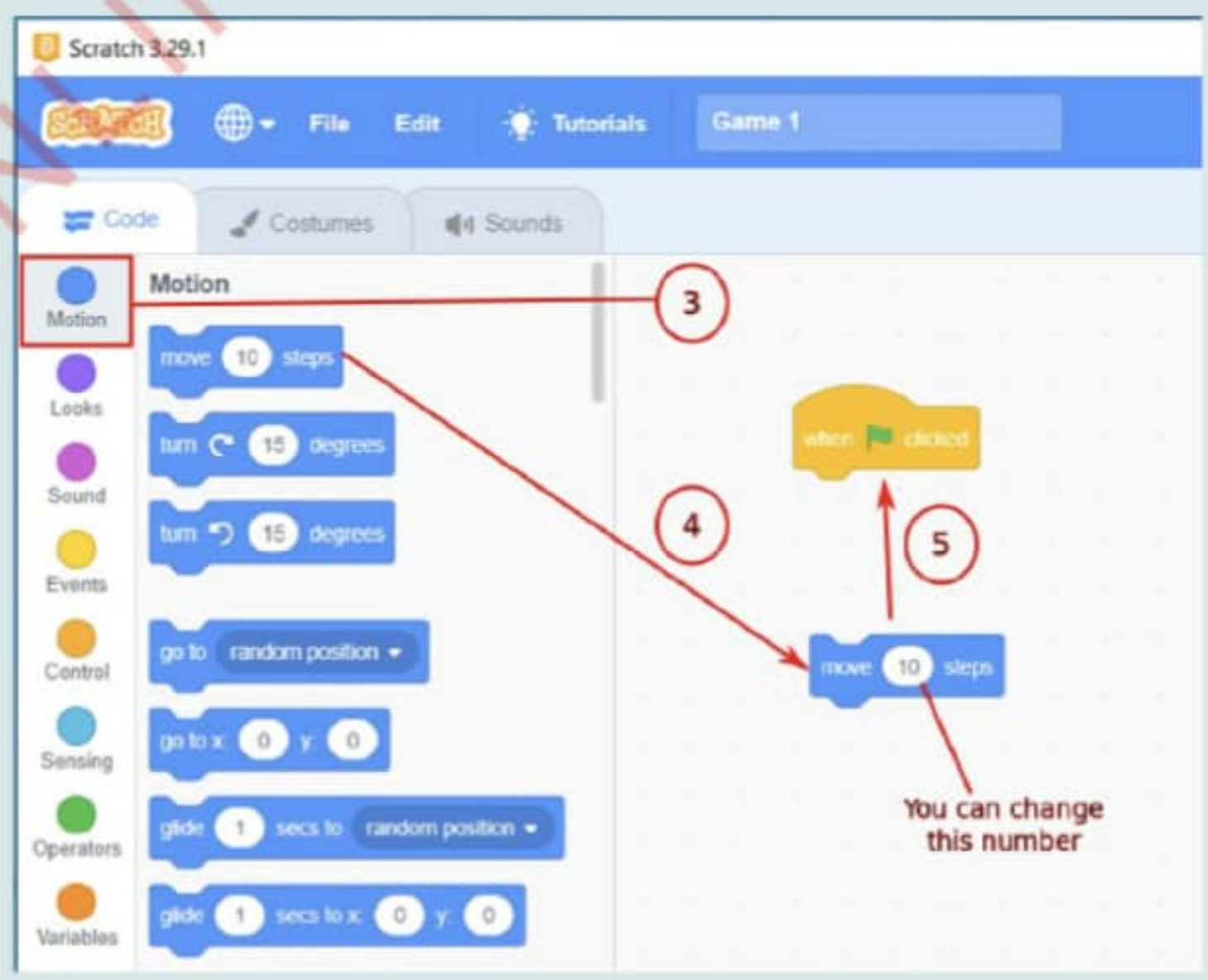


Figure 22: *Dragging and a ttaching blocks*

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6. Click on the *'Green flag'* as shown in Figure 23 to start a game.

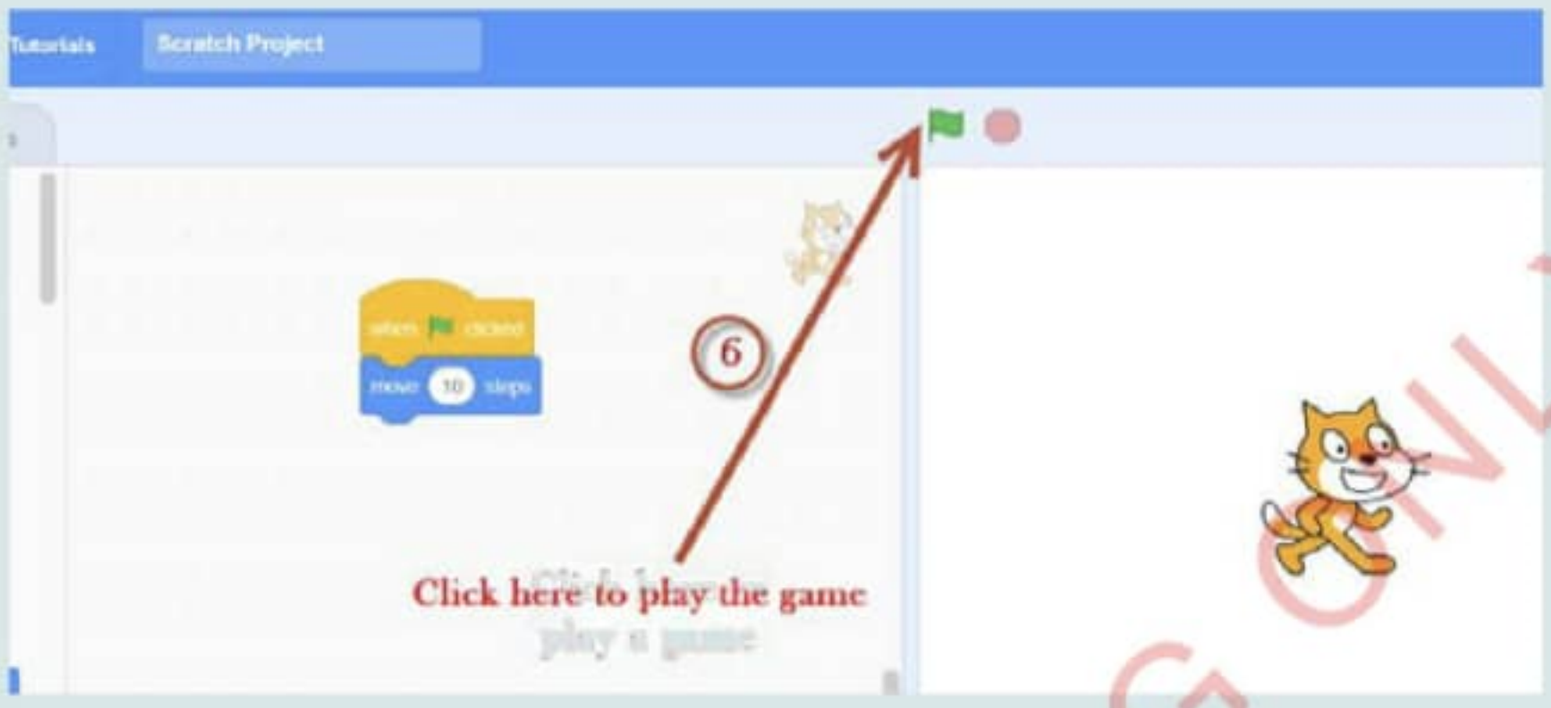


Figure 23: *Playing the game*

7. Change the number of steps from 10 to 70, then play the game. Change again from 70 to 130 steps, then play the game. What did you notice after increasing the number of steps?

Saving your work

To save your work using Scratch program, follow the steps below.

1. Click on the *'File'* menu.
2. Select *'Save to your computer'* as shown in Figure 24.

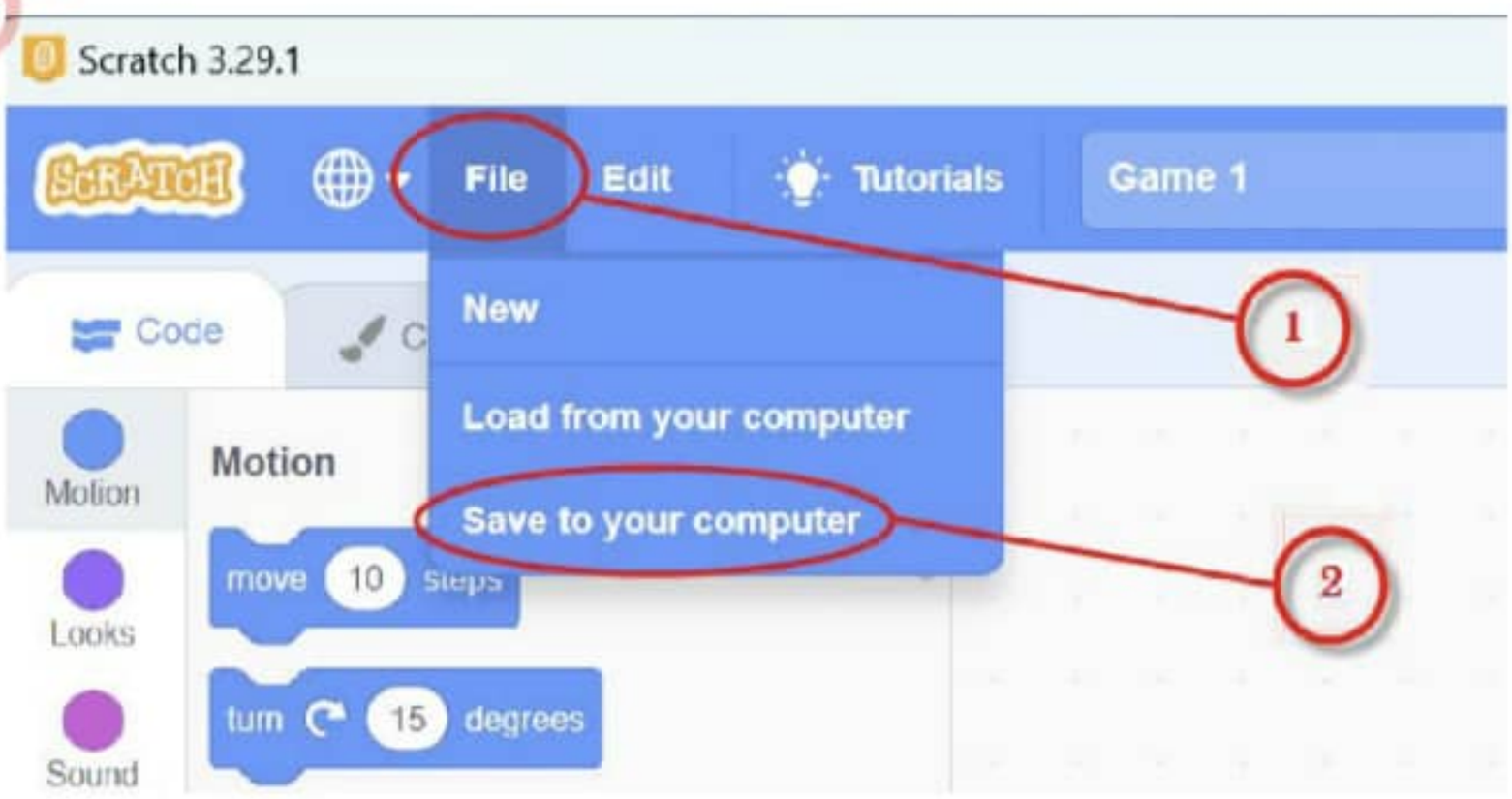


Figure 24: *Initial steps in saving your work on a computer*

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3. The '*Save as*' dialog box will appear as shown in Figure 25.

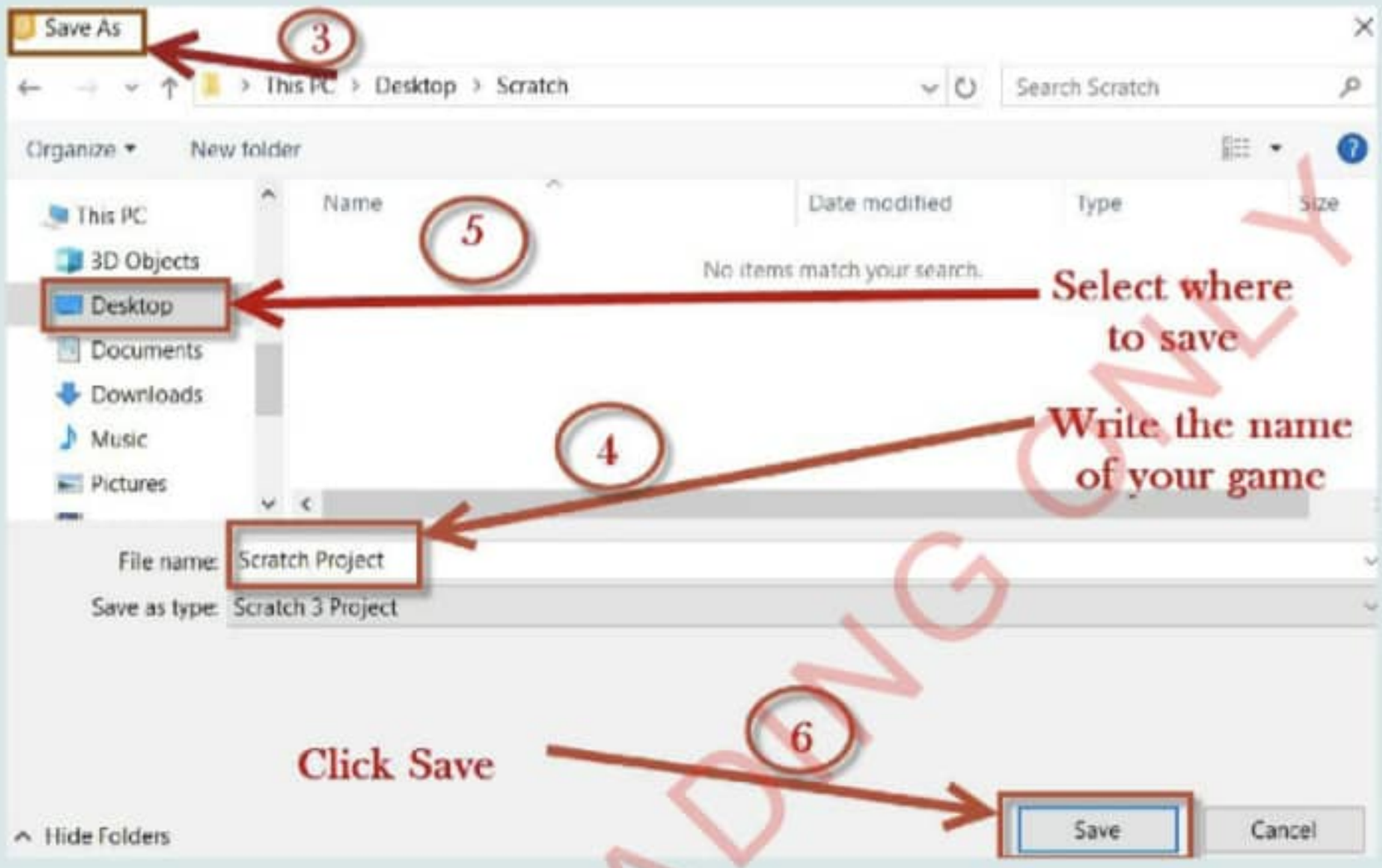


Figure 25: *Final steps to save your work in the computer*

4. For file name, delete Scratch Project and write the name of your Project.
5. Select where to save your Project for example, Desktop.
6. Click 'Save' as shown on Figure 25.

Activity 7

Creating a game, which makes a Sprite rotate

To create a program that rotates the 'Sprite'(Cat), follow these steps:

Steps

1. Click on the '*Events'* block menu.
2. Drag and drop the '*when clicked'* block on the script area.

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3. Click on the *'Motion'* block menu.

4. Drag and drop the '*turn (↻) degrees*' block on the script area.

5. Attach the '*turn (↻) degrees*' block to the '*when clicked*' block as shown in Figure 26.

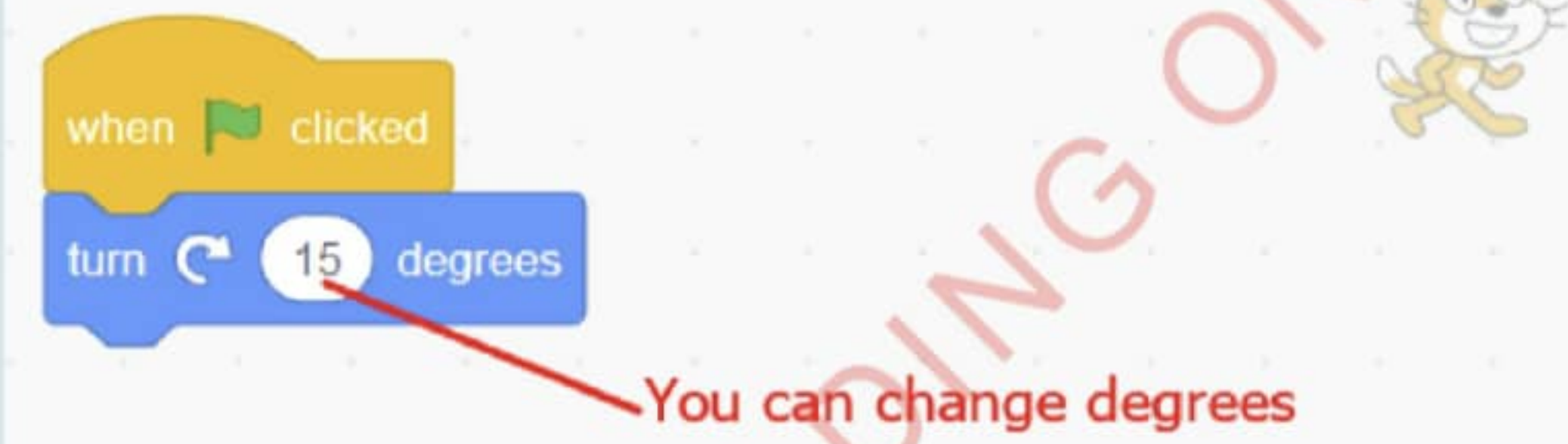


Figure 26: *After attaching the 'turn degrees()' block to the 'when clicked' block*

6. Click on the *'Green flag'* to play your program.

7. Continue playing to see how the *Sprite* rotates. You can change the degrees from 15 to 45 to see more about how the Sprite rotates. What did you notice after changing the degrees?

Activity 8

Creating a 'go to anywhere' game

To create a *'go to anywhere'* game, follow these steps:

Steps

1. Click on the *'Events'* block menu.

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2. Drag and drop the *'when clicked'* block on the script area.

3. Click on the *'Motion'* block menu.

4. Drag and drop the '*go to* (*random position*)' block on the script area.

5. Attach the '*go to* (*random position*)' block to the 'when clicked' block as shown in Figure 27.



Figure 27: *After attaching the 'go to' block to 'when clicked' block*

6. Click on the *'Green flag'* to play the game.

7. Repeat playing the game to see how the Sprite goes to a *'random position'* each time. What did you notice in this game?

Combining multiple movement

Activity 9

Use Scratch program to combine multiple movements

In this activity, you will combine three Sprite movements. First,

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the Sprite will '*go to a random position*'. Second, the Sprite will '*rotate 30 degrees(anticlockwise)*'. Third, if the Sprite is at '*the edge*', it will be '*returned*'. The following steps will guide you in creating this game. Refer to Figure 28.

Steps

1. Click on the '*Events*' block menu, drag and drop the '*when clicked*' block on the script area.

2. Click on the '*Motion*' block menu, drag and drop the '*go to (random position)*' block on the script area.

3. Attach the '*go to (random position)*' block to the 'when clicked' block.

4. Drag and drop the '*turn (↻) 15 degrees*' block on the script area.

5. Attach the *'turn (↻) 15 degrees*' block to the '*go to (random position)*' block.

6. Change the degrees from 15 to 30 degrees.

7. Drag and drop the '*if on edge, bounce'* block on the script area.

8. Attach the '*if on edge, bounce'* block to the "*turn () degrees'* block."

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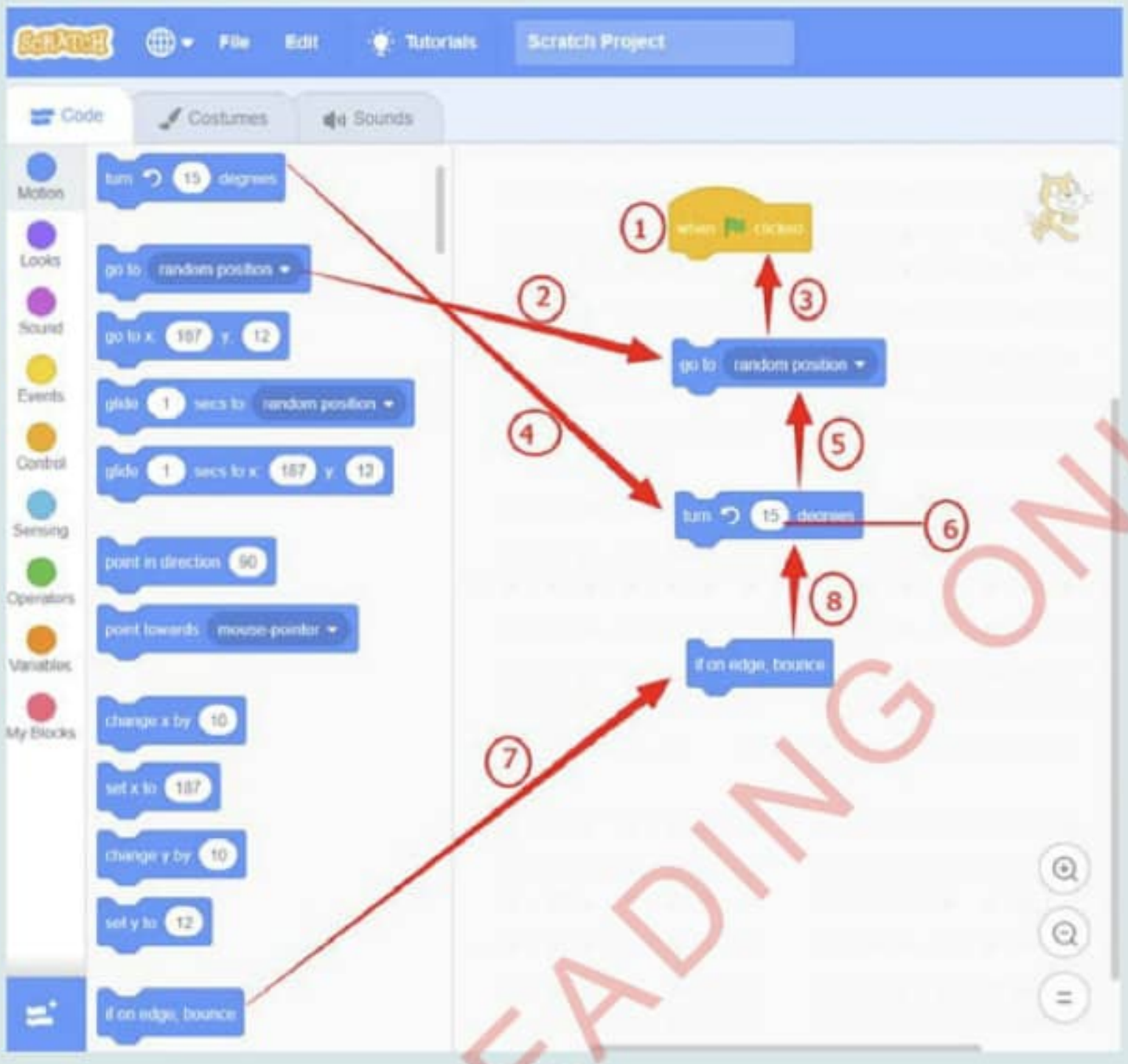


Figure 28: *Combining multiple movements*

9. The complete program will look as shown in Figure 29.

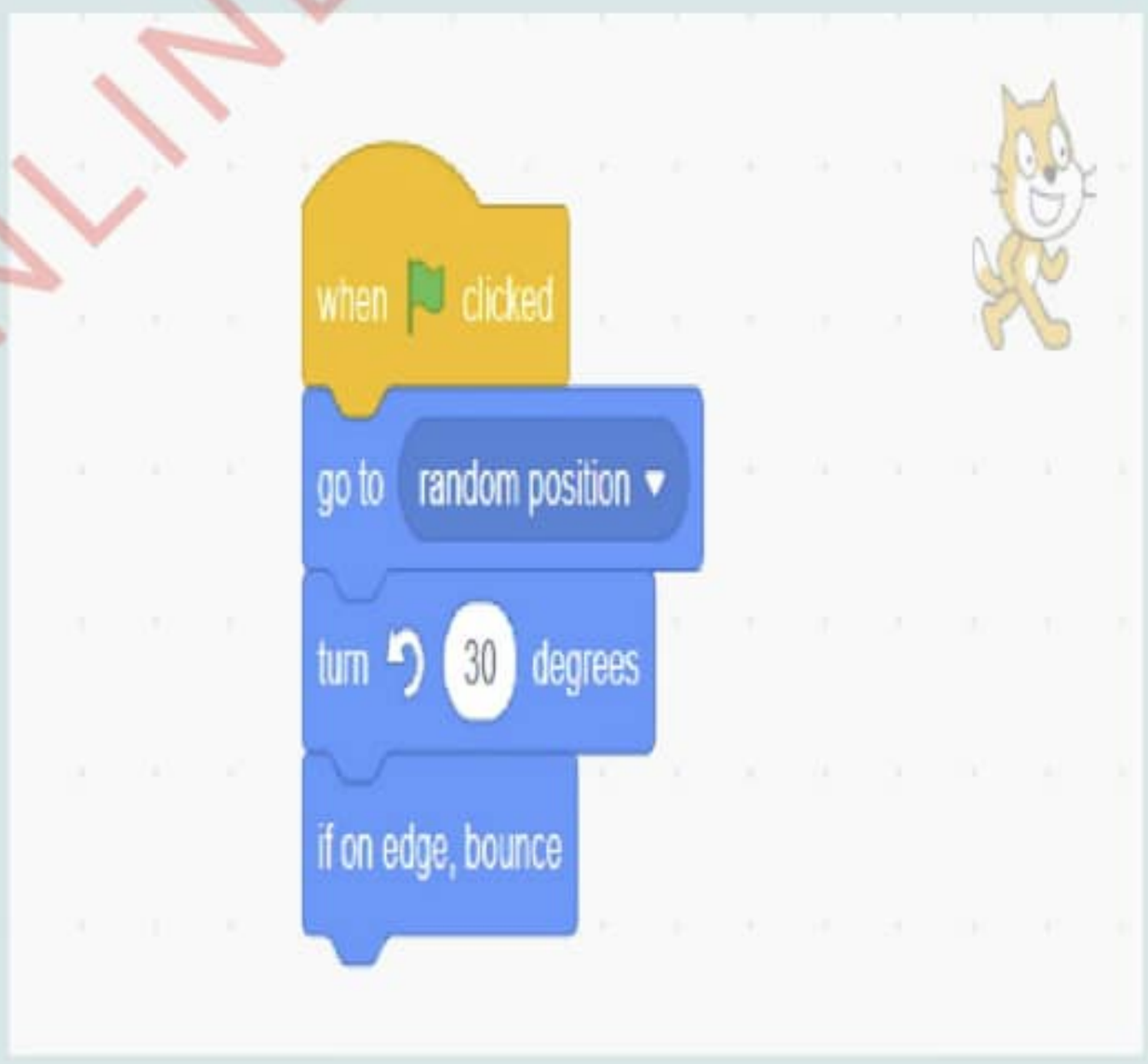


Figure 29: *A completed program*

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10. Play the game. Repeat playing the game as many times as you like. What did you notice when you played this game?

Game with sound

You can create a program that makes a 'Sprite'(Cat) produce sound or add sound effects into the game. Sound blocks are used to play sounds.

Activity 10

Game with sound

In this activity, you will prepare a program that makes a 'Sprite'(Cat) produce sound on the script area.

Steps

1. Click on the '*Events*' block menu. Drag and drop the '*when clicked*' block.

2. Click on the '*Sound'* block menu. Drag and drop the '*play sound until done*' block on the script area.

3. Attach the '*play sound until done*' block to the '*when clicked*' block, as shown in Figure 30.

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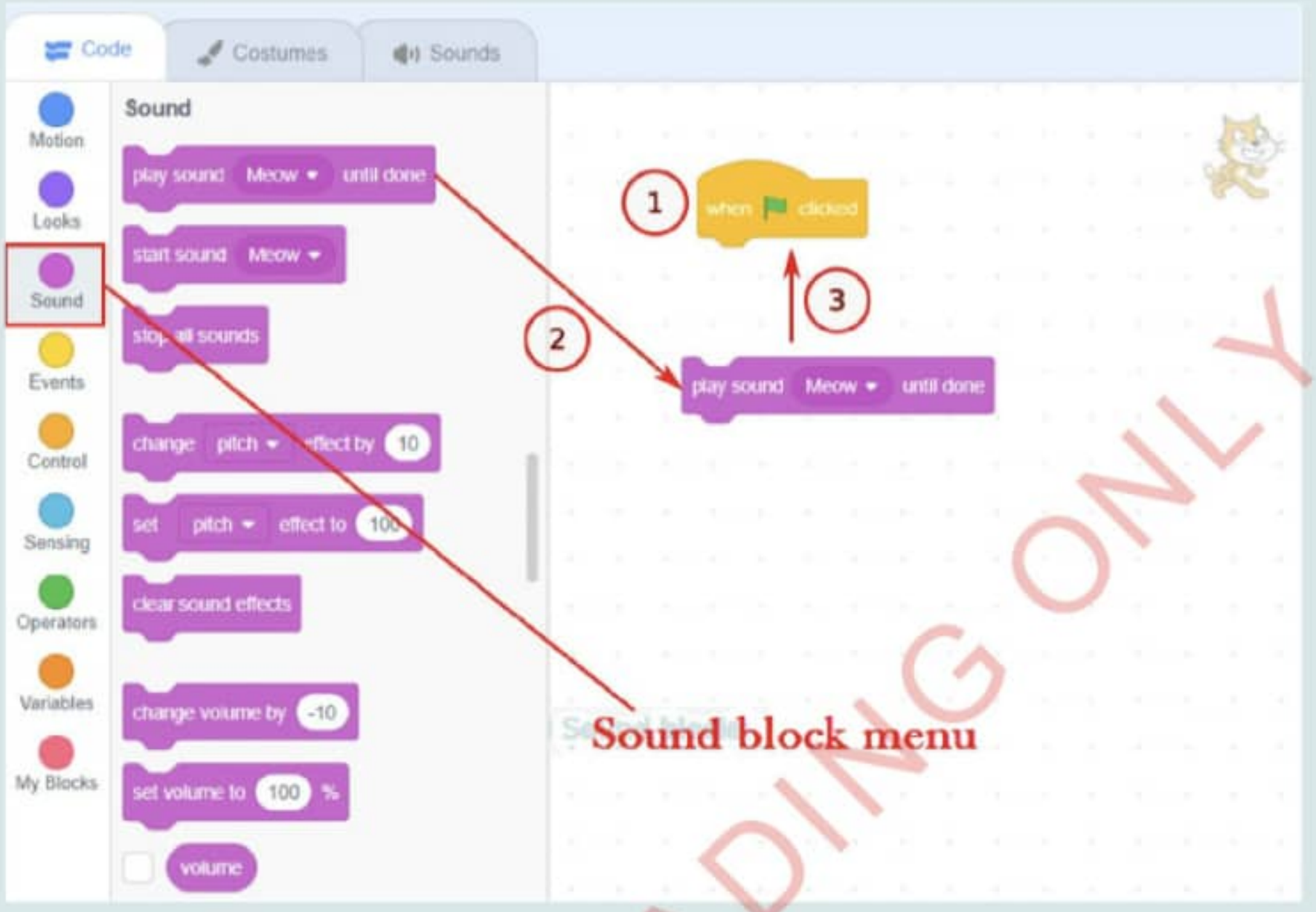


Figure 30. *Attaching the 'play sound' block to the 'when' clicked block*

The program will look as shown in Figure 31.

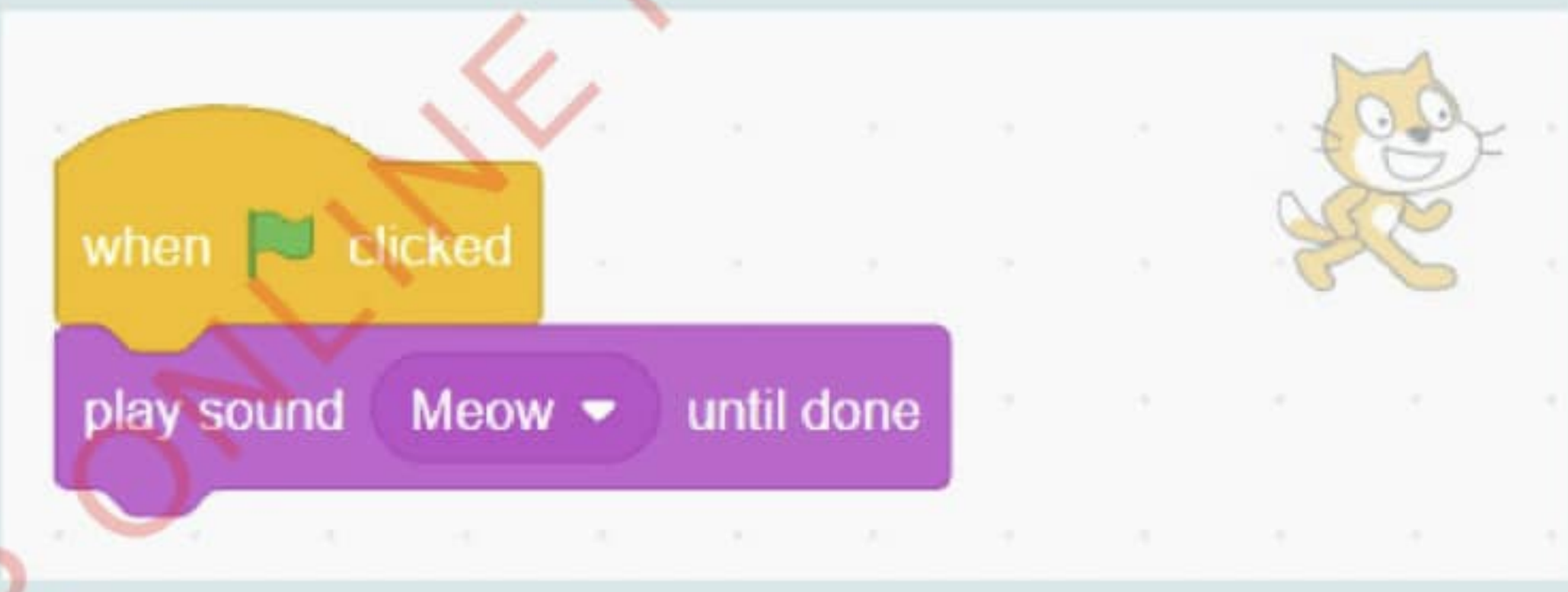


Figure 31: *Attached 'play sound' block to the 'when clicked' block*

4. Click on the '*when clicked*' block to start playing the game and listen to the sound. What did you hear when you played it?"

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Program control structures

Program control structures refer to how the flow of computer instructions is managed. There are three types of program control structures namely sequence, repetition and selection.

Sequence structure in programming

In programming, a sequence structure follows the order of flow of instructions or blocks from top to bottom. The first block is executed before the next block. For example, in Figure 32, blocks will be executed as follows.

1. The '*when clicked*' block is executed first.
2. The '*go to (random position)*' block is executed next.
3. The '*turn (↻) 15 degrees*' block is then executed
4. Lastly, the '*if on edge, bounce*' block is executed.



Figure 32: *Sequence structure in programming*

Thus, the flow becomes as follows: '*When clicked → go to* (*random position*) → *turn* (15) *degrees → if on edge, bounce*'.

Repetition structure in programming

In computer programming, you might need some action to repeat several times. For example, you might need movement

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or sound to repeat without clicking the *Start* button. This ability is known as repetition. When using repetition, you instruct the computer to repeat some action as many times as you want.

Activity 11

Playing sound repeatedly

You can play sound by making it repetitive according to the number of times you want. The following steps will guide you in a program that play sound repeatedly.

Steps

1. Click on the '*Events'* block menu.
2. Drag and drop the '*when clicked*' block on the script area.
3. Click on the *'Control'* block menu.
4. Drag and drop the '*repeat (10)*' block on the script area. Next, attach it to the '*when clicked'* block as shown in Figure 33.

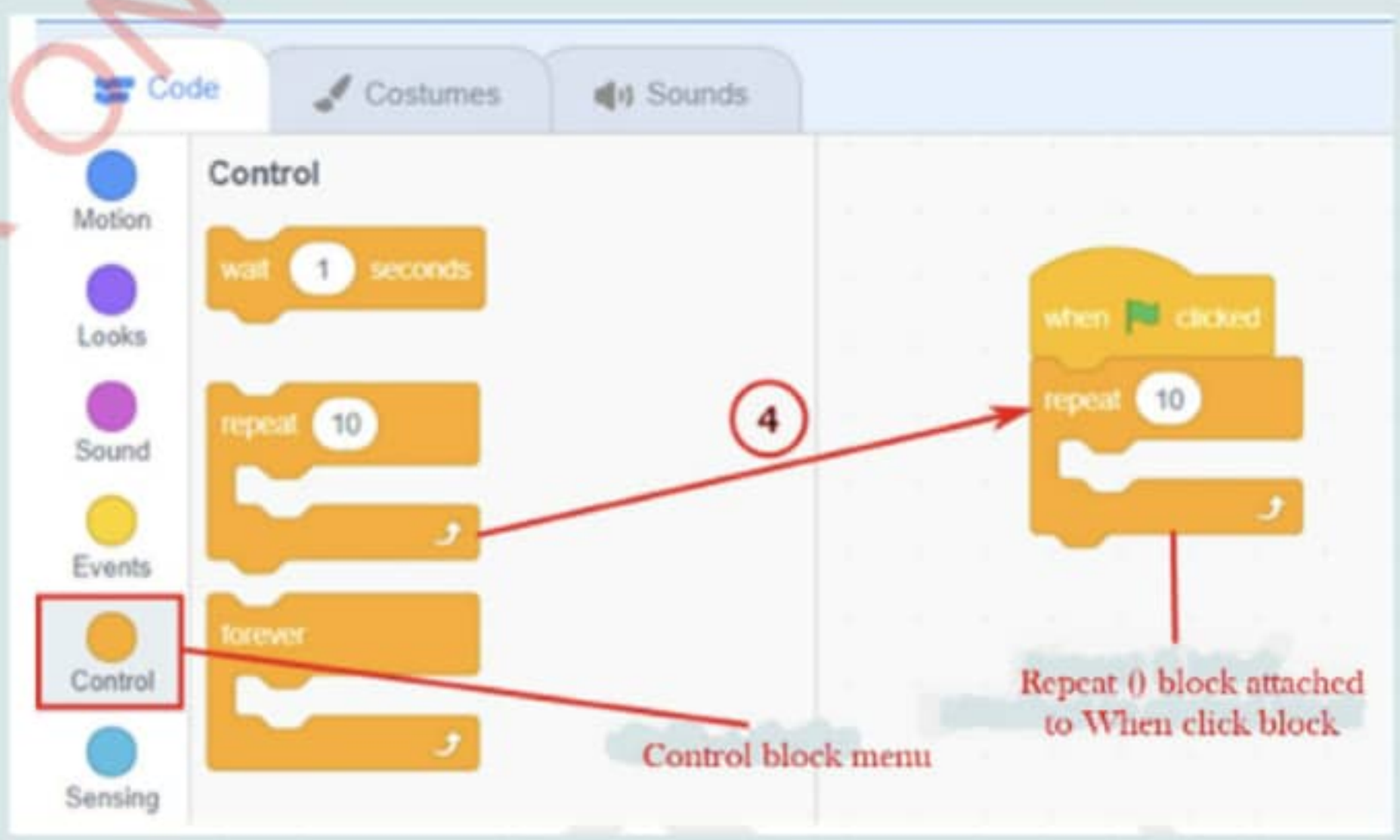


Figure 33: *Using the repeat block in a program*

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5. Click on the '*Sounds*' menu.

6. Insert the '*play sound () until done*' block inside the '*repeat*' block as shown in Figure 34.



Figure 34: *Inserting the sound block inside the repeat block*

7. You will get a script as shown in Figure 35.

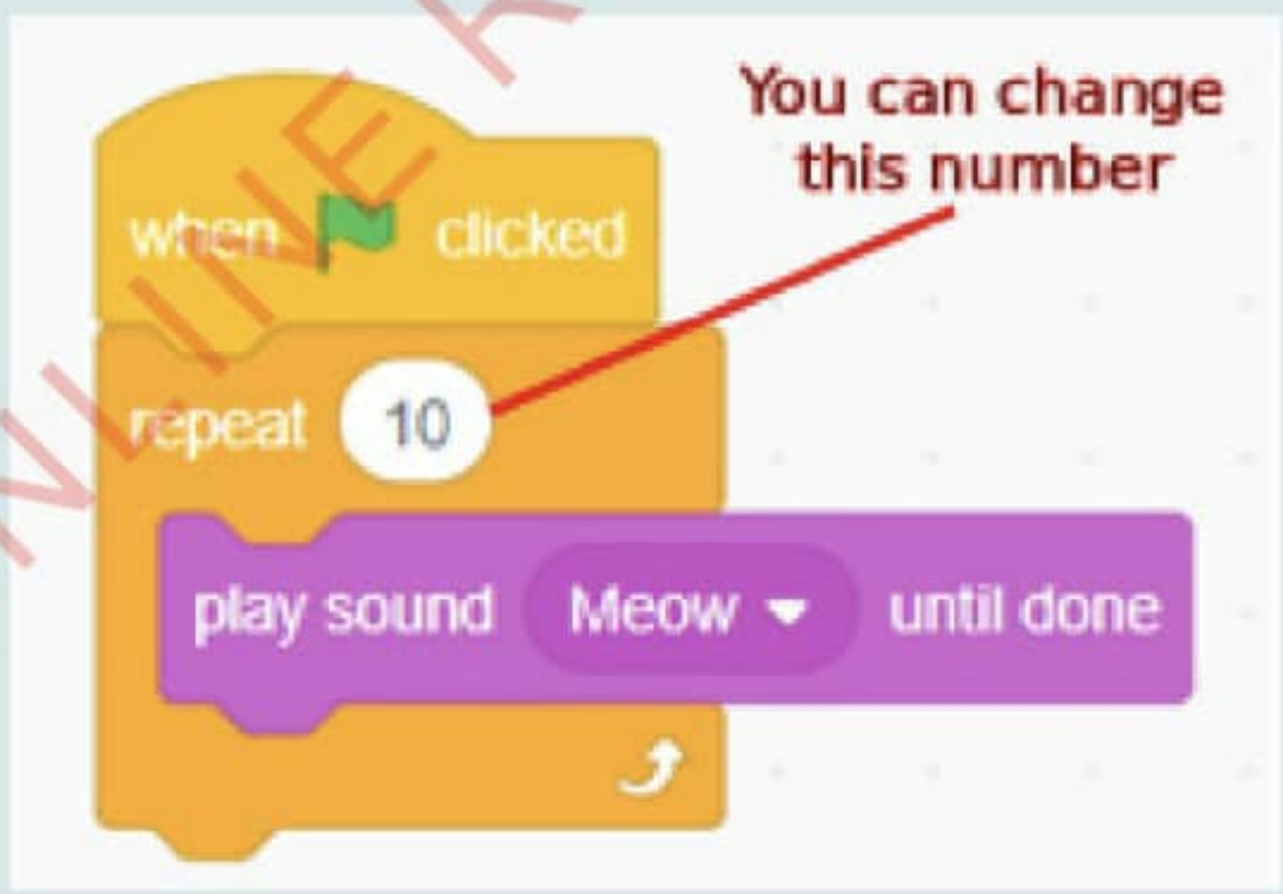


Figure 35: *Blocks for playing sound repeatedly*

8. Play the game and count how many times the sound plays.

9. Change the number from 10 to 15. Play the game again and count how many times the Sprite is heard.

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Activity 12

Playing sound continuously

To play sound repeatedly without end, follow these steps:

Steps

1. Click on the '*Events*' block menu.
2. Drag and drop the '*when clicked*' block on the script area.
3. Click on the '*Control*' block menu.
4. Drag and drop the '*forever*' block on the specific area. Attach the forever block to the '*when clicked*' block as shown in Figure 36.
5. Click on the '*Sound*' block menu.
6. Drag and drop the '*play sound until done*' block inside the '*forever*' block. You will get the blocks as shown in Figure 36.

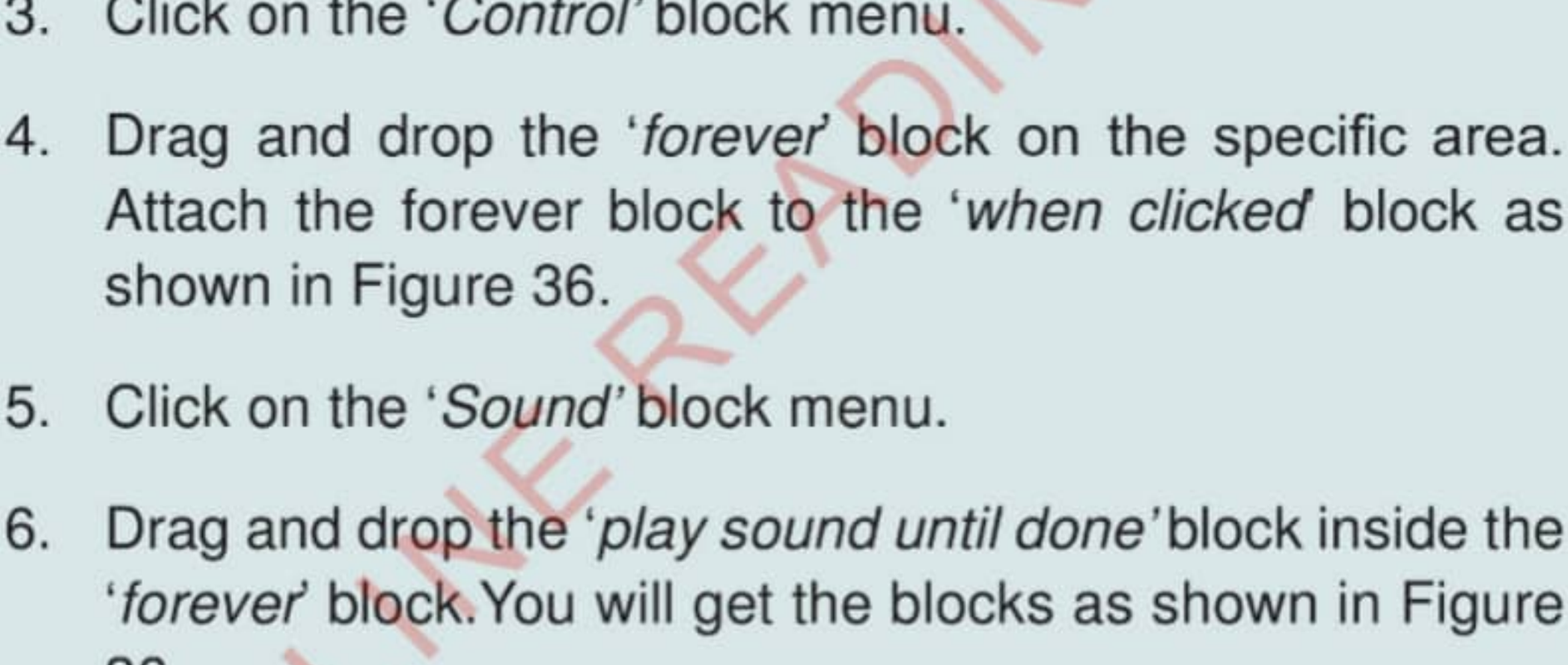


Figure 36: *Using the forever block*

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7. Play the game as many times as you like. How many times did you hear the sound?

8. Click the stop button if you want to stop the program.

Decision structure in programming

Sometimes it happens that, we need to do something if a certain condition is true. If the condition is false, we choose not to do it. The following activities will guide you in making decisions while preparing a computer program.

Activity 13

Using decisions in a sound program

The following steps will guide you in using decisions when creating a sound game.

Steps

1. Click on the '*Events*' block menu.

2. Drag and drop the '*when clicked*' block on the script area.

3. Click on the '*Control*' block menu.

4. Drag and drop the '*if () then*' block on the script area. Attach the 'if () then' block to the '*when clicked*' block, as shown in Figure 37.

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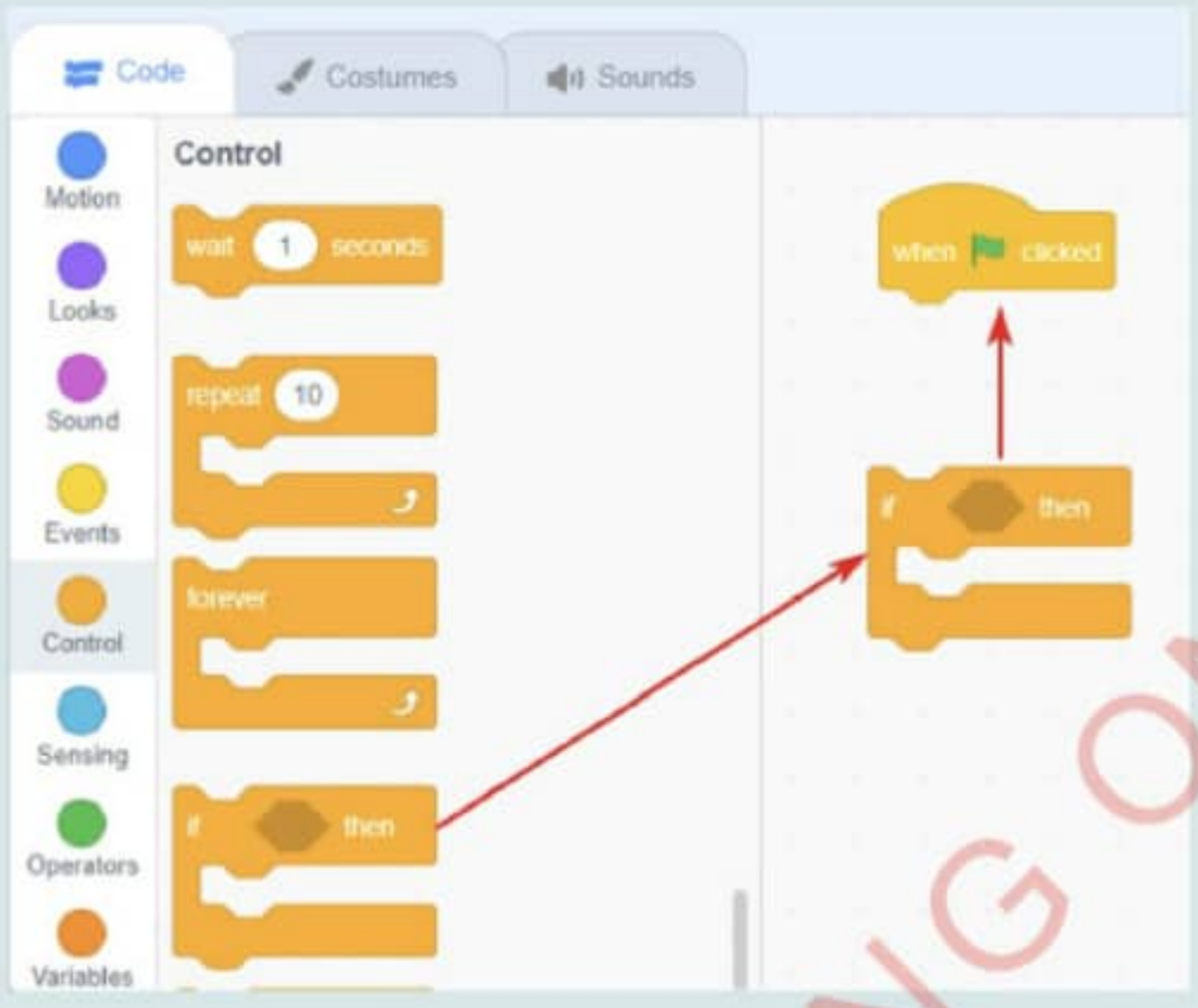


Figure 37: *Using the if then block*

5. Click on the '*Operators'* block menu.

6. Drag and drop the '*greater than'* block on the script area, as shown in Figure 38.

7. Insert the '*greater than*' block into the 'if () then' block as shown in Figure 38.

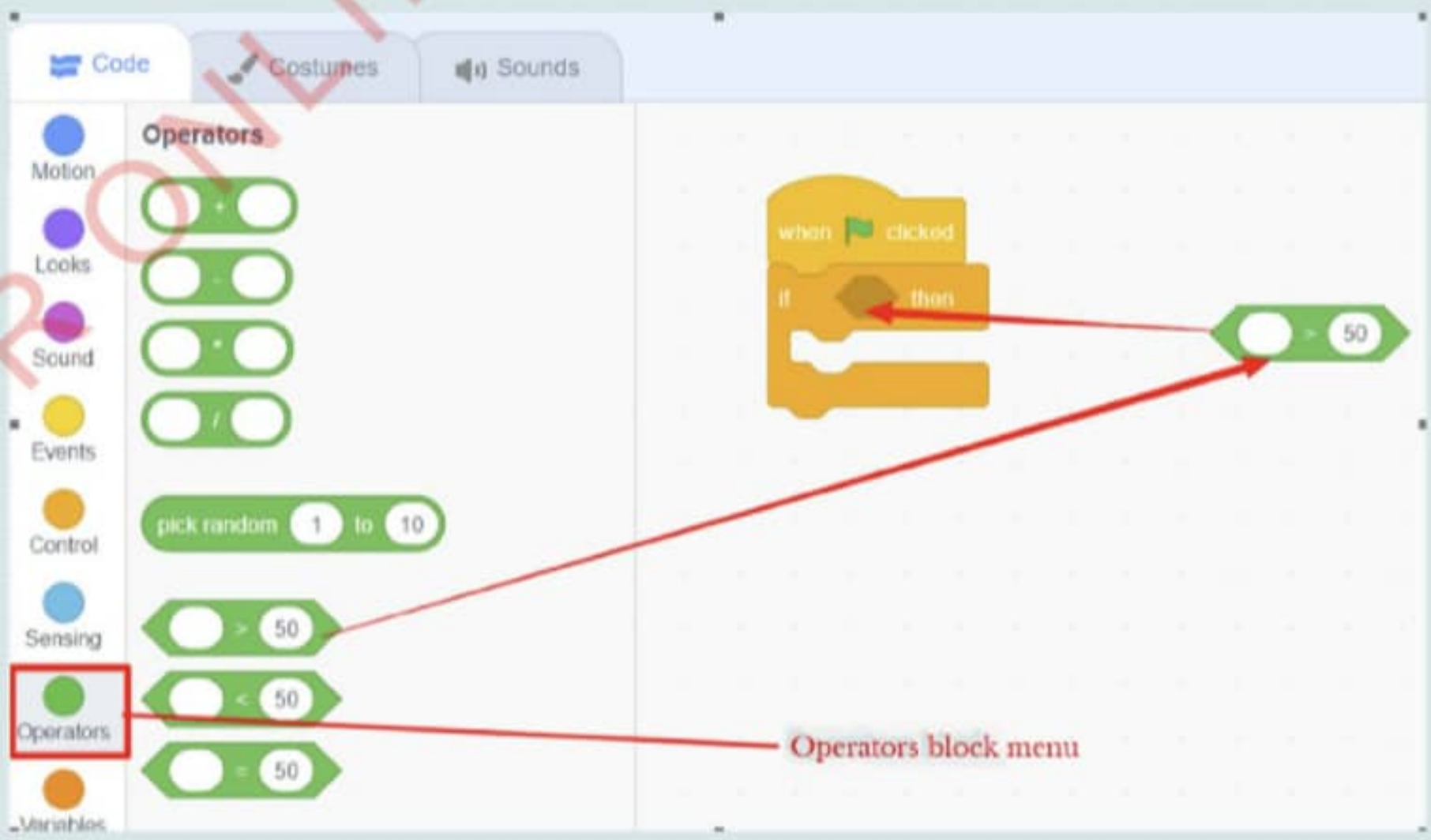


Figure 38: *Using the decision structure in a program*

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8. In the first slot of the '*greater than'* block, enter the number 10.

9. In the second slot of the '*greater than*' block, enter the number 6.

10. Click on the '*Sound'* block menu.

11. Drag and drop the '*play sound ( ) until done*' block inside the '*if ( ) then*' block. See Figure 39.

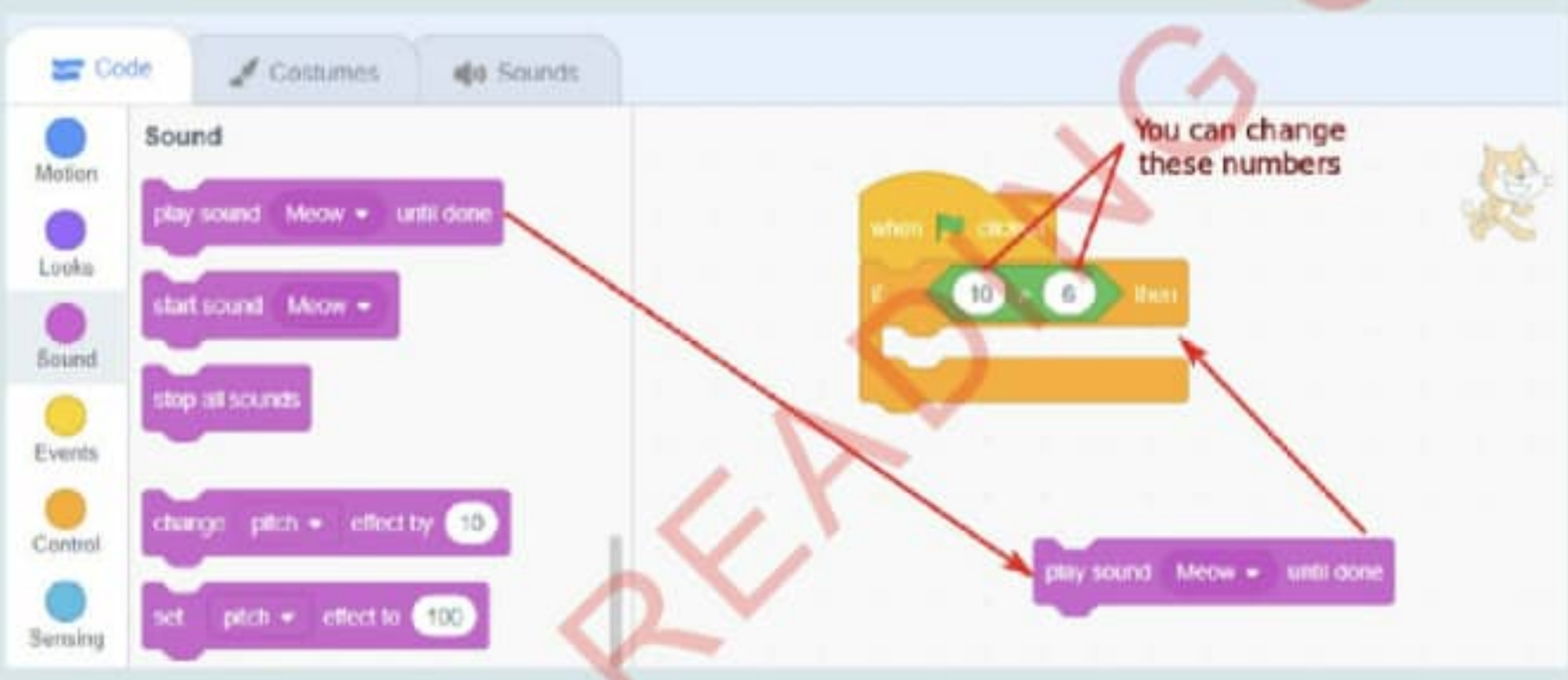


Figure 39: *Inserting the sound block inside the if then block*

You will get the script as shown in Figure 40.

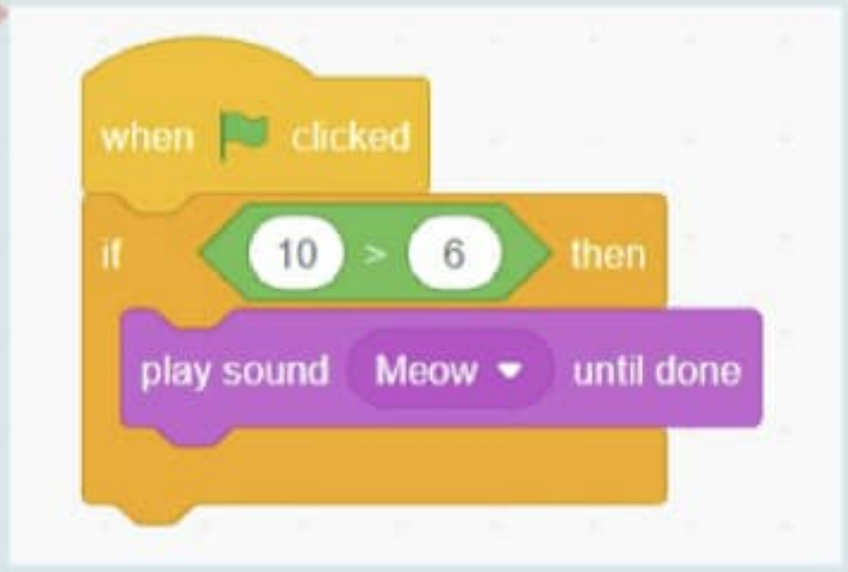


Figure 40: *A computer program that plays sound when a condition is true*

12. Play the game when the number in the first slot is 10 and the number in the second slot is 6. This is a true condition because 10 is greater than 6. What did you hear in this step?

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13. Change the numbers so that the first slot has the number 6 and the second slot has the number 10. Play the game again. This is not a true condition because 6 is not greater than 10. What did you hear this time?

Activity 14: Creating a game to make a 'Sprite'(Cat) go up and down while producing a sound

The following steps will guide you in creating a game that will enable a player to move a Sprite up or down. Also, the Sprite will produce a sound.

Steps

1. Click on the *'Events'* block menu.
2. Drag and drop the '*when key pressed*' block to the script area, as shown in Figure 41. Then, select the '*up arrow*'.

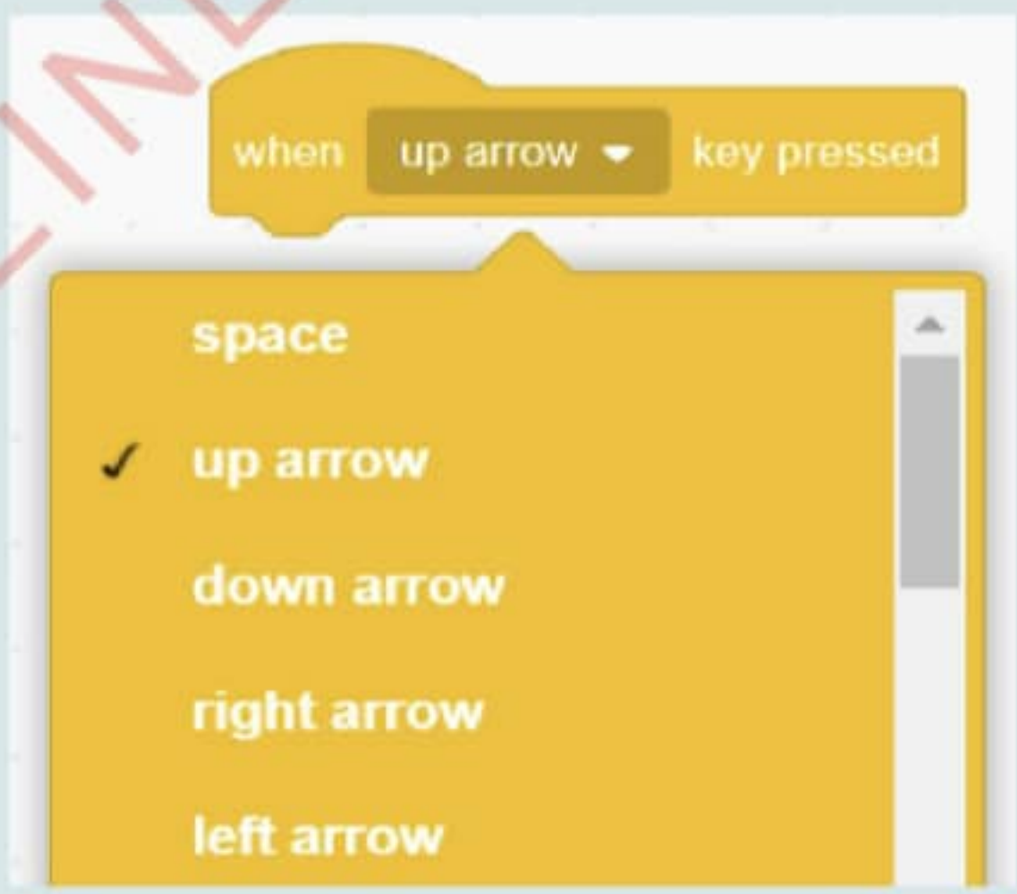


Figure 41: *Selecting 'up arrow'*

3. Click on the *'Motion'* block menu.
4. Drag and drop the *'move steps'* block into the script area.

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Then, connect the *'move steps'* block to the *'when key pressed'* block, as shown in Figure 42.

5. Click on the *'Sound'* block menu.

6. Drag and drop the block *'play sound until done'* into the script area. Then, connect that block to the *'move steps'* block, as shown in Figure 42.

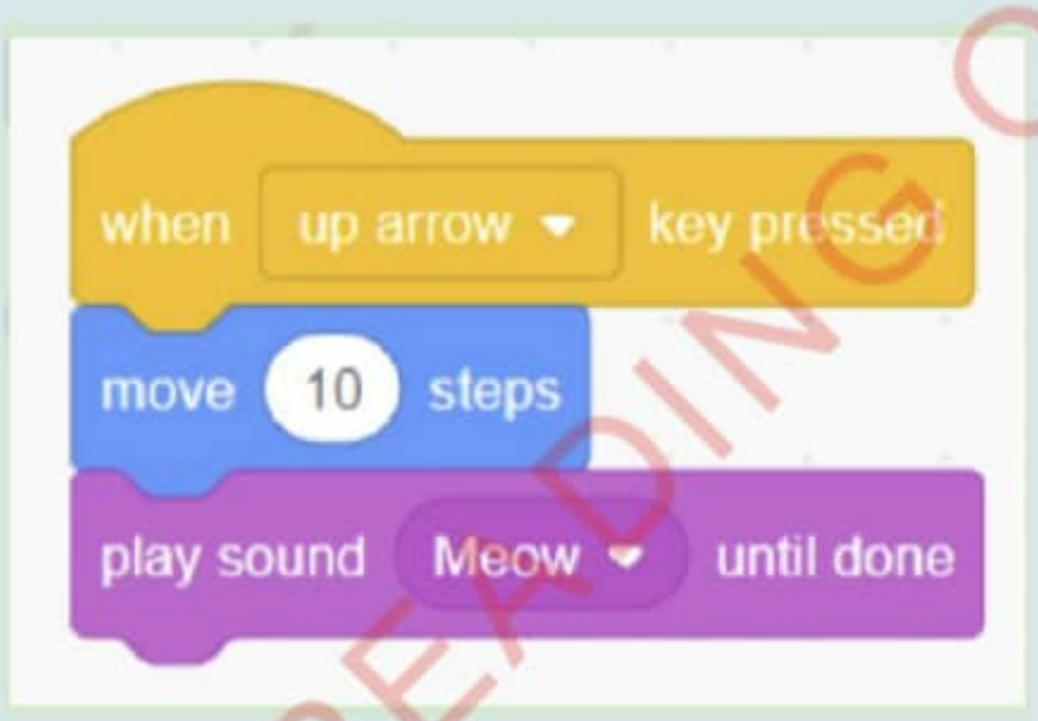


Figure 42: *Adding the 'play sound until done' block*

7. After that, repeat the previous steps to create a program for the 'arrow down', as guided in step 8.

8. Click the *'Event'* block menu.

9. Drag and drop the block *'when key pressed'* into the writing into the script area shown in Figure 43. Place that block at the bottom of the script area. Then select *'down arrow'*.



Figure 43: *Selecting 'arrow down'*

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10. Click the **'Motion'** block menu.

11. Drag and drop the **'turn degrees'** block into the script area. Then, connect the **'turn degrees'** block to the **'when key pressed'** block, as shown in Figure 44.

12. Click the **'Sound'** block menu.

13. Drag and drop the 'play sound until done' block into the script area. Then connect that block to the 'rotate degrees' block as shown in Figure 44.

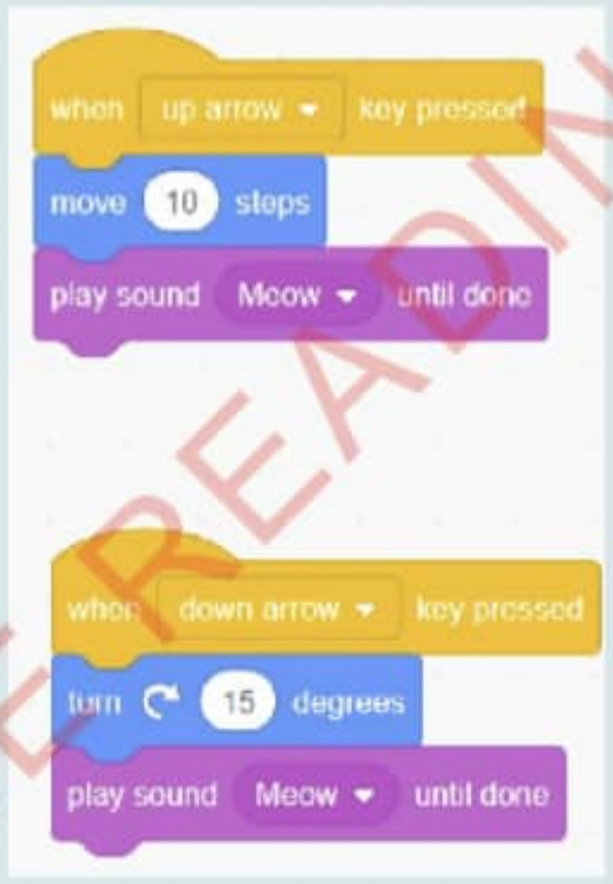


Figure 44: **A program 'up arrow' and 'down arrow'**

14. Play your game using the up arrow and down arrow keys. These keys are found on your computer keyboard. What have you seen and heard?

15. In the **'move steps'** block, change from 10 steps to 50 steps.

16. In the **'turn degrees'** block, change from 10 to 30 degrees.

17. What changes do you observe when you compare the current findings to the ones you observed before adjusting the steps and degrees?

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Activity 15

Creating a game to make a 'Sprite'(Cat) move in four directions

The following steps will guide you to create a game that makes a Sprite to move up, down, left, and right while making sounds.

Steps

1. Click on the *'Events'* block menu.
2. Drag and drop the *'when key pressed'* block into the script area. Then select *'up arrow'*.
3. Click on the *'Motion'* block menu.
4. Drag and drop the *'point in direction'* block into the script area. Then attach this block to the *'when key pressed'* block as shown in in Figure 45.

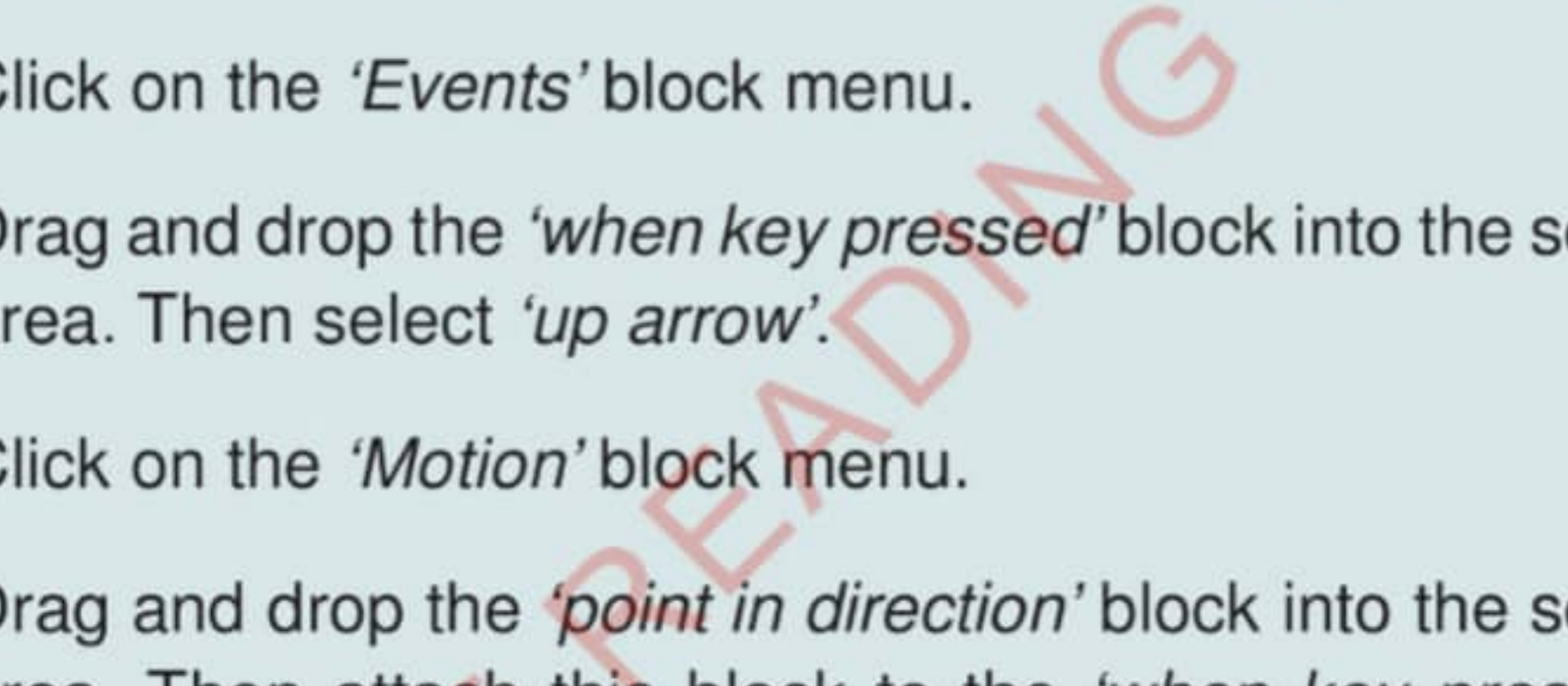


Figure 45: *The 'point in direction' block attached to the 'when key pressed' block.*

5. Change the degrees in the *'point in direction'* block from 90 to 0, as shown in Figure 46. This will make the Sprite move up when the *'up arrow'* key is pressed.

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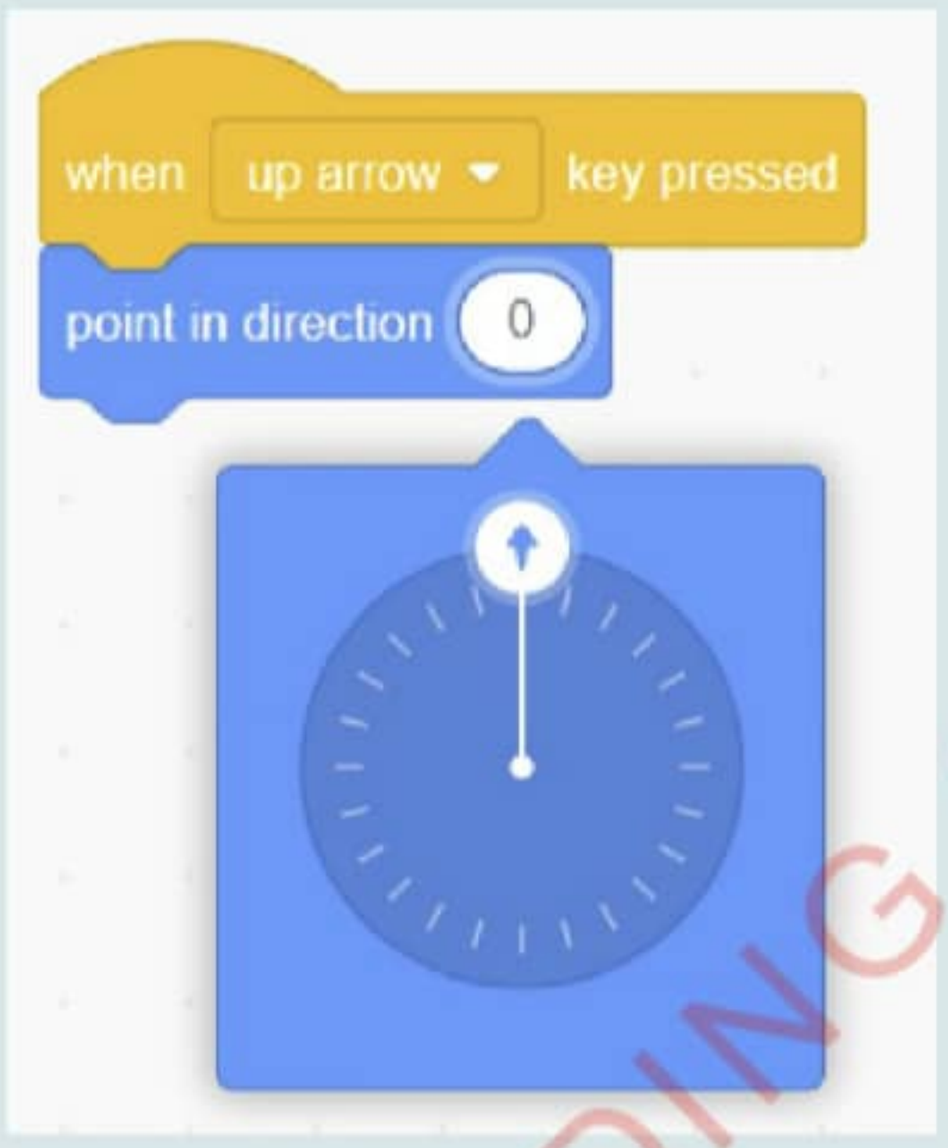


Figure 46: *Changing degrees*

6. Drag and drop the *'move steps'* block into the script area. Then, connect it to the *'point in direction'* block.

7. Click on the *'Sound'* block menu.

8. Drag and drop the *'play sound until done'* block into the script area. Then, connect it to the *'move steps'* block. You will get the first program, as shown in Figure 47.

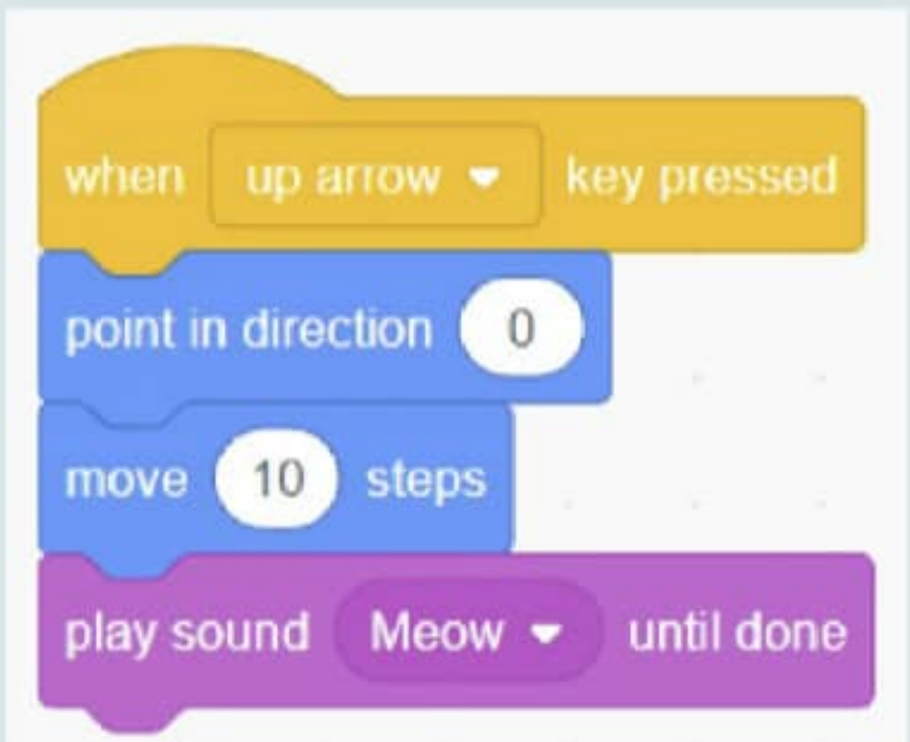


Figure 47: *A program to make the 'Sprite' move up*

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9. Place the mouse on the program you created, then right-click the mouse. You will see a menu as shown in Figure 48.

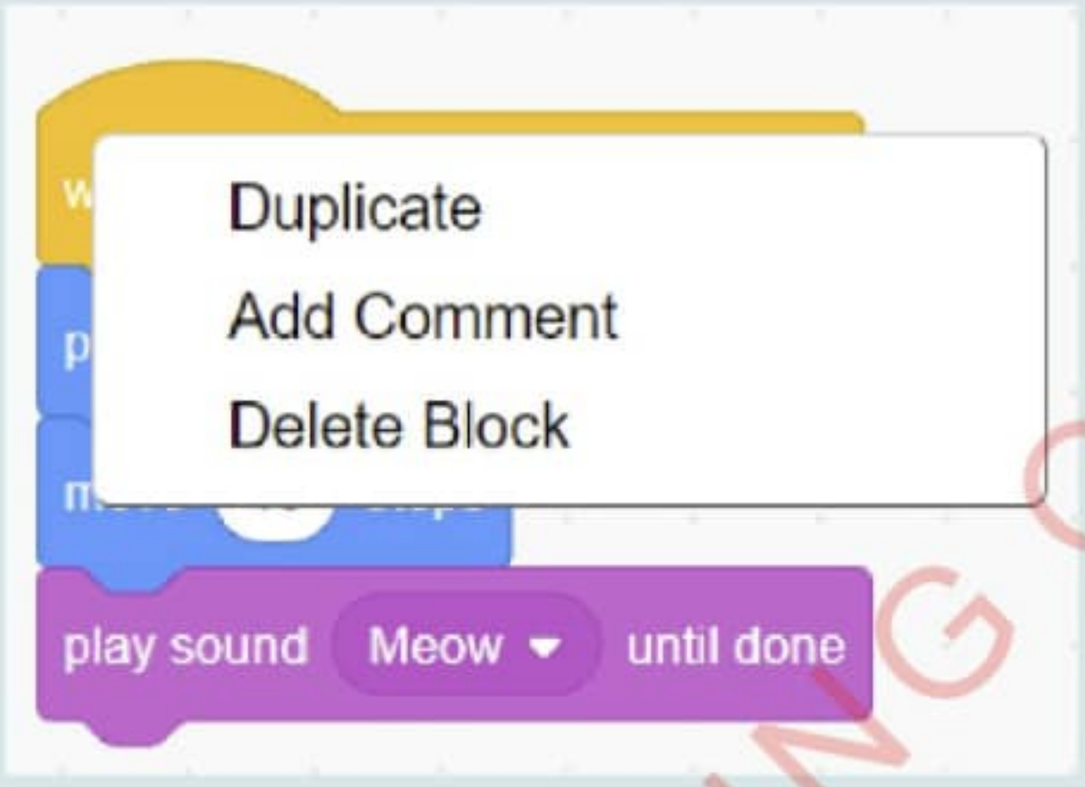


Figure 48: *A program to make the 'Sprite' move up*

10. Select *'Duplicate'*, then move the mouse down and click the left button once. You will have created a second program by duplicating the first one, as shown in Figure 49.

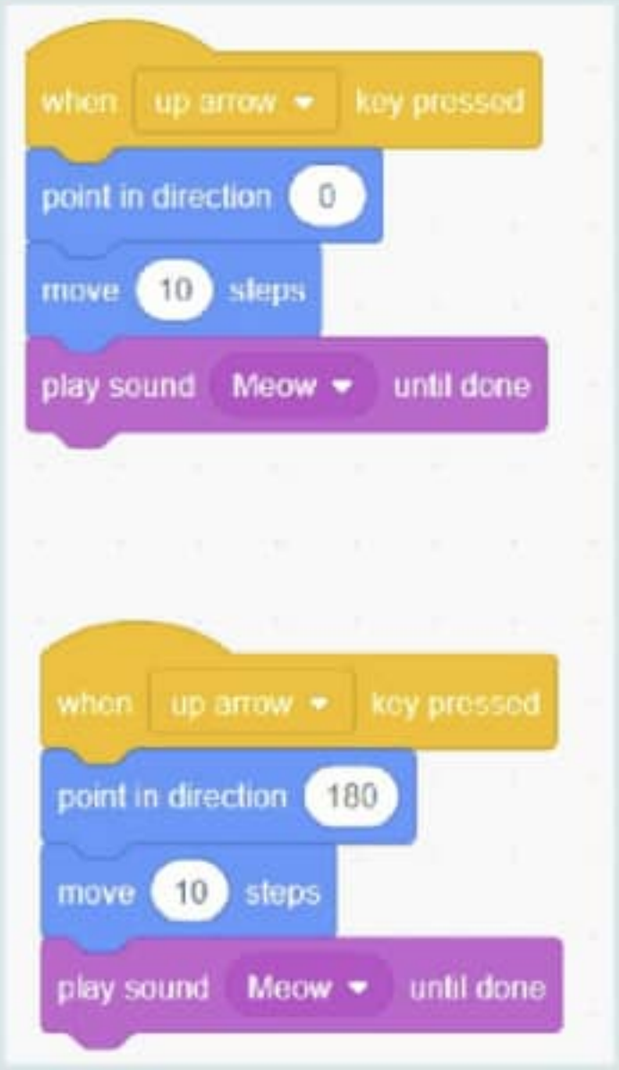


Figure 49: *A duplicated program*

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11. In the second program, inside the *'when key pressed'* block, change the arrow to the *'down arrow'*. Then, inside the *'point in direction'* block, change the degrees to 180. The program will change, as shown in Figure 50.

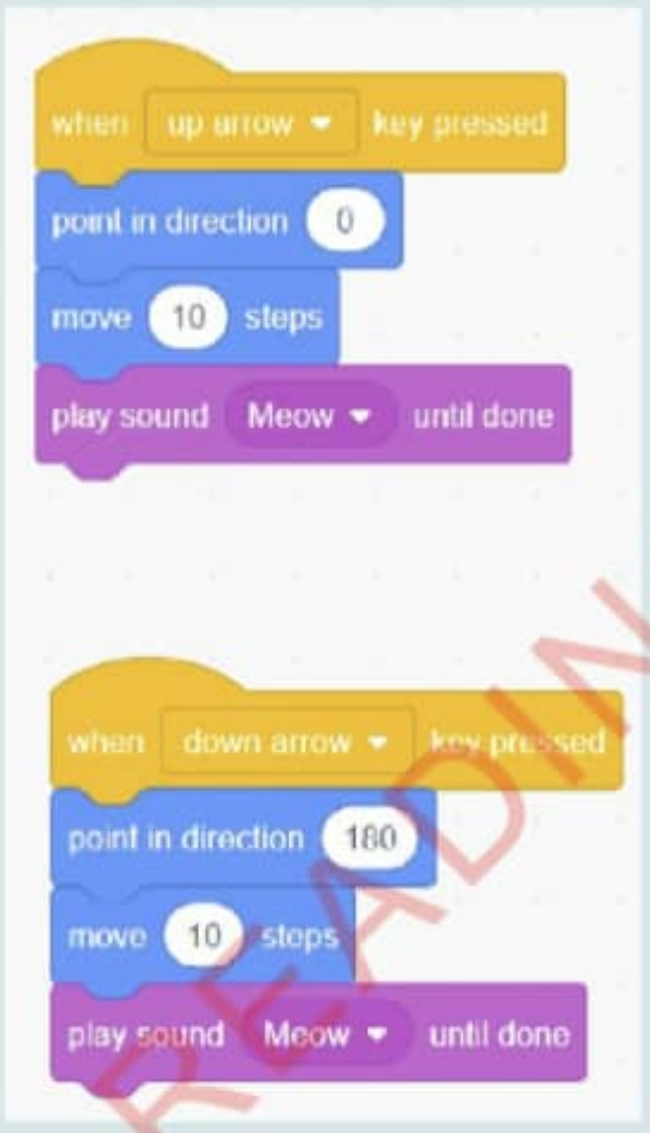


Figure 50: *A duplicated program with modified degrees and steps*

12. Make another duplicate (as you did in step number 9). Place that program on the right side. Then, change the arrow to the *'right arrow'* and set the direction to 90 degrees.

13. Make another duplicate (as you did in step number 11). Place that program on the left side. Then, change the arrow to the *'left arrow'* and set the direction to 270 degrees. Those degrees will automatically adjust to '-90'.

14. You will have the completed programs, as shown in Figure 51.

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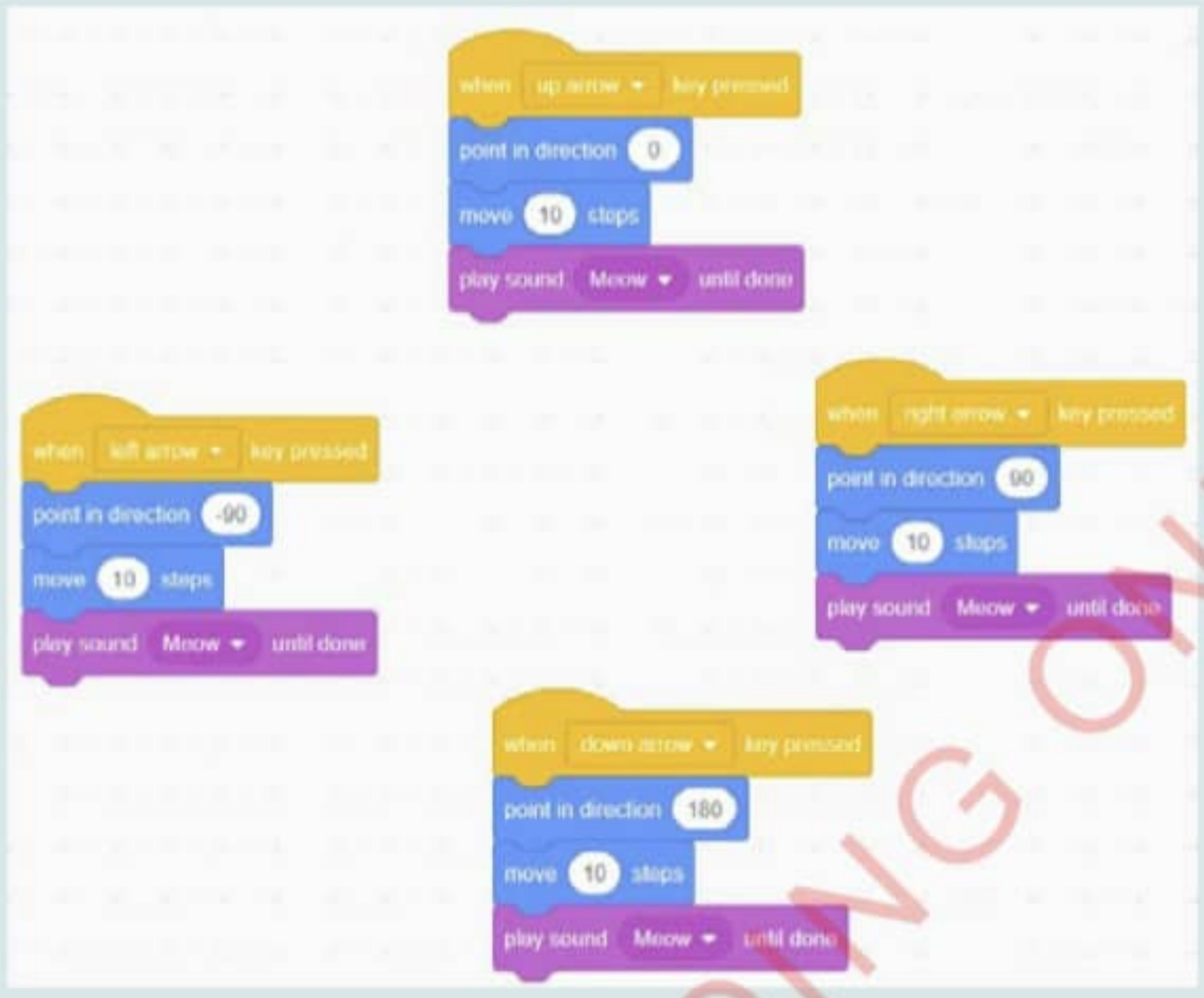


Figure 51: *Duplicated program*

15. You can start playing your game using the *'up'*, *'down'*, *'left'*, and *'right'* arrow keys on your computer keyboard.

16. What do you see and hear? How many directions were you able to move the 'Sprite'?

17. You can increase the steps in the *'move steps'* block, then play the game again.

Activity 16: *Creating a game to make the 'Sprite' (City Bus) move in different directions*.

The following steps will guide you in creating a game to make the 'City Bus' move in different directions.

1. Choose a 'Sprite' by clicking on the sprite icon as shown in Figure 52.

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Figure 52: *Adding a 'Sprite'*

2. Search for 'Sprite' by typing the word 'car' as shown in Figure 53.

3. Then select the 'Sprite', for example, choose 'City Bus' by clicking on that 'Sprite'.

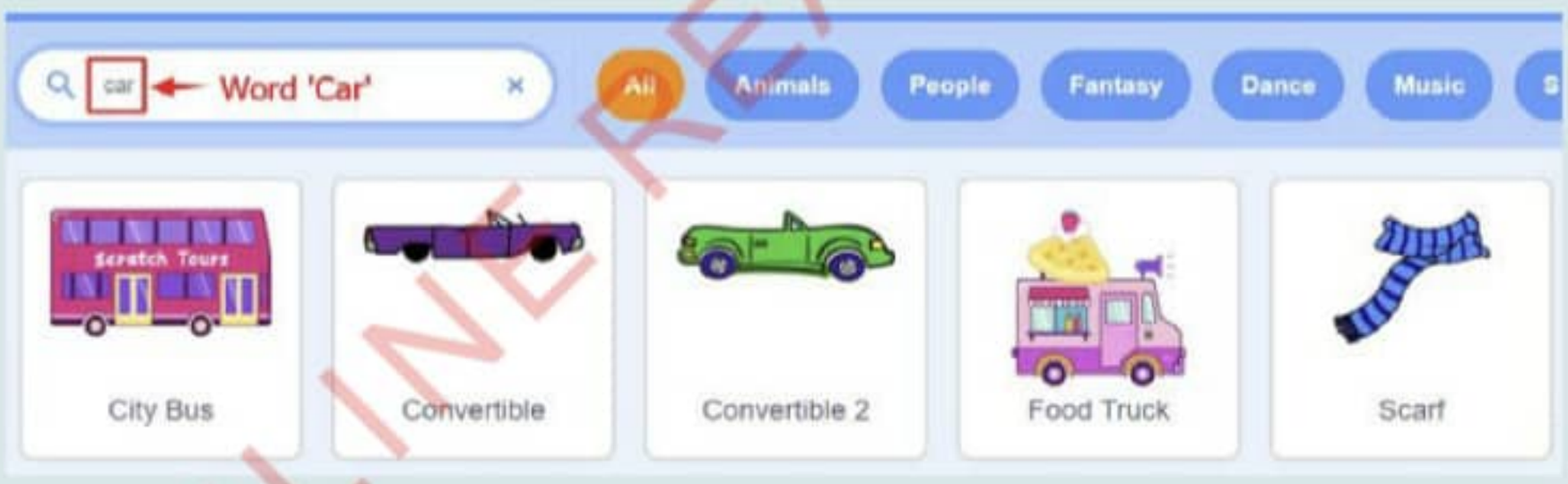


Figure 53: *Finding and selecting a 'Sprite'*

4. You will find two Sprites as shown in Figure 54.



Figure 54: *After adding a second 'Sprite'*

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5. To remove one Sprite, click on the 'Cat' Sprite, then click the delete icon as shown in Figure 55.

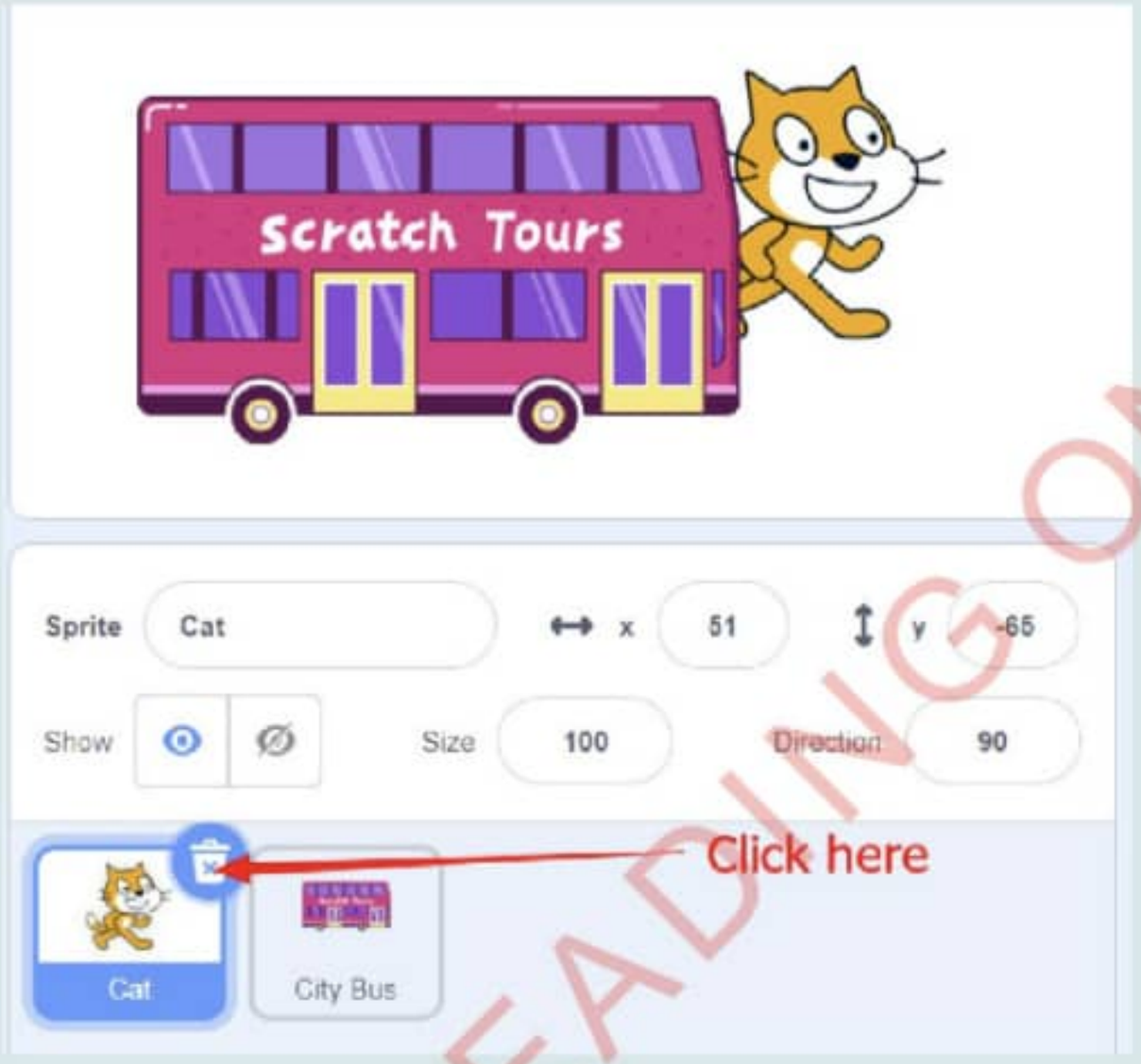


Figure 55: *Removing a 'Sprite' 'Cat'*

6. After that, you will remain with one Sprite called 'City Bus,' as shown Figure 56.



Figure 56: *Remained with one Sprite 'City Bus'*

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7. Click on the *'Events'* block menu.
8. Drag and drop the *'when clicked'* block into the script area.
9. Click on the *'Motion'* block menu.
10. Drag and drop the *'set rotation style'* block into the script area. Then, connect it to the *'when clicked'* block. Choose *'don't rotate'* as shown in Figure 57.

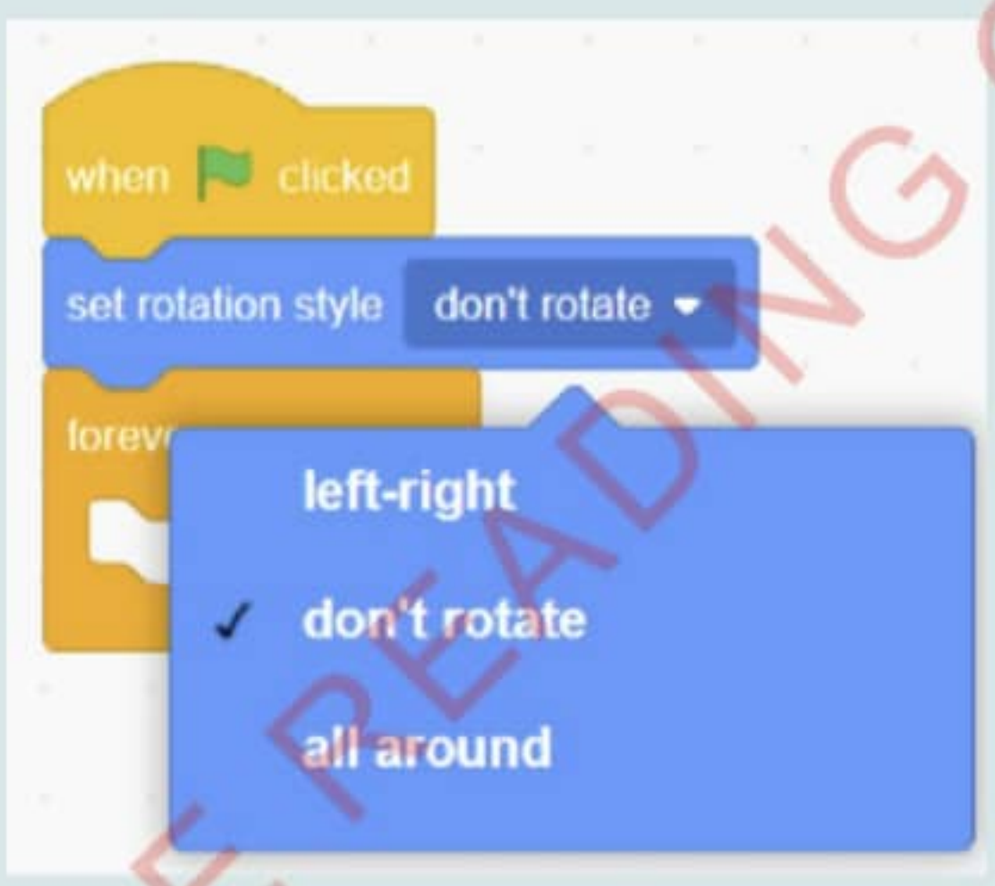


Figure 57: *Set "rotation style" block*

11. Click on the *'Control'* block menu.
12. Drag and drop the *'forever'* block into the script area. Then, connect it to the *'set rotation style'* block.
13. Drag and drop the *'if then'* block into the script area. Then, place it inside the *'forever'* block.
14. Click on the *'Sensing'* block menu.
15. Drag and drop the *'key pressed'* block into the script area. Then insert it inside the *'if then'* block as shown in Figure 58.

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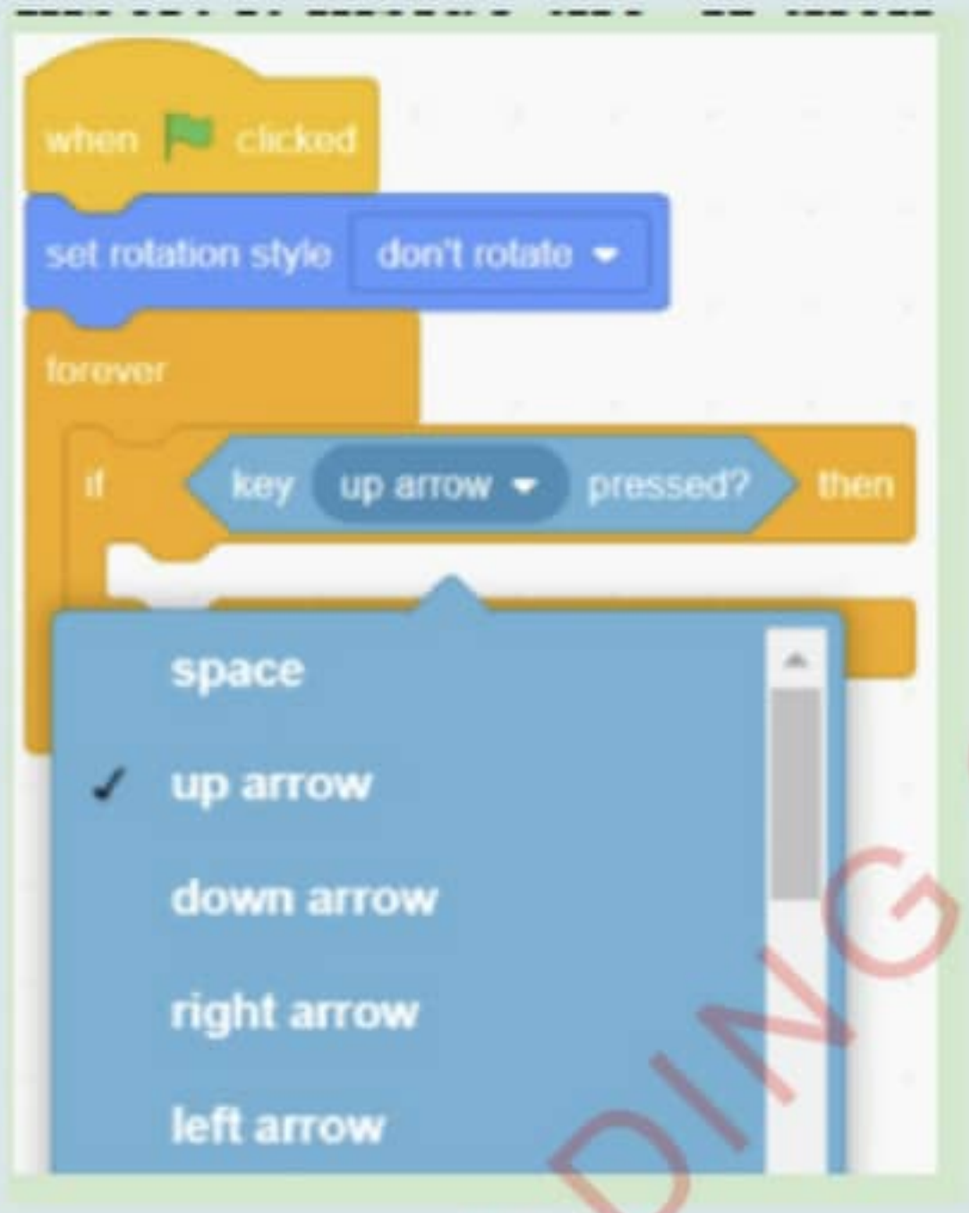


Figure 58: **Select 'up arrow'**

16. Click on the **'Motion'** block menu.

17. Drag and drop the **'point in direction'** block into the script area. Then insert it inside the **'if then'** block, and change the degrees from 90 to 0 as shown in Figure 59.

18. Drag and drop the **'move steps'** block into the script area. Then insert it inside the **'if then'** block and connect it to the **'point in direction'** block as shown in Figure 59.



Figure 59: **The 'point in direction' and 'move steps' blocks when connected**

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19. Click on the *'Control'* block menu.

20. Drag and drop the *'if then'* block into the script area. Then, connect it inside the *'forever'* block, as shown in Figure 60.



Figure 60: *The 'if then' block is connected for the second time*

21. Click on the *'Sensing'* block menu.

22. Drag and drop the 'key pressed' block into the script area. Then insert it inside the 'if then' block, and choose 'right arrow' as shown in Figure 61.

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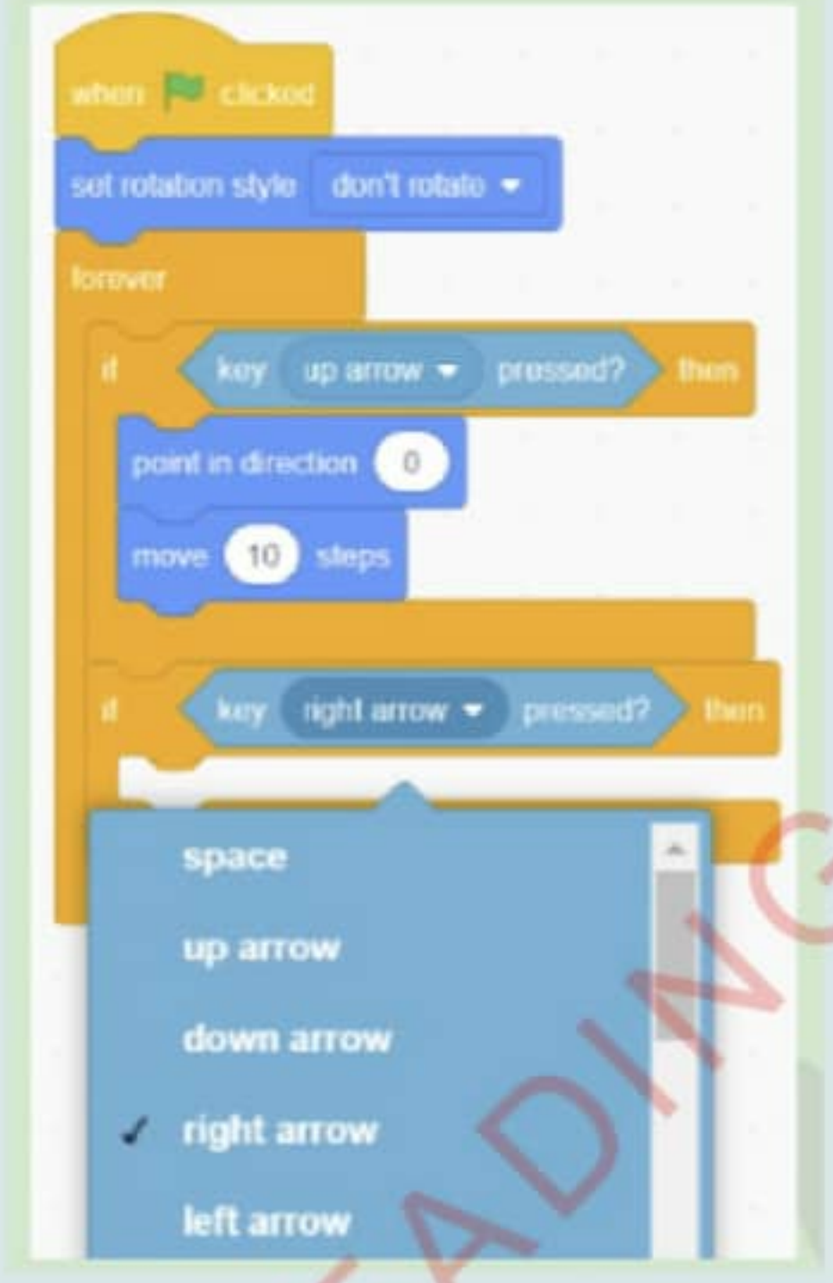


Figure 61: *The 'if then' block is connected for the second time*

23. Click on the *'Motion'* block menu.

24. Drag and drop the *'point in direction'* block into the script area. The insert it inside the *'if then'* block as shown in Figure 62.

25. Drag and drop the *'move steps'* block into the script area. Then insert it inside the *'if then'* block and connect it to the *'point in direction'* block, as shown in Figure 62.

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Figure 62: *The 'if then' block is connected for the second time*

26. To play your game, click *'Go'*. Then, use the up arrow and right arrow keys on your computer keyboard.

27. You can change the steps and degrees to see changes in the movement and direction of the 'Sprite' (car).

Revision exercise

Section A: Choose the correct answer

1. What software can you use to prepare computer program?

   (a) Scratch

   (b) Sprite

   (c) Paint

   (d) Script

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2. What happens when you click on the block with the green flag?

   (a) The program runs

   (b) The sprite is guided/led to a specific direction

   (c) If conditions are correct, the blocks inside will be executed

   (d) Script execution stops

3. What happens when you click 'move 10 steps?

   (a) Sprite moves once specific key clicked

   (b) Sprite moves forward

   (c) Sprite moves backward

   (d) Sprite stops

4. In Scratch, what is the uses of 'forever' block?

   (a) Creating images

   (b) Listening to sound

   (c) Repeating commands without limit

   (d) Repeating commands with limit

5. Which block menu would you use to make the Sprite rotate?

   (a) Events

   (b) Sound

   (c) Looks

   (d) Motion

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6. Which block menu guides the Sprite in the game?

(a) Operator

(b) Motion

(c) Control

(d) Sensing

7. What will happen to a sprite if the number in the "go to (step)" block increases?

(a) The sprite will move faster

(b) The sprite stops

(c) The sprite moves backward

(d) The sprite turns right

8. What will happen if you arrange Scratch instruction blocks in the wrong order?

(a) The program will function slowly

(b) The program will not function correctly

(c) The program stops

(d) The speed will increase

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Section B: Match the word in Section A with the statement from Section B.

9.

Section A	Section B
(i) Sound	(a) Go or move or turn or change according to the instruction
(ii) Events	(b) Stop playing the game
(iii) Control	(c) Applying mathematical operations and statements according to the instruction
(iv) Sensing	(d) Continue playing the game
(v) Operators	(e) Display or execute the blocks of the game
(vi) Motion	(f) Play or change or stop sounds
	(g) Wait or repeat according to the instruction
	(h) Touching mouse-pointer or colour

Section C: Practical questions

10. Prepare a program that can play a sound and move the Sprite 35 steps.

11. Prepare a program that allows the Sprite to play a sound and move 15 steps, then have the program repeat 12 times.

12. (a) Prepare a program that enables the Sprite to go to any position using the "greater than" operator. Enter

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the number 8 in the first box and 5 in the second box. Then, try running your program. What happens after you try it?

(b) Repeat question 12(a) by entering the number 9 in the first box and 13 in the second box. What happens?

13. Draw a rectangle and place it on top of two circles of equal size. One circle should be on the left side and the other on the right side. Colour the drawings.

Vocabulary

Drag	select and move to a particular location
Logic	something that shows clear and relevant reasons leading to a valid conclusion
Sequence	set of items arranged in a particular order

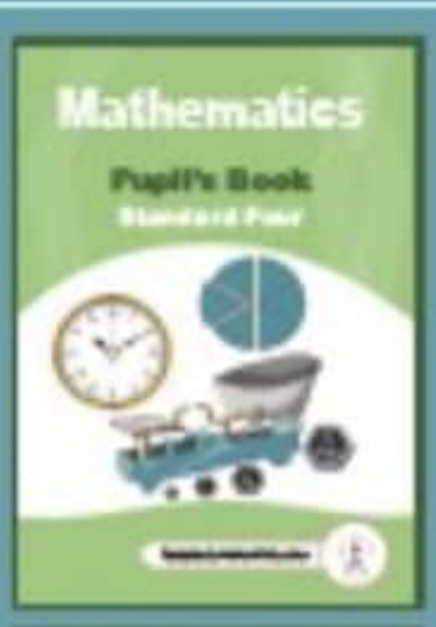
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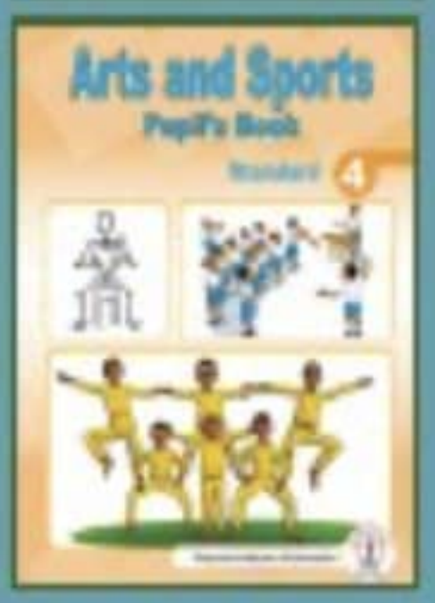


















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